Notes on Algebraic Number Theory

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0. Introduction

This is a set of notes I am writing to help me study Algebraic Number Theory. The main goal is just to speed through all the content as fast as possible so I can remember everything easily. I will also occasionally reference results from later in the textbook, since I have prioritized conceptual organization over a linear flow of results. Despite this I have taken care that there is no circular reasoning. I have also included some "problems", which are mainly just things I don't want to write out.

"Wait, it's all lattices?"

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1. Commutative Algebra

In this section I will speed through a lot of the basic commutative algebra and background knowledge needed to do Algebraic Number Theory.

1.1. Fractional Ideals

Definition 1.1 (Fractional Ideals)**.** Let R be an integral domain and K its field of fractions. A *fractional ideal* I is an R-submodule of K such that there exists some nonzero $r \in R$ with $rI \subseteq R$. We call I *invertible* if there exists a fractional ideal J with $IJ = R$.

In general it is hard to understand multiplication in interesting rings. However if we can instead understand multiplication in the *group of invertible fractional ideals* we begin making progress.

Theorem 1.2.

(a) If I is an invertible fractional ideal with $IJ = R$ then $J = R :_K I$.

(b) We have the implications

$$I \text{ non-zero principal} \quad \Rightarrow \quad I \text{ invertible} \quad \Rightarrow \quad I \text{ fractional}$$

Proof.

(a) Since $IJ = R$, we have $J \subseteq (R :_K I)$ and $I(R :_K I) \subseteq R$. Therefore $R = IJ \subseteq I(R :_K I) \subseteq R$, so $(R :_K I)$ is an inverse. Therefore it suffices to prove that inverses are unique. Suppose J, J' invert I and write:

$$J = JR = J(J'I) = J'(JI) = J'$$

(b) A principal R-submodule of K, (x) has inverse (x^{-1}). If I is an invertible fractional ideal then any nonzero element $x \in (R :_K I)$ must have $xI \subseteq R$ as desired.

□

In general for an R-submodule of K we will denote $(R :_K I)$ as I^{-1}.

Theorem 1.3. *Any finitely generated R-submodule of K is a fractional ideal. Moreover if R is Noetherian then the opposite implication holds.*

Proof.

$\Rightarrow$**:** Suppose that $I = (a_1/b_1, \dots, a_n/b_n)$ for $a_i, b_i \in R$. Then if $x = b_1 \cdots b_n$ we see $xI \subseteq R$.

$\Leftarrow$**:** By assumption we may find nonzero x with $xI \subseteq R$. Then by the Noetherian condition $xI = (a_1, \dots, a_n)$ so $I = (a_1/x, \dots, a_n/x)$ as desired.

□

Theorem 1.4. *Let I be an invertible fractional ideal of R. Then I is finitely generated, and moreover if R is local then I is principal.*

Proof. Suppose that $II^{-1} = R$. Then since $1 \in II^{-1}$ we may find $a_i \in I$ and $b_i \in I^{-1}$ so that

$$\sum_{i=1}^{n} a_i b_i = 1$$

and $\langle a_1, \dots, a_n \rangle I^{-1} = R$. However by Theorem 1.2 I is the unique inverse to I^{-1} so $I = \langle a_1, \dots, a_n \rangle$ is finitely generated. In the case that R is local let $\mathfrak{m}$ be its maximal ideal and take modulo $\mathfrak{m}$ to see at least one of $a_k b_k$ is not in $\mathfrak{m}$. However since R is local this means that $a_k b_k = u$ is a unit. Therefore $\langle a_k \rangle I^{-1} = R$ so by the same argument $I = \langle a_k \rangle$. □

Lemma 1.5. *If R is a 1-dimensional Noetherian local domain with maximal ideal $\mathfrak{m}$, and I is a nonzero proper ideal, then I^{-1} strictly contains R.*

Proof. Pick some nonzero $x \in I$. Since $\sqrt{(x)} = \mathfrak{m}$ and I is finitely generated, there is some n with $I^n \subseteq (x)$. Let n be minimal, so $x^{-1}I^n \subseteq R$ but $x^{-1}I^{n-1} \nsubseteq R$. Then any element $y \in x^{-1}I^{n-1} \setminus R$ satisfies $yI \subseteq R$ and $y \notin R$, showing that $R \subsetneq I^{-1}$. □

1.2. Discrete Valuation Rings

Definition 1.6 (Valuation)**.** A *valuation* on a field K is a mapping $v : K \to G \cup \{\infty\}$, where G is a totally ordered abelian group, satisfying:

(a) $v(a) = \infty \iff a = 0$

(b) $v(ab) = v(a) + v(b)$

(c) $v(a + b) \geq \min\{v(a), v(b)\}$

(d) The image of $K^\times$ under v is nontrivial

We call v a *discrete valuation* if $G \cong \mathbb{Z}$.

It is worth noting some people do not enforce the last condition. We mainly do it to disallow the trivial valuation $v(K^\times) = 0$. To give some intuition for a valuation, here are some familiar examples:

- The p-adic valuation on $\mathbb{Q}$. For a prime p, $v_p(x)$ is the exponent of the highest power of p dividing x.
- The a-adic valuation on $k(x)$. For $a \in k$, $v_a(f)$ is the order of the zero (if positive) or pole (if negative) of f at $x = a$.
- The degree valuation on $k(x)$. Defined as $v_\infty(P/Q) = \deg(Q) - \deg(P)$.
- The Laurent series valuation. On $k((t))$, $v(f)$ is the smallest n such that the coefficient a_n of t^n is non-zero.
- The monomial valuation on $k(x, y)$. For a nonzero weight $w = (w_1, w_2) \in \mathbb{R}^2$, $v(\sum a_{ij}x^i y^j) = \min\{iw_1 + jw_2 : a_{ij} \neq 0\}$. Extend this to rational functions by $v(P/Q) = v(P) - v(Q)$.

In general, thinking of a valuation as a way to count the order of a zero or a pole of a function on a point gives very good intuition. The reason we care about valuations is their connection to the concept of a valuation ring.

Definition 1.7 (Valuation Rings)**.** We call an integral domain R which is not a field, with fraction field K, a *valuation ring* if one of the two equivalent conditions is satisfied:

(a) For every $x \in K^\times$ either x or x^{-1} is in R.

(b) There exists a valuation on K such that $R = \{x \in K : v(x) \geq 0\}$

Proof.

$(a) \Rightarrow (b)$**:** We define the valuation group $G = K^\times/R^\times$ to be the multiplicative group of K quotient the units of R. We write its operation additively. The valuation is $v(x) = \bar{x}$ for $x \neq 0$ and $v(0) = \infty$. We need to show three things:

G **is totally ordered:** We claim that the ideals of R are totally ordered by inclusion. This induces a total order on G by $\bar{a} \geq \bar{b}$ if $a/b \in R$, equivalently $aR \subseteq bR$. Suppose FTSOC that I and J are ideals such that we can find some $x \in I \setminus J$ and $y \in J \setminus I$. By assumption either x/y or y/x is in R, so suppose WLOG that $x/y \in R$. Then $x = y \cdot (x/y)$ which implies that $x \in (y) \subseteq J$, a contradiction.

$v(ab) = v(a) + v(b)$**:** This follows directly from the definitions.

$v(a + b) \geq \min(v(a), v(b))$**:** This translates to the statement that

$$(a + b) \subseteq \max((a), (b)) = (a, b)$$

where the equality uses the fact that either $(a) \subseteq (b)$ or $(b) \subseteq (a)$. The same argument applies to principal fractional ideals after clearing denominators. Since $R \neq K$, the valuation is nontrivial.

$(b) \Rightarrow (a)$**:** For $x \in K^\times$ we either have $v(x) \geq 0$ or $v(x) \leq 0$. If $v(x) \geq 0$, then $x \in R$. If $v(x) \leq 0$, then $v(x^{-1}) = -v(x) \geq 0$, so $x^{-1} \in R$.

□

Definition 1.8 (Discrete Valuation Ring)**.** We call a valuation ring *discrete* if its valuation is discrete.

The concept of a discrete valuation ring (often abbreviated as DVR) is extremely important in algebraic number theory. Therefore try very hard to internalize the concept via the following examples:

- The ring of p-adic integers, denoted $\mathbb{Z}_p$. The uniformizer is the prime number p, and the unique maximal ideal is $p\mathbb{Z}_p$.
- Formal power series rings, denoted $k[[t]]$ where k is a field. The uniformizer is the variable t. Elements are of the form $\sum_{n=0}^{\infty} a_n t^n$.
- The localization of $\mathbb{Z}$ at a prime p, denoted $\mathbb{Z}_{(p)}$. This consists of rational numbers a/b where $p \nmid b$. The uniformizer is p.
- The local ring of a curve at a smooth point. If C is a non-singular algebraic curve and $P \in C$ is a closed point, the local ring $\mathcal{O}_{C,P}$ is a DVR. The uniformizer is any local parameter at P.

It is worth noting that a ring having a valuation does not make it a valuation ring. However if R has a valuation, then $\mathfrak{p} = \{r \in R : v(r) > 0\}$ is a prime ideal of the subring $R' = \{r \in R : v(r) \geq 0\}$. If $R = K$ is a field, then $R' = \{x \in K : v(x) \geq 0\}$ is already a valuation ring with maximal ideal $\mathfrak{p}$. For an arbitrary subring of K, the localization of R' at $\mathfrak{p}$ need not be a valuation ring.

Problem 1.9. Prove that a discrete valuation ring is a Noetherian local domain

Now we have some characterizations of a discrete valuation ring.

Theorem 1.10. *For a Noetherian local ring R the following statements are equivalent:*

(a) R is a discrete valuation ring.

(b) R is a principal ideal domain, but not a field.

(c) R is a 1-dimensional unique factorization domain.

(d) R is a 1-dimensional normal domain.

(e) R is a 1-dimensional regular ring.

Proof. Through the proofs we denote $\mathfrak{m}$ as the unique maximal ideal of R.

$(a) \Rightarrow (b)$**:** By the Noetherian condition every ideal can be written as $I = (a_1, \ldots, a_n)$. Then $I = (a_i)$ where a_i is the generator with minimal valuation. Since the valuation is nontrivial there must be a non-unit element.

$(b) \Rightarrow (c)$**:** Since a PID is a UFD we only have to show that R is 1-dimensional. Suppose FTSOC that there is a nonzero prime $\mathfrak{p} \subsetneq \mathfrak{m}$. We can write $\mathfrak{p} = (\rho)$, $\mathfrak{m} = (\pi)$ and find some element x with $x\pi = \rho$. However since ρ generates a prime ideal it is irreducible, a contradiction.

$(c) \Rightarrow (d)$**:** It is a general result that every UFD is normal.

$(d) \Rightarrow (e)$**:** We will show that $\mathfrak{m}$ is principal by showing it is invertible, so consider $\mathfrak{m}\mathfrak{m}^{-1}$ (where $\mathfrak{m}^{-1} := (R :_K \mathfrak{m})$). By definition this is an ideal in R containing $\mathfrak{m}$, therefore it is either $\mathfrak{m}$ or R.

$\mathfrak{m}\mathfrak{m}^{-1} = \mathfrak{m}$**:** In this case every element $x \in \mathfrak{m}^{-1}$ satisfies $x\mathfrak{m} \subseteq \mathfrak{m}$ which implies by the determinant trick on the finitely generated faithful module $\mathfrak{m}$ that x is integral over R and therefore in R by assumption. However this is a contradiction with Lemma 1.5.

$\mathfrak{m}\mathfrak{m}^{-1} = R$**:** In this case $\mathfrak{m}$ is an invertible ideal. Now by Theorem 1.4 every invertible ideal is principal as desired.

$(e) \Rightarrow (a)$**:** By Nakayama $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$ implies that $\mathfrak{m}$ is principal with some generator π. Since a regular local ring is a domain, we claim that every nonzero element $x \in R$ is of the form $u\pi^n$ for $u \in R^\times$, at which point it follows that R is a DVR. By Krull intersection theorem let n be maximal so that $x \in (\pi^n)$ so $x = y\pi^n$. By maximality of n we see $y \notin (\pi)$, and since (π) is the unique maximal ideal this means y is a unit.

□

These might seem hard to remember, but the last 3 conditions essentially say the same thing. That a DVR looks like the functions near a smooth point on a curve. The slogan is that for a *curve* the following are the same:

- There is *one* codimension-1 direction through a smooth point.
- Near a smooth point there is a single local coordinate (a *uniformizer*).
- Every nonzero function has a well-defined *order of vanishing* which is just an integer.

This is exactly the discrete valuation story. Geometrically, the last three conditions in Theorem 1.10 are three different ways of saying

"near P the curve looks like one smooth branch, i.e. like a line."

In dimension 1 these points of view collapse; in higher dimension they genuinely diverge.

Here is a useful mental dictionary for a local ring $R = \mathcal{O}_{C,P}$ of an integral curve C over a perfect field at a closed point P:

- *Regular (smoothness / tangent space).* At a smooth point you can choose one local parameter t cutting out P. Algebraically this means the maximal ideal is generated by one element:

$$\mathfrak{m}_P = (t)$$

and infinitesimally there is only one direction along the curve:

$$\dim_{\kappa(P)}(\mathfrak{m}_P/\mathfrak{m}_P^2) = 1$$

A singular point forces at least two independent first-order directions (or "too many tangent vectors"), which is why $\mathfrak{m}_P$ stops being principal.

- *Normal (no hidden functions / normalization).* The normalization $\nu : \widetilde{C} \to C$ is the operation that "pulls apart" branches and "fills in" missing integral functions. Saying R is normal means that locally nothing changes under normalization: there are no missing integral elements. On a curve, the only way to fail normality is precisely to have a singularity like a cusp or a node: there is some rational function that is integral over R which should be considered regular on the curve, but is not actually in R.

- *UFD (divisors become a single integer).* On a smooth curve, any non-zero rational function f has a well-defined integer $\mathrm{ord}_P(f)$ measuring how many times it vanishes at P (negative if it has a pole). The key point is that for a smooth point this is *one integer*, not a vector of integers. This corresponds to the fact that every nonzero ideal is just a power of (t):

$$0 \neq I \subseteq R \quad \Longleftrightarrow \quad I = (t^n) \text{ for a unique } n \geq 0$$

So locally there is a single "prime divisor" $[P]$, and everything factors as a multiple of it. If the point has multiple branches, you would instead get a *tuple* of orders (one per branch), and there is no longer a single generator controlling all vanishing.

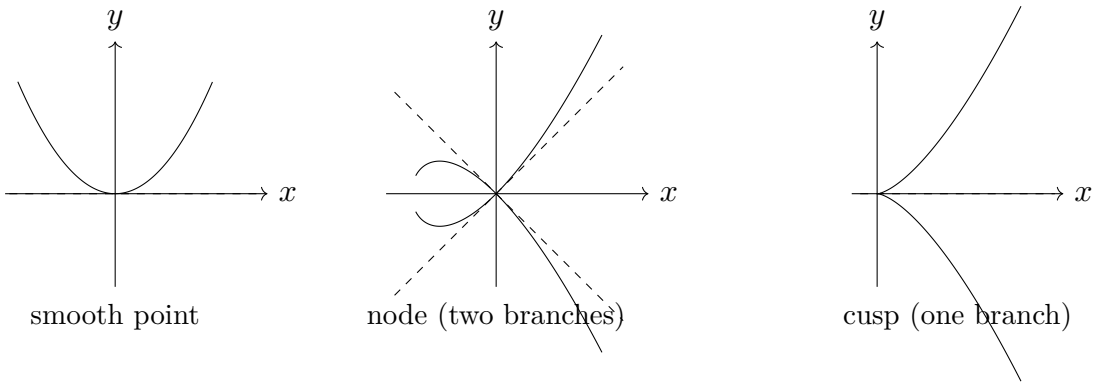


Figure 1: Singularities force "too many" first-order directions. At a smooth point there is one local parameter t; at a node there are two branches; at a cusp there is one branch but the tangent is "thick". In both singular pictures the maximal ideal cannot be generated by one element, and normalization changes the local ring.

It is helpful to keep two toy local models in mind (purely as intuition):

- *Cusp: one branch but not normal.* The parametrization $t \mapsto (t^2, t^3)$ suggests that (after completing) the local ring looks like

$$k[[t^2, t^3]] \subsetneq k[[t]]$$

The element t is "the missing uniformizer": adjoining it is exactly what normalization does. This is why normality fails, and consequently regularity/UFD fail as well.

- *Node: two branches.* Near a node there are two ways to approach P along the curve, so vanishing wants to be measured by

$$(\mathrm{ord}_{P,1}(f),\ \mathrm{ord}_{P,2}(f)) \in \mathbb{Z}^2$$

rather than a single integer. Normalization separates the branches, and locally you should think of it as producing (roughly) two DVRs, one for each branch. Their intersection inside the function field is the integral closure of the original local ring. The original ring also requires the values on the two branches to agree at the node, which explains why it is smaller.

So, on a curve, *regular* says "one parameter" (one tangent direction), *normal* says "no hidden integral functions" (normalization does nothing), and *UFD* says "orders of vanishing are controlled by a single integer." These are three shadows of the same geometric phenomenon.

1.3. Dedekind Domains

A DVR is a very local notion, so you might wonder if there is a corresponding global notion. The answer is yes, and it is a Dedekind domain. Just like with a DVR, there are many equivalent conditions for a ring to be a Dedekind domain. Now buckle up, because we're about to start making the rounds.

Definition 1.11 (Dedekind Domain)**.** We call an integral domain R that is not a field a Dedekind domain if it satisfies any of the equivalent conditions:

(a) R is Noetherian, and the localization at every maximal ideal is a discrete valuation ring.

(b) R is a normal (or regular), 1-dimensional Noetherian domain.

(c) Every nontrivial quotient R/I for $I \neq 0$ is a principal ideal ring.

(d) Every nonzero proper ideal factors into maximal ideals.

(e) Every nonzero fractional ideal of R is invertible.

(f) For any ideals I, J in R, I is contained in J if and only if J divides I. That is, there is an ideal H with $I = JH$.

Proof.

$(a) \Leftrightarrow (b)$**:** Firstly, (a) implies that R is 1-dimensional since otherwise localizing at a maximal ideal above a nontrivial prime does not give a DVR. Recall from Theorem 1.10 that a DVR is a Noetherian, 1-dimensional, normal/regular *local* ring. Therefore we only need to show that normality and regularity are local properties, at least when every nonzero prime is maximal.

- Note that there is the standard inclusion map $\psi : R \to \bar{R}$ where $\bar{R}$ is the integral closure of R in its field of fractions. By commutative algebra ψ is an isomorphism iff $\psi_{\mathfrak{m}}$ is an isomorphism for all $\mathfrak{m} \in \operatorname{Spm} R$. However since integral closure commutes with localization this is exactly the condition that $R_{\mathfrak{m}}$ is integrally closed for all $\mathfrak{m}$.
- Regularity means that $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$ for every $\mathfrak{m} \in \operatorname{Spm} R$. However $R/\mathfrak{m}$ is local, and in particular $(R/\mathfrak{m})_{\mathfrak{m}} = R/\mathfrak{m}$ so this is a local condition.

$(a) \Rightarrow (d)$**:** Suppose that I is a nonzero proper ideal of R. Since R is Noetherian by the Noether-Lasker theorem we may take a primary decomposition $I = Q_1 \cap \cdots \cap Q_n$. We want to show that $Q_i = \mathfrak{p}_i^k$ for the prime ideal $\mathfrak{p}_i = \sqrt{Q_i}$. For each i, write $\mathfrak{p} = \mathfrak{p}_i$ and $S = R \setminus \mathfrak{p}$ and note:

$$(Q_i)_{\mathfrak{p}} \cap R = \{x \in R | (Q_i : x) \cap S \neq \emptyset\}$$

However by the primary condition if $sx \in Q_i$ for $s \notin \mathfrak{p}$ then $x \in Q_i$. Therefore $(Q_i)_{\mathfrak{p}} \cap R = Q_i$. Since $R_{\mathfrak{p}}$ is a DVR we see that $(Q_i)_{\mathfrak{p}} = \mathfrak{p}^k R_{\mathfrak{p}}$. The ideal $\mathfrak{p}^k$ is also $\mathfrak{p}$-primary, so contracting gives $Q_i = \mathfrak{p}^k$. Since every nonzero prime is maximal by (b) we can use CRT to write this as $I = \mathfrak{p}_1^{k_1} \cdots \mathfrak{p}_n^{k_n}$

$(d) \Rightarrow (f)$**:** Let $\mathfrak{m}$ be a maximal ideal and choose nonzero $a \in \mathfrak{m}$. Factor (a) into maximal ideals. One factor must be contained in $\mathfrak{m}$, so it is $\mathfrak{m}$ itself. Thus $(a) = \mathfrak{m}J$ for an ideal J, and $\mathfrak{m}(a^{-1}J) = R$. Every maximal ideal is therefore invertible, and by (d) so is every nonzero ideal. If $I \subseteq J$ with $J \neq 0$, then $H = IJ^{-1} \subseteq R$ and $I = JH$. The converse is immediate, and the case $J = 0$ is trivial. Invertibility also makes the factorization in (d) unique: a prime containing a product contains one factor, and we can cancel equal maximal factors.

$(f) \Rightarrow (e)$**:** Firstly we claim integral ideals are invertible. Let I be a nonzero integral ideal of R, and choose any nonzero $a \in I$. Since $(a) \subseteq I$, (f) implies there exists an integral ideal H such that:

$$IH = (a)$$

Multiplying both sides by the fractional ideal (a^{-1}) yields $I(a^{-1}H) = R$. Since H is an integral ideal, $a^{-1}H$ is a fractional ideal. Thus, I is invertible.

Now let M be an arbitrary nonzero fractional ideal. By definition, there exists a nonzero $d \in R$ such that dM is an integral ideal. From Part 1, the integral ideal dM is invertible, meaning there exists a fractional ideal J such that:

$$(dM)J = R \implies M(dJ) = R$$

Since dJ is a fractional ideal, M is invertible.

$(e) \Rightarrow (a)$**:** Let $\mathfrak{m}$ be a maximal ideal of R and note that

$$I_{\mathfrak{m}} I_{\mathfrak{m}}^{-1} = (II^{-1})_{\mathfrak{m}} = R_{\mathfrak{m}}$$

so the localization retains the property of every nonzero ideal being invertible. Moreover every ideal of R is finitely generated by Theorem 1.4, so R is Noetherian. Therefore $R_{\mathfrak{m}}$ is a PID by Theorem 1.4, so by Theorem 1.10 (a) we are done.

$(a) \Leftrightarrow (c)$**:** To prove $(a) \Rightarrow (c)$, note that any nonzero quotient decomposes according to the Chinese remainder theorem.

$$\frac{R}{\mathfrak{m}_1^{k_1} \cdots \mathfrak{m}_n^{k_n}} \cong \prod_i \frac{R_{\mathfrak{m}_i}}{\mathfrak{m}_i^{k_i} R_{\mathfrak{m}_i}}$$

Each term is a quotient of a DVR, hence a principal ideal ring, and a finite product of principal ideal rings is a principal ideal ring.

In the other direction, choose nonzero $a \in I$ for any nonzero ideal I. Since $R/(a)$ is a principal ideal ring, $I = (a, b)$ for some b, so R is Noetherian. The quotient condition persists after localization, so work in the local domain $R_{\mathfrak{m}}$. Choose nonzero $x \in \mathfrak{m}^2 R_{\mathfrak{m}}$. Since $R_{\mathfrak{m}}/(x)$ is a principal ideal ring, $\mathfrak{m}R_{\mathfrak{m}} = (\pi, x)$ for some π. Nakayama gives $\mathfrak{m}R_{\mathfrak{m}} = (\pi)$, and the principal ideal theorem gives $\dim R_{\mathfrak{m}} = 1$. Now Theorem 1.10 shows that $R_{\mathfrak{m}}$ is a DVR.

□

In general the standard way to check whether a ring is a Dedekind domain is via (b). That is that a Dedekind domain is a normal 1-dimensional Noetherian domain. One may think of (a)-(c) as expressing the notion that a Dedekind domain looks locally like a DVR. On the other hand (d)-(f) express the fact that the fractional ideals behave like a group. The connection is in algebraic geometry:

$$\text{nonzero fractional ideals} \qquad \cong \qquad \text{divisors}$$

and ideal classes correspond to line bundles up to isomorphism, so being locally a DVR (smooth curve) means that divisors are just sums of points. As before, here are some examples of a Dedekind domain:

- The ring of integers $\mathbb{Z}$ is a principal ideal domain and the archetypal example of a Dedekind domain.
- The polynomial ring $k[x]$ over any field k is a Euclidean domain and therefore a Dedekind domain.
- The ring of integers $\mathcal{O}_K$ of a number field K (a finite extension of $\mathbb{Q}$) is a Dedekind domain.
- The Gaussian integers $\mathbb{Z}[i]$ form a principal ideal domain, which implies they are a Dedekind domain.

- Any discrete valuation ring (DVR), such as the ring of p-adic integers $\mathbb{Z}_p$, is a local Dedekind domain.
- The coordinate ring $k[C]$ of a smooth (non-singular) integral affine algebraic curve C over an algebraically closed field is a Dedekind domain.
- The ring $\mathbb{Z}[\sqrt{-5}]$ is a Dedekind domain that is not a principal ideal domain, illustrating that element factorization need not be unique even when ideal factorization is.

Now we can define the main exact sequence of study in algebraic number theory.

Construction 1.12. Let $\mathcal{O}_K$ be a Dedekind domain with fraction field K. We define the following objects:

$\mathcal{I}$**:** The *ideal group* $\mathcal{I}$ is the group of nonzero fractional ideals under multiplication.

Cl_K**:** The *class group* $\mathrm{Cl}_K := \mathcal{I}_K/\mathcal{P}_K$ is the quotient of $\mathcal{I}_K$ by the *principal divisors* $\mathcal{P}_K$ which are the group of nonzero principal fractional ideals.

Then there is the following exact sequence

$$1 \to \mathcal{O}_K^\times \to K^\times \to \mathcal{I}_K \to \mathrm{Cl}_K \to 1 \tag{1}$$

Proof. We will show exactness at each point.

- The map $\mathcal{O}_K^\times \to K^\times$ is by inclusion and the map $K^\times \to \mathcal{I}_K$ is by $x \mapsto (x)$. The elements which generate $\mathcal{O}_K$ (the identity element in $\mathcal{I}_K$) are exactly the units.
- The map $\mathcal{I} \to \mathrm{Cl}_K$ is the quotient map, and the kernel consists of the principal ideals. These are exactly the ideals of the form (x) for $x \in K^\times$.

□

Remark 1.13. Note that

$$\mathcal{I}_K \cong \bigoplus_{\mathfrak{p} \in \mathrm{Spm}\, \mathcal{O}_K} \mathbb{Z}$$

Proof. Any fractional ideal can be characterized by the power of each prime appearing in the decomposition. □

Remark 1.13 gives us a very good understanding of the ideal group. We will eventually better understand $\mathcal{O}_K^\times$ and Cl_K so that we may understand multiplication in $K^\times$ and $\mathcal{O}_K$ via the exact sequence in Eq. (1).

2. Computational Tools

There are some other tools one might want to know before diving into the algebraic number theory. The reader who is bored of preliminaries should feel free to skip this section for now, until its results become needed.

2.1. The Trace Pairing and Discriminant

In this section I will define the discriminant of a field extension, and how it comes from the trace pairing.

Definition 2.1 (Bilinear Form)**.** A k-linear *bilinear form* on a vector space V is a bilinear map $V \times V \to k$. If V is finite dimensional, we define the *discriminant* of a bilinear form relative to a basis e_i as $\det\langle e_i, e_j\rangle$.

Lemma 2.2. *Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be two bases of V and T the change of basis map with $Ta_i = b_i$. Then the discriminants with respect to each basis are related by the formula:*

$$\det\langle b_i, b_j\rangle = (\det T)^2 \cdot \det\langle a_i, a_j\rangle$$

Proof. Let A denote the matrix $\langle a_i, a_j\rangle$. Note that A is specified by the property that

$$\langle x, y\rangle = x^T A y$$

when x and y are written in the a_i-basis. If x and y are instead written in the b_i-basis then we would need to convert them to the a_i-basis to evaluate the pairing:

$$\langle x, y\rangle = (Mx)^T A(My) = x^T(M^T AM)y$$

where M is the matrix of T in the a_i-basis. Therefore

$$\langle b_i, b_j\rangle = M^T AM$$

and discriminant is $\det(M^T AM) = (\det T)^2 \cdot \det A$. □

Definition 2.3. On a finite dimensional vector space V, we call a bilinear form *non-degenerate* or *perfect* if it satisfies the following equivalent conditions:

(a) it has nonzero discriminant with respect to one (hence every) basis;

(b) for every nonzero $x \in V$ there is a $y \in V$ with $\langle x, y\rangle \neq 0$;

(c) for every nonzero $y \in V$ there is an $x \in V$ with $\langle x, y\rangle \neq 0$.

Proof. The fact that (a) does not care about the basis follows from Lemma 2.2. It suffices to prove $(a) \Leftrightarrow (b)$, since transposing the matrix gives the equivalence with (c). Let $A = \langle e_i, e_j\rangle$ for a basis $e_1, \dots, e_n$.

$(a) \Rightarrow (b)$**:** If $x \neq 0$, choose a coordinate $x_i \neq 0$ and set $y = A^{-1}e_i$. Then

$$\langle x, y\rangle = x^T Ay = x_i \neq 0$$

$(b) \Rightarrow (a)$**:** Suppose in the contrapositive that $\det A = 0$. Then there is a nonzero $x \in V$ with $A^T x = 0$. It is now clear that $x^T Ay = 0$ for every $y \in V$.

□

Note that the discriminant of a bilinear form is defined up to the square of a number, due to the change of basis formula. A good way to think about the discriminant is the following. A positive definite real symmetric bilinear form $\langle\rangle$ gives some way to measure distance, angles, and volume. The discriminant of a basis $e_1, \dots, e_n$ is exactly the square of the covolume of the lattice it generates.

Definition 2.4 (R-lattice)**.** Let R be an integral domain with fraction field K, V a finite dimensional K-vector space, and M an R-submodule of V. We call a set of elements $\{\beta_1, \dots, \beta_n\}$ a *basis* for M if they generate M over R and are simultaneously a basis for V over K. M is called an R-*lattice* if it has a basis. Given a bilinear form on V, we define the *discriminant* of M/R to be its discriminant with respect to a basis of M. When $V = L$ is a finite field extension of K, we will use the trace pairing defined below.

Note that $\operatorname{disc} V/K$ is an element of $K/K^{\times 2}$ (defined up to the square of an element) by our analysis in Definition 2.3. The field discriminant does not carry much information. On the other hand two bases of M over R must be related by an R-matrix, and $\operatorname{disc} M/R$ is therefore defined up to the square of a unit of R. If the pairing takes values in R on M, its discriminant lies in R. This allows it to carry much more information since R might not have that many units. The most important example of this is when $R = \mathbb{Z}$ and M is the ring of integers in a finite separable extension of $\mathbb{Q}$.

Construction 2.5. Let R be an integral domain with fraction field K, V a finite dimensional K-vector space, and $\langle\rangle$ a perfect pairing $V \times V \to K$. For any R-submodule M of V, we can define the *dual* of M as:

$$M^\sharp := \{x \in V : \langle x, y\rangle \in R \text{ for all } y \in M\}$$

Suppose moreover that M is an R lattice. Then $M^\sharp$ is an R-lattice generated by the *dual basis*.

Proof. Since the pairing is perfect it induces an isomorphism (via dimension counting)

$$V \xrightarrow{\sim} V^\vee \qquad x \mapsto \langle x, -\rangle$$

and $M^\sharp$ is the preimage of $M^\vee = \operatorname{Hom}_R(M, R)$ (sitting inside $V^\vee = \operatorname{Hom}_K(V, K)$) under this isomorphism. Suppose that $e_1, \dots, e_n$ is an R-basis of M that is also a K-basis of V, we can use this isomorphism to induce a unique *dual basis* $e_1^*, \dots, e_n^*$ which has the property that:

$$\langle e_j^*, e_i\rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

We claim that this dual basis $e_1^*, \dots, e_n^*$ generates $M^\sharp$. Suppose that $x \in M^\sharp$ and write $x = x_1 e_1^* + \cdots + x_n e_n^*$ for $x_i \in K$. By the assumption that $x \in M^\sharp$ we see $\langle x, e_i\rangle = x_i$ is in R as desired. □

Geometrically suppose that $R = \mathbb{Z}$ in Construction 2.5. Then the conditions we put on M exactly translate to it being a full-rank *lattice* in $\mathbb{Q}^n$. The theorem tells us that the set of points whose pairing with the lattice gives only integers is another lattice. The *dual lattice* if you will. This geometric picture is very important to keep in mind.

Lemma 2.6. *Suppose the pairing is symmetric and recall that it defines an isomorphism $\tau_{V/K} : V \to V^\vee$ which identifies $M^\vee$ with $M^\sharp$ inside V. Then the discriminant is (up to units) the determinant of any lattice map sending $M^\sharp$ onto M.*

Proof. Let $e_1, \dots, e_n$ be a basis for M and $e_1^*, \dots, e_n^*$ be the dual basis generating $M^\sharp$. Since

$$e_i = \sum_j \langle e_i, e_j \rangle e_j^*$$

the matrix $\langle e_i, e_j \rangle$ is the change of basis matrix from $e_1, \dots, e_n$ to $e_1^*, \dots, e_n^*$. Therefore as a matrix in the dual basis it sends $M^\sharp$ to M. On the other hand the determinant is $\operatorname{disc} M/R$ by definition. □

Definition 2.7. Suppose that L/K is a finite extension, then L has the structure of a K vector space. Therefore any element $x \in L$ induces a K-linear transformation on L by multiplication by x. We can then define:

- the *norm* $\mathrm{N}_{L/K}(x)$ to be the determinant of the transformation;
- the *trace* $\mathrm{Tr}_{L/K}(x)$ to be the trace of the transformation.

Problem 2.8. Suppose $L/K/F$ is a tower of finite field extensions. Then:

$$\mathrm{N}_{L/F} = \mathrm{N}_{K/F} \circ \mathrm{N}_{L/K} \qquad \mathrm{Tr}_{L/F} = \mathrm{Tr}_{K/F} \circ \mathrm{Tr}_{L/K}$$

Definition 2.9 (Trace Pairing)**.** The *trace pairing* of a finite extension L/K is the bilinear form:

$$\langle x, y \rangle = \mathrm{Tr}(xy)$$

The *discriminant* of the field extension is the discriminant of this pairing.

Proposition 2.10. *Suppose that L/K is a separable degree n extension and that $\beta_1, \dots, \beta_n$ is a basis for L over K. Then*

$$(\mathrm{Tr}(\beta_i \beta_j))_{ij} = VV^T \qquad V := \begin{bmatrix} \sigma_1(\beta_1) & \cdots & \sigma_n(\beta_1) \\ \vdots & & \vdots \\ \sigma_1(\beta_n) & \cdots & \sigma_n(\beta_n) \end{bmatrix}$$

where V is the embedding matrix and the σ_i are the n embeddings of L into the algebraic closure of K. In particular, the discriminant of L/K is $(\det V)^2$.

Proof. We note that:

$$(VV^T)_{ij} = \sum_{k=1}^n V_{ik} V_{jk} = \sum_{k=1}^n \sigma_k(\beta_i)\sigma_k(\beta_j) = \mathrm{Tr}(\beta_i \beta_j)$$

□

Corollary 2.11. *The trace pairing of a finite extension L/K is non-degenerate if and only if the extension is separable.*

Proof. We consider the separable and inseparable cases separately.

- If L/K is *separable*, the embeddings σ_i are distinct. Each one may be thought of as a homomorphism from the multiplicative group $L^\times$ to $\overline{K}^\times$. Therefore by Dedekind's theorem on the linear independence of characters $\det V \neq 0$.
- If L/K is *inseparable*, let E be its maximal separable subextension. The trace of the nontrivial purely inseparable extension L/E is zero. By transitivity $\mathrm{Tr}_{L/K} = 0$, so the pairing is degenerate.

□

At this point the reader is encouraged to read Example 7.8 through Proposition 7.11. Now we will use the trace pairing to understand Dedekind domains.

Lemma 2.12. *Let R be a normal domain with fraction field K, and let S be the integral closure of R in a finite separable extension L of K. There exist free R-submodules M and $M^\sharp$ with:*

$$M \subseteq S \subseteq M^\sharp$$

Proof. Since L/K is separable, the trace pairing $(x, y) \mapsto \mathrm{Tr}_{L/K}(xy)$ is non-degenerate. We may find a basis for L over K contained in S and let M be the R-module generated by this basis. Then Construction 2.5 tells us that

$$M^\sharp := \{x \in L : \mathrm{Tr}(xy) \in R \text{ for all } y \in M\}$$

is finitely generated by the dual basis. Note that

$$\mathrm{Tr}_{L/K}(SM) \subseteq \mathrm{Tr}_{L/K}(S) \subseteq R$$

since $M \subseteq S$ and the trace of something integral over R is in R (by normality). Therefore $S \subseteq M^\sharp$ by definition and $M \subseteq S \subseteq M^\sharp$ as desired. □

2.2. The Resultant

I will first give a few equivalent definitions of the resultant. Then we will see its importance. For the remainder R is an integral domain, K is its field of fractions, and Ω is an algebraic closure of K.

Definition 2.13 (Resultant)**.** Let $f(x) = a_n x^n + \cdots + a_0$ and $g(x) = b_m x^m + \cdots + b_0$ be nonzero polynomials in $R[x]$ of degrees n and m respectively. The *resultant* of f and g can be computed in the following ways:

(a) Write $[h]$ for the column of coefficients of h in the ordered basis $1, x, \ldots, x^{n+m-1}$, padding with zeros. The *Sylvester matrix* of f and g is

$$S_{f,g} = \begin{pmatrix} [g] & [xg] & \cdots & [x^{n-1}g] & [f] & [xf] & \cdots & [x^{m-1}f] \end{pmatrix}$$

which consists of n shifted copies of g followed by m shifted copies of f. Now $\mathrm{Res}(f, g)$ is defined as the determinant of $S_{f,g}$.

(b) We can define the resultant by looking at the roots

$$\mathrm{Res}(f,g) = a_n^m b_m^n \prod_{i=1}^{n}\prod_{j=1}^{m}(\alpha_i - \beta_j)$$

where α_i and β_j are the roots of f and g in Ω, counted with multiplicity.

(c) In the case that f is monic, the resultant is the determinant of the R-linear map

$$\frac{R[x]}{(f)} \xrightarrow{\cdot g} \frac{R[x]}{(f)}$$

We will initially take (a) as the definition. Intuitively, the resultant should express how coprime two polynomials are. The order of the columns fixes the sign, so it is useful to keep the simplest case in mind

$$S_{x-a,x-b} = \begin{pmatrix} -b & -a \\ 1 & 1 \end{pmatrix} \qquad \mathrm{Res}(x-a, x-b) = a - b$$

For a nonzero constant c the definition gives

$$\mathrm{Res}(f,c) = c^n \qquad \mathrm{Res}(c,g) = c^m$$

The resultant of two nonzero constants is 1. We also set $\mathrm{Res}(f,0) = \mathrm{Res}(0,f) = 0$ when f has positive degree. Since R injects into K and Ω, we can compute the same determinant over either field.

Lemma 2.14. $\mathrm{Res}(f,g) = 0$ *if and only if* f *and* g *share a root in* Ω.

Proof. The Sylvester matrix represents the map

$$K[x]_{<n} \oplus K[x]_{<m} \longrightarrow K[x]_{<n+m} \qquad (B, A) \longmapsto Bg + Af$$

where $K[x]_{<d}$ denotes the polynomials of degree less than d. If $d = \gcd(f,g)$ has positive degree then $A = g/d$ and $B = -f/d$ give a nonzero vector in the kernel. If $\gcd(f,g) = 1$ and $Af = -Bg$, then g divides A. The degree bound forces $A = 0$, and then $B = 0$. Thus the determinant vanishes exactly when f and g have a common factor of positive degree over K, equivalently a common root in Ω. □

Theorem 2.15. *The definitions in Definition 2.13 are equivalent.*

Proof. First assume that f is monic. Polynomial division gives the following basis of $R[x]_{<n+m}$

$$1, x, \ldots, x^{n-1}, f, xf, \ldots, x^{m-1}f$$

Its change of basis matrix has determinant 1. In this basis the Sylvester map has the block matrix

$$\begin{pmatrix} \mu_g & 0 \\ * & I_m \end{pmatrix}$$

where μ_g is multiplication by g on $R[x]/(f)$. Taking determinants proves (a)=(c).

Now work over Ω. The operator μ_x on $\Omega[x]/(f)$ has characteristic polynomial f, so after triangularizing its diagonal entries are the α_i, counted with multiplicity. Therefore $\mu_g = g(\mu_x)$ has diagonal entries $g(\alpha_i)$ and

$$\operatorname{Res}(f,g) = \det \mu_g = \prod_{i=1}^{n} g(\alpha_i) = b_m^n \prod_{i,j} (\alpha_i - \beta_j)$$

This proves (b) in the monic case, including when f has repeated roots. For arbitrary f, work over K and use

$$\operatorname{Res}(f,g) = a_n^m \operatorname{Res}(f/a_n, g)$$

which follows by scaling the m columns coming from f. □

Corollary 2.16. *The resultant can be evaluated using either set of roots*

$$\operatorname{Res}(f,g) = a_n^m \prod_{i=1}^{n} g(\alpha_i) = (-1)^{mn} b_m^n \prod_{j=1}^{m} f(\beta_j)$$

In particular

$$\operatorname{Res}(x-a, g) = g(a) \qquad \operatorname{Res}(f, x-b) = (-1)^n f(b)$$

Theorem 2.17. *The resultant satisfies the formula*

$$\operatorname{Res}(f,g) = (-1)^{mn} \operatorname{Res}(g,f)$$

Proof. Swapping f and g moves n columns past m columns, contributing the sign $(-1)^{mn}$. □

Theorem 2.18. *Let $h \neq 0$ be the remainder of g after division by f in $K[x]$. Then*

$$\operatorname{Res}(f,g) = \operatorname{lc}(f)^{m - \deg h} \operatorname{Res}(f,h)$$

where $\operatorname{lc}(f)$ *denotes the leading coefficient of f. If $h = 0$ and* $\deg f > 0$, *then* $\operatorname{Res}(f,g) = 0$.

Proof. At each root α_i of f we have $g(\alpha_i) = h(\alpha_i)$. Now apply Corollary 2.16 and compare the powers of the leading coefficient. □

Remark 2.19. Combining Theorem 2.17 with Theorem 2.18 gives

$$\operatorname{Res}(f,g) = (-1)^{mn} \operatorname{lc}(g)^{n - \deg r} \operatorname{Res}(g,r)$$

where $r \neq 0$ is the remainder of f after division by g. This gives a fast method of computing the resultant via the Euclidean algorithm. We stop at a nonzero constant using $\operatorname{Res}(f,c) = c^{\deg f}$, or at a zero remainder, which gives resultant 0 when the divisor has positive degree.

Example 2.20. Let $f = x^2 - a$ and $g = x^2 - b$. The remainder of g after division by f is $a - b$, so

$$\operatorname{Res}(x^2 - a, x^2 - b) = \operatorname{Res}(x^2 - a, a - b) = (a-b)^2$$

Theorem 2.21. *The resultant is multiplicative in both arguments*

$$\operatorname{Res}(ff', g) = \operatorname{Res}(f, g)\operatorname{Res}(f', g)$$

$$\operatorname{Res}(f, gg') = \operatorname{Res}(f, g)\operatorname{Res}(f, g')$$

Proof. Work over K and first take f monic. Multiplication by gg' on $K[x]/(f)$ is the composition of multiplication by g and by g', so its determinant is the product of their determinants. Scaling f gives the general case, and Theorem 2.17 gives multiplicativity in the first argument. □

Now that we understand the resultant, we will list some useful theorems.

Lemma 2.22 (Resultant Adjugate Identity)**.** *For $f, g \in R[x]$ of positive degrees n, m and $k \in R[x]$ with $\deg k < n + m$, there exist $A, B \in R[x]$ such that*

$$Af + Bg = \operatorname{Res}(f, g)k \qquad \deg A < m \qquad \deg B < n$$

If $\operatorname{Res}(f, g) \neq 0$, then A and B are unique. Moreover if every coefficient of k lies in an ideal $J \subseteq R$, the adjugate construction gives A and B with every coefficient in J.

Proof. Let $[k]$ be the coefficient vector of k and set

$$\begin{pmatrix} [B] \\ [A] \end{pmatrix} = \operatorname{adj}(S_{f,g})[k]$$

Then $S_{f,g}\operatorname{adj}(S_{f,g}) = \operatorname{Res}(f, g)I_{n+m}$ gives the identity and the degree bounds. The adjugate has entries in R, so it preserves the condition that coefficients lie in J. When the resultant is nonzero, the Sylvester matrix is invertible over K, giving uniqueness. □

The reader who is interested in how this formula is used should check out the strong Hensel's lemma in the lifting roots section.

Corollary 2.23. *The resultant belongs to the ideal generated by f and g*

$$\operatorname{Res}(f, g) \in (f, g) \cap R$$

In particular, if $\operatorname{Res}(f, g)$ is a unit then $(f, g) = R[x]$. For monic f, g the converse also holds.

Proof. Take $k = 1$ in Lemma 2.22. For the converse, if $Af + Bg = 1$ then multiplication by B is an inverse to multiplication by g on the free R-module $R[x]/(f)$. Thus its determinant $\operatorname{Res}(f, g)$ is a unit. □

Over $K[x]$, a nonzero resultant is enough to give coprimality. Over $R[x]$, being a unit matters: x and $x + 2$ have resultant 2 and generate the proper ideal $(x, 2)$ in $\mathbb{Z}[x]$.

Theorem 2.24 (Reduction of the resultant)**.** *The resultant is a polynomial with integer coefficients in the coefficients of f and g. If $\phi : R \to T$ is a ring homomorphism which preserves their degrees, then*

$$\phi(\operatorname{Res}(f, g)) = \operatorname{Res}(\phi(f), \phi(g))$$

In particular this always applies to monic polynomials.

Proof. Apply ϕ to every entry of the Sylvester matrix and use the determinant formula. If a degree drops, the same calculation still gives the determinant of the specialized matrix with the original sizes, which need not be the resultant formed using the smaller degrees. □

Corollary 2.25. *Let $f, g \in R[x]$ be monic and $\mathfrak{p}$ a prime ideal of R. Then*

$$\operatorname{Res}(f,g) \in \mathfrak{p} \qquad \Longleftrightarrow \qquad \bar{f}, \bar{g} \text{ share a root over } \overline{\operatorname{Frac}(R/\mathfrak{p})}$$

For $R = \mathbb{Z}$ this says that a prime p divides the resultant exactly when the reductions have a common factor of positive degree in $\mathbb{F}_p[x]$.

Theorem 2.26 (Resultants and norms)**.** *Let $f \in K[x]$ be monic and irreducible, let α be a root, and set $L = K(\alpha)$. For every $g \in K[x]$ we have*

$$\operatorname{Res}(f,g) = \mathrm{N}_{L/K}(g(\alpha))$$

Proof. The isomorphism $K[x]/(f) \cong L$ identifies multiplication by g with multiplication by $g(\alpha)$. Their determinants are the resultant and the norm respectively. □

Corollary 2.27. *If $f, g \in \mathbb{Z}[x]$, f is monic, and $\operatorname{Res}(f,g) \neq 0$, then*

$$\left|\frac{\mathbb{Z}[x]}{(f,g)}\right| = |\operatorname{Res}(f,g)|$$

Proof. The quotient is the cokernel of multiplication by g on the free abelian group $\mathbb{Z}[x]/(f)$. Its order is the absolute value of the determinant. □

Definition 2.28 (Discriminant of a polynomial)**.** For $f = a_n \prod_{i=1}^{n}(x-\alpha_i)$ of positive degree define

$$\operatorname{disc}(f) = a_n^{2n-2} \prod_{i<j} (\alpha_i - \alpha_j)^2$$

Theorem 2.29. *If f is monic, the discriminant of the trace pairing on $R[x]/(f)$ in the basis $1, x, \ldots, x^{n-1}$ is*

$$\operatorname{disc}(f) = (-1)^{n(n-1)/2} \operatorname{Res}(f, f')$$

For arbitrary f with $\deg f' = n-1$, this becomes

$$\operatorname{disc}(f) = \frac{(-1)^{n(n-1)/2}}{a_n} \operatorname{Res}(f, f')$$

Proof. For monic f, the trace matrix is the product of the Vandermonde matrix of the roots with its transpose, as in Proposition 2.10. The same trace calculation holds with repeated roots by triangularizing multiplication by x. Its determinant is therefore $\prod_{i<j}(\alpha_i - \alpha_j)^2$. Also

$$\prod_i f'(\alpha_i) = a_n^n \prod_i \prod_{j \neq i} (\alpha_i - \alpha_j) = (-1)^{n(n-1)/2} a_n^n \prod_{i<j} (\alpha_i - \alpha_j)^2$$

Now apply Corollary 2.16. □

Theorem 2.30. *For polynomials f, g of positive degree*

$$\operatorname{disc}(fg) = \operatorname{disc}(f)\operatorname{disc}(g)\operatorname{Res}(f,g)^2$$

Proof. The root differences of fg consist of differences between two roots of f, two roots of g, and one root of each. The first two groups give the root products in $\operatorname{disc}(f)$ and $\operatorname{disc}(g)$. The remaining factors, including the leading coefficients, give

$$a_n^{2m} b_m^{2n} \prod_{i,j} (\alpha_i - \beta_j)^2 = \operatorname{Res}(f,g)^2$$

This proves the formula with its sign. □

3. Ideals

In this section you will learn how to determine what primes do in extensions of Dedekind domains (such as number fields).

3.1. Prime Behaviour

For the remainder let R be a Dedekind domain with fraction field K, let L/K be a finite separable extension of degree n, and let S be the integral closure of R in L.

Theorem 3.1. *S is a Dedekind domain.*

Proof. To prove that S is a Dedekind domain, we must verify three conditions:

- S is integrally closed in L. By definition, S is the integral closure of R in L. Since the integral closure of a ring in a field is always integrally closed, S is integrally closed in $L = \operatorname{Frac}(S)$.
- S is Noetherian. Note that Lemma 2.12 implies that S is a submodule of a finitely generated module. Since R is Noetherian, S is a finitely generated R-module and satisfies the ascending chain condition on R-submodules, hence on ideals.
- Every non-zero prime ideal of S is maximal. Let $\mathfrak{q}$ be a non-zero prime ideal of S, and let $\mathfrak{p} = \mathfrak{q} \cap R$. Choose $0 \neq x \in \mathfrak{q}$. The minimal polynomial of x over K has coefficients in R and a nonzero constant term in $(x) \cap R$, so $\mathfrak{p}$ is nonzero. Because R is a Dedekind domain, $\mathfrak{p}$ is maximal. The domain $S/\mathfrak{q}$ is integral over the field $R/\mathfrak{p}$, so it is a field and $\mathfrak{q}$ is maximal. Thus, S has Krull dimension 1.

□

Theorem 3.2. *If R is a Principal Ideal Domain then S is an R-lattice.*

Proof. Note that S is a finitely generated torsion-free R-module, therefore by the structure theorem for finitely generated modules over a PID $S \cong R^n$ as an R-module. We need to show the R-basis for S is also a K-basis for L. For linear independence, note that a

linear dependence with coefficients in K leads to one with coefficients in R after clearing denominators. To show that this basis spans L, note any $x \in L$ satisfies a monic polynomial with coefficients in K. By clearing denominators we can find $r \in R$ with rx satisfying a monic polynomial with coefficients in R. Therefore $rx \in S$ as desired. □

Definition 3.3. For any nonzero prime $\mathfrak{p} \in \operatorname{Spec} R$ we may note

(a) $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k}$;

(b) The $\mathfrak{q}_i$ are exactly the primes with $\mathfrak{q}_i \cap R = \mathfrak{p}$.

We call the e_i the *ramification indices*, we call $\dim_{R/\mathfrak{p}} S/\mathfrak{q}_i$ the *inertia degree* of $\mathfrak{q}_i$ and denote it as $f(\mathfrak{q}_i|\mathfrak{p})$. We call the number of primes g. There are the following special cases:

- if $f(\mathfrak{q}_i|\mathfrak{p}) = e(\mathfrak{q}_i|\mathfrak{p}) = 1$ for every i then $\mathfrak{p}$ is *totally split*;
- if $f(\mathfrak{q}_1|\mathfrak{p}) = g = 1$ then $\mathfrak{p}$ is *totally ramified*;
- if $e(\mathfrak{q}_1|\mathfrak{p}) = g = 1$ then $\mathfrak{p}$ is *inert*;
- if $e(\mathfrak{q}_i|\mathfrak{p}) = 1$ for every i then $\mathfrak{p}$ is *unramified*, otherwise it is *ramified*.

Proof.

(a) S is a Dedekind domain so we can factorize any ideal into primes.

(b) A prime $\mathfrak{q}$ lies above $\mathfrak{p}$ if and only if it contains $\mathfrak{p}S$. Such a prime contains one of the factors $\mathfrak{q}_i$, and maximality gives $\mathfrak{q} = \mathfrak{q}_i$.

□

Theorem 3.4. *For any nonzero prime ideal $\mathfrak{p} \subset R$, let the factorization of $\mathfrak{p}S$ in S be given by $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_g^{e_g}$, and let $f_i = [S/\mathfrak{q}_i : R/\mathfrak{p}]$ be the residue degree. Then:*

$$\sum_{i=1}^{g} e_i f_i = n$$

Proof. We analyze the dimension of the $R/\mathfrak{p}$-vector space $S/\mathfrak{p}S$ in two different ways. The localization $R_\mathfrak{p}$ is a DVR, and hence a PID. Then by Theorem 3.2 $S_\mathfrak{p}$ is an $R_\mathfrak{p}$-lattice and $L \cong S \otimes_R K \cong S_\mathfrak{p} \otimes_{R_\mathfrak{p}} K$, the rank of this free module is:

$$\dim_K(S_\mathfrak{p} \otimes_{R_\mathfrak{p}} K) = [L : K] = n$$

Thus, S is locally free of rank n. This implies S is projective.

Dimension via Rank: We compute the dimension of $S/\mathfrak{p}S$ over the field $k = R/\mathfrak{p}$. Since S is locally free of rank n, we can localize at $\mathfrak{p}$ without changing the quotient:

$$S/\mathfrak{p}S \cong S_\mathfrak{p}/\mathfrak{p}S_\mathfrak{p} \cong S_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p}S_\mathfrak{p}$$

Because $S_\mathfrak{p} \cong R_\mathfrak{p}^{\oplus n}$, we have:

$$S_\mathfrak{p}/\mathfrak{p}S_\mathfrak{p} \cong (R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p})^{\oplus n} \cong k^{\oplus n}$$

Therefore,

$$\dim_k(S/\mathfrak{p}S) = n$$

Dimension via Factorization: Alternatively, use the factorization $\mathfrak{p}S = \prod_{i=1}^{g} \mathfrak{q}_i^{e_i}$. By the Chinese Remainder Theorem, since the powers $\mathfrak{q}_i^{e_i}$ are pairwise comaximal:

$$S/\mathfrak{p}S \cong \bigoplus_{i=1}^{g} S/\mathfrak{q}_i^{e_i}$$

The dimension is additive:

$$\dim_k(S/\mathfrak{p}S) = \sum_{i=1}^{g} \dim_k(S/\mathfrak{q}_i^{e_i})$$

Consider the filtration of $S/\mathfrak{q}_i^{e_i}$ given by the powers of $\mathfrak{q}_i$:

$$S \supset \mathfrak{q}_i \supset \mathfrak{q}_i^2 \supset \cdots \supset \mathfrak{q}_i^{e_i}$$

Each successive quotient $V_j = \mathfrak{q}_i^j/\mathfrak{q}_i^{j+1}$ is a vector space over the field $S/\mathfrak{q}_i$. Since $\mathfrak{q}_i$ is invertible in the Dedekind domain S, there are no proper ideals between $\mathfrak{q}_i^j$ and $\mathfrak{q}_i^{j+1}$, implying V_j is a one-dimensional vector space over $S/\mathfrak{q}_i$.

Thus, as a k-vector space:

$$\dim_k(\mathfrak{q}_i^j/\mathfrak{q}_i^{j+1}) = [S/\mathfrak{q}_i : k] \cdot \dim_{S/\mathfrak{q}_i}(V_j) = f_i \cdot 1 = f_i$$

Summing over the chain of length e_i, we get $\dim_k(S/\mathfrak{q}_i^{e_i}) = e_i f_i$.

Equating the two calculations of the dimension yields $n = \sum_{i=1}^{g} e_i f_i$. □

Now that we understand what primes might do, it is natural to ask how we might figure out what a specific prime actually does in more detail. There are many tools for this which we will investigate in these sections. The most immediately natural one is the *Dedekind-Kummer* theorem.

Theorem 3.5. *Suppose that $L = K(\alpha)$ with $\alpha \in S$, and let $f(x) \in R[x]$ be its monic minimal polynomial. If $\mathfrak{p}$ is a nonzero prime ideal of R such that evaluation at α gives a natural isomorphism:*

$$\phi : (R/\mathfrak{p})[x]/(\bar{f}(x)) \xrightarrow{\sim} S/\mathfrak{p}S \tag{1}$$

Then the factorization into monic irreducibles in $(R/\mathfrak{p})[x]$

$$\bar{f}(x) = \bar{g}_1(x)^{e_1} \cdots \bar{g}_k(x)^{e_k}$$

controls the prime factorization of $\mathfrak{p}S$ in the following way

$$\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k} \qquad \mathfrak{q}_i := (\mathfrak{p}, g_i(\alpha)) \qquad f(\mathfrak{q}_i|\mathfrak{p}) = \deg(\bar{g}_i)$$

where $g_i(x)$ is any lift of $\bar{g}_i(x)$ to $R[x]$.

Proof. We utilize the assumed structural isomorphism to transport the factorization from the polynomial ring to the Dedekind domain S.

Since the factors $\bar{g}_i(x)$ are distinct and irreducible in $(R/\mathfrak{p})[x]$, they are pairwise coprime. Applying the Chinese Remainder Theorem to the quotient ring $(R/\mathfrak{p})[x]/(\bar{f}(x))$ yields the decomposition:

$$\frac{(R/\mathfrak{p})[x]}{(\bar{f}(x))} \cong \prod_{i=1}^{k} \frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x)^{e_i})}$$

Via the isomorphism ϕ, we obtain the corresponding structure for the quotient:

$$S/\mathfrak{p}S \cong \prod_{i=1}^{k} \frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x)^{e_i})}$$

The prime ideals of S containing $\mathfrak{p}S$ correspond to the prime ideals of the quotient $S/\mathfrak{p}S$. Under the decomposition above, the maximal ideals of the product ring are generated by the classes of $\bar{g}_i(x)$.

Let $\mathfrak{q}_i$ be the ideal in S generated by $\mathfrak{p}$ and $g_i(\alpha)$. Under ϕ, the image of $\mathfrak{q}_i$ in $S/\mathfrak{p}S$ corresponds to the ideal $(\bar{g}_i(x))$. Since $(R/\mathfrak{p})[x]/(\bar{g}_i(x))$ is a field (as $\bar{g}_i$ is irreducible), the ideal $\mathfrak{q}_i$ is maximal in S. Thus, $\mathfrak{q}_1, \dots, \mathfrak{q}_k$ are the distinct prime ideals lying over $\mathfrak{p}$. The local factor $(R/\mathfrak{p})[x]/(\bar{g}_i(x)^{e_i})$ is a local ring with a unique maximal ideal which is nilpotent of index e_i. This implies that in the factorization of $\mathfrak{p}S$, the prime ideal $\mathfrak{q}_i$ must appear with exponent e_i. Therefore, $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_k^{e_k}$. Finally, the inertia degree is given by the dimension of the residue field extension:

$$f(\mathfrak{q}_i|\mathfrak{p}) = [S/\mathfrak{q}_i : R/\mathfrak{p}] = \dim_{R/\mathfrak{p}} \left(\frac{(R/\mathfrak{p})[x]}{(\bar{g}_i(x))} \right) = \deg(\bar{g}_i)$$

□

While this theorem is very strong, and in particular tells us exactly how each prime behaves, the condition in Eq. (1) is very opaque. We will therefore spend some time talking about how to actually see whether this is true in the computational sections.

Problem 3.6. Understand quadratic extensions.

(a) Suppose that K is a field with characteristic not equal to 2. Prove that any degree 2 extension of K is $K[\sqrt{d}]$ where $d \in K^\times$ is not a square.

(b) Suppose that $R = \mathbb{Z}$, $K = \mathbb{Q}$, and $L = \mathbb{Q}[\sqrt{d}]$ with $d \in \mathbb{Z}$ squarefree and $d \neq 0, 1$. Compute the ring of integers S, that is the integral closure of R in L.

(c) For every prime $p \in \mathbb{Z}$, determine whether it is inert, ramified, or split in the extension $L/\mathbb{Q}$.

3.2. Ideal Norms

In this section, I will define some machinery that will allow us to reason more smoothly with ideals. That is the ideal norm, a generalization of the norm.

Theorem 3.7. *An element $x \in S$ is a unit if and only if $N(x)$ is a unit.*

Proof.

$\Rightarrow$ In the case that x is a unit we have $N(x^{-1}) \in R$. Therefore

$$N(x)N(x^{-1}) = N(xx^{-1}) = N(1) = 1$$

implies that $N(x)$ is a unit.

$\Leftarrow$ By integrality of x write

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

to be the minimal polynomial of x and recall that $N(x)$ is a power of $(-1)^n a_0$. Therefore $N(x)$ is a unit if and only if a_0 is a unit. If a_0 is a unit we can write

$$x^{-1} = \frac{-1}{a_0}(x^{n-1} + a_{n-1}x^{n-2} + \cdots + a_1)$$

which is an element of S.

□

If R is a PID recall Theorem 3.2 that S is an R-lattice. In this case multiplication by some $s \in S$ sends this lattice to a sub-lattice of itself via some R-matrix. The determinant of this matrix is by definition the norm of s (since the lattice is spanned by a K-basis of L).

By looking at the image of the lattice under s, we can detect the norm of s up to a unit. Essentially, we can see the ideal that the norm of s is generating. This motivates us to try to generalize the concept to ideals.

Definition 3.8. Let M be an R-lattice and N a sub-lattice. We call

$$[M : N]$$

the *index* of N in M and define it as the ideal generated by the determinant of any R-linear map sending M onto N.

Proof. To prove this is well defined, choose maps A, B sending M onto N. If N is not full-rank then $\det(A) = \det(B) = 0$. Otherwise choose a basis $M \cong R^n$ and consider the matrices $B^{-1}A$ and $A^{-1}B$. Since they both send M to M they have entries in R, hence determinants in R. However they are inverses of each other, implying that their determinants are units in R. Therefore $\det(A)/\det(B)$ is a unit, so $\det(A), \det(B)$ are the same up to a unit, and generate the same ideal as desired. □

The most important example of this is when $R = \mathbb{Z}$. Identifying an ideal of $\mathbb{Z}$ with its nonnegative generator, the absolute determinant of the transition matrix is exactly the index of the sub-lattice

$$[M : N] = |M/N|$$

which motivates our definition.

Definition 3.9 (Ideal Norm)**.** Suppose that R is a PID and that I is a nonzero ideal of S. Then I is a full-rank sub-lattice of S. We define the *ideal norm* of I as the index $[S : I]$.

Proof. There are a few things to check:

- This definition works since R being a PID does indeed make S an R-lattice;
- The ideal norm is never 0 since we may find some $x \in K$ such that $S \subseteq xI$. Therefore the sub-lattice must have full rank.

□

Unfortunately when R is not a PID this definition breaks down. Luckily the localization of a Dedekind domain at a nonzero prime is a PID (indeed, it is a DVR). Therefore we define the ideal norm by requiring its localization at each nonzero prime $\mathfrak{p}$ to be $[S_\mathfrak{p} : I_\mathfrak{p}]$. Only finitely many of these local indices are nontrivial, since some nonzero $d \in R$ satisfies $dS \subseteq I$.

Remark 3.10. The ideal norm agrees with the element norm on principal ideals

$$\mathrm{N}((s)) = (\mathrm{N}(s))$$

where $0 \neq s \in S$.

Proof. The action of s on S (considered as an R-lattice) is an R-linear map sending S to (s). □

Theorem 3.11. *The ideal norm is multiplicative. For nonzero ideals $\mathfrak{a}, \mathfrak{b} \subseteq S$:*

$$\mathrm{N}_{L/K}(\mathfrak{ab}) = \mathrm{N}_{L/K}(\mathfrak{a})\,\mathrm{N}_{L/K}(\mathfrak{b})$$

Proof. Working locally, we may assume that R is a DVR, therefore S is semi-local and a PID. Then for ideals (a) and (b) the result follows from the multiplicativity of the element norm

$$\mathrm{N}(ab) = \mathrm{N}(a)\,\mathrm{N}(b)$$

and Remark 3.10. □

Lemma 3.12. *Let M be an R-lattice, N a full-rank sub-lattice, and D the index $[M : N]$. Then $DM \subseteq N$.*

Proof. Let A be a matrix sending M to N. Choose a basis $M \cong R^n$, so A and $\mathrm{adj}(A)$ have entries in R. Then

$$\det(A)M = A\,\mathrm{adj}(A)M \subseteq AM = N$$

Since $D = (\det A)$, this gives $DM \subseteq N$. □

Corollary 3.13. *If I is a nonzero ideal of S then $I \cap R$ contains $N(I)$.*

Theorem 3.14. *The ideal norm is explicitly*

$$\mathrm{N}(\mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_n^{e_n}) = \prod_i (R \cap \mathfrak{q}_i)^{e_i f_{\mathfrak{q}_i}}$$

where $f_{\mathfrak{q}_i}$ is the inertia degree of $\mathfrak{q}_i$.

Proof. By Theorem 3.11 it suffices to prove that $\mathrm{N}(\mathfrak{q}) = \mathfrak{p}^f$ where f is the inertia of $\mathfrak{q}$ and $\mathfrak{p} = \mathfrak{q} \cap R$. Working locally, let R be a DVR, $\mathfrak{q} = (\lambda)$, and $\mathfrak{p} = (\pi)$. Write the exact sequence

$$0 \longrightarrow S \xrightarrow{m_\lambda} S \longrightarrow S/\mathfrak{q} \longrightarrow 0$$

where m_λ is multiplication by λ. By changing the source and target bases, we may put this map into *Smith normal form*

$$\operatorname{diag}(\pi^{c_1}, \dots, \pi^{c_n}) \qquad S/\mathfrak{q} \cong \bigoplus_{i=1}^{n} R/(\pi^{c_i})$$

Since $\mathfrak{p}$ kills $S/\mathfrak{q}$, each c_i is 0 or 1. Therefore

$$(\det m_\lambda) = \mathfrak{p}^{c_1 + \cdots + c_n} \qquad f = \dim_{R/\mathfrak{p}} S/\mathfrak{q} = c_1 + \cdots + c_n$$

as desired. □

It is worth noting that all this machinery also works for fractional ideals just as well.

3.3. The Different Ideal

In this section, we will investigate further prime ramification. We will need the assumption that $R/\mathfrak{p}$ is a perfect field for every nonzero prime $\mathfrak{p}$ (which is always true for number fields), in order to proceed.

Lemma 3.15 (Nilpotents force degeneracy)**.** *Let k be a field and A a finite-dimensional* commutative *k-algebra. If A has a nontrivial nilpotent element then the trace pairing $\mathrm{Tr}_{A/k}$ is degenerate.*

Proof. If $x \in A$ is nilpotent then xy is nilpotent for every y, hence has trace 0. Thus $\mathrm{Tr}_{A/k}(xy) = 0$ for all y and x lies in the left kernel of $\mathrm{Tr}_{A/k}$. □

Definition 3.16 (Discriminant ideal)**.** The *discriminant ideal* $\operatorname{disc}(S/R) \subseteq R$ is the invertible ideal characterized by the rule: for every nonzero prime $\mathfrak{p} \subset R$ and every $R_\mathfrak{p}$-basis $e_1, \dots, e_n$ of $S_\mathfrak{p}$,

$$\operatorname{disc}(S/R)_\mathfrak{p} \;=\; \left(\det(\mathrm{Tr}_{L/K}(e_i e_j))\right) \subseteq R_\mathfrak{p}$$

If R is a PID (for instance $\mathbb{Z}$) then S is a free R-module and we can use Definition 2.9 to compute $\operatorname{disc}(S/R)$. However S isn't always a free R-module so we might run into problems. On the other hand the localizations of R are DVRs, so each $S_\mathfrak{p}$ is indeed free over $R_\mathfrak{p}$ (by the structure theorem for fingen modules over PIDs) and we can compute the discriminant locally.

Lemma 3.17. *Suppose that $\mathfrak{p}$ is a nonzero prime of R and $\mathfrak{q} \in \operatorname{Spm} S$ lies over it. Set $k = R/\mathfrak{p}$, $A = S/\mathfrak{p}S$, and $A_\mathfrak{q} = S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}$, and let $\tau : S \to \mathrm{Hom}_R(S, R)$ be the trace-duality map. Then the trace pairing controls ramification in the following way*

$$\mathfrak{p} \text{ ramifies} \iff \mathrm{Tr}_{A/k} \text{ is degenerate} \iff \tau_\mathfrak{p} \text{ is not an isomorphism}$$

$$\mathfrak{q} \text{ ramifies} \iff \mathrm{Tr}_{A_\mathfrak{q}/k} \text{ is degenerate} \iff \tau_\mathfrak{q} \text{ is not an isomorphism}$$

Here $\mathrm{Hom}_R(S, R)$ is an S-module by $(s\ell)(x) = \ell(sx)$, so we can localize τ at $\mathfrak{q}$.

Proof. Write $\mathfrak{p}S = \mathfrak{q}_1^{e_1} \cdots \mathfrak{q}_g^{e_g}$. By Chinese remainder theorem

$$S/\mathfrak{p}S \cong \prod_{i=1}^{g} S/\mathfrak{q}_i^{e_i}$$

The trace pairing of this product is the direct sum of the pairings on its factors. If $e_i > 1$, then $S/\mathfrak{q}_i^{e_i}$ has a nonzero nilpotent, so its trace pairing is degenerate by Lemma 3.15. If $e_i = 1$, this factor is a finite separable extension of the perfect field k, so its trace pairing is non-degenerate. Localizing at $\mathfrak{q}_i$ isolates this factor. This gives the first column of equivalences.

The reduction of $\tau_{\mathfrak{p}}$ modulo $\mathfrak{p}$ is the trace-duality map of $S/\mathfrak{p}S$, since trace commutes with reduction of a matrix. The cokernel of $\tau_{\mathfrak{p}}$ is finitely generated, and Nakayama shows that it vanishes exactly when its reduction does. The same argument after localizing this cokernel at $\mathfrak{q}$ gives the second equivalence. Finally τ is injective because the trace pairing on L/K is non-degenerate. □

This gives us a very interesting connection. Prime ramification comes from the failure of the trace pairing to be perfect. Therefore we try to understand when this occurs.

Construction 3.18 (Trace-duality map)**.** The trace pairing on L/K restricts to an R-bilinear pairing on S, and hence an R-linear map

$$\tau_{S/R} : S \longrightarrow S^\vee := \mathrm{Hom}_R(S, R) \qquad x \longmapsto \mathrm{Tr}(x-)$$

Over K we have shown that the trace duality is an isomorphism $L \xrightarrow{\sim} \mathrm{Hom}_K(L, K)$ because L/K is separable. However over R the pairing may fail to be perfect. In particular $\tau_{S/R}$ is injective and $\mathrm{coker}(\tau_{S/R})$ is a torsion R-module. The *different* is a tool that will allow us to understand "how much" this pairing fails to be perfect over R.

Definition 3.19 (Codifferent / trace dual)**.** The *codifferent* (or *trace dual*) of S/R is the fractional S-ideal

$$S^\sharp := \{x \in L : \mathrm{Tr}_{L/K}(xS) \subseteq R\}$$

as in Construction 2.5.

Remark 3.20. Recall that $S^\sharp$ is identified with $S^\vee$ through the trace-duality map. Then the statement that the pairing is perfect on S is exactly that $S = S^\sharp$ (so that $S \to S^\vee$ is an isomorphism). Unfortunately the map is not necessarily surjective when restricted to S, so $S^\sharp$ is generally bigger than S. We measure the failure of perfectness by exactly how much bigger it is. However since it's annoying to work with ideals bigger than S, we may as well just invert it.

Definition 3.21 (Different)**.** The *different* of S/R is the inverse fractional ideal

$$\mathcal{D}_{S/R} := (S^\sharp)^{-1} = \{y \in L : \ yS^\sharp \subseteq S\}$$

Since $S \subseteq S^\sharp$ (trace of an integral element is integral), we have $\mathcal{D}_{S/R} \subseteq S$.

Now the failure of perfectness is measured by how much smaller than S the different is.

Proposition 3.22. *The discriminant is the ideal norm of the different.*

Proof. We work locally so that R is a DVR. The ideal norm of $S^\sharp$ is by definition the determinant of the map of lattices sending S to $S^\sharp$. Therefore $\mathrm{N}(\mathfrak{D}_{S/R}) = \mathrm{N}(S^\sharp)^{-1}$ is the determinant of the map of lattices sending $S^\sharp$ to S. This is $\operatorname{disc} S/R$ by Lemma 2.6. □

Theorem 3.23. *Let $\mathfrak{p} \subset R$ be a nonzero prime. The following are equivalent*

(a) $\mathfrak{p} \mid \operatorname{disc}(S/R)$;

(b) $S^\sharp_\mathfrak{p} \neq S_\mathfrak{p}$ *(equivalently,* $\mathcal{D}_{S/R,\mathfrak{p}} \neq S_\mathfrak{p}$*);*

(c) $\mathfrak{p}$ *is ramified.*

Moreover if $\mathfrak{q} \in \operatorname{Spm} S$ lies over $\mathfrak{p}$, then the following are equivalent

(d) $\mathfrak{q} \mid \mathcal{D}_{S/R}$ *(equivalently* $(S^\sharp)_\mathfrak{q} \neq S_\mathfrak{q}$ *or* $\mathcal{D}_{S/R,\mathfrak{q}} \neq S_\mathfrak{q}$*);*

(e) $\mathfrak{q}$ *is ramified.*

Proof. By Remark 3.20 (b) is the statement that the localized trace pairing $S_\mathfrak{p} \times S_\mathfrak{p} \to R_\mathfrak{p}$ is imperfect. Moreover (a) is the statement that the reduced trace pairing $S/\mathfrak{p}S \times S/\mathfrak{p}S \to R/\mathfrak{p}$ is degenerate. The equivalence $(a) \Leftrightarrow (b) \Leftrightarrow (c)$ is Lemma 3.17 .

By Remark 3.20, (d) says that the trace-duality map fails to be an isomorphism after localization at $\mathfrak{q}$. The equivalence $(d) \Leftrightarrow (e)$ follows from Lemma 3.17. □

Theorem 3.24 (Monogenic formula for the different)**.** *Suppose $S = R[\alpha]$ for some $\alpha \in S$, and let $f(x) \in R[x]$ be the minimal polynomial of α over K. Then*

$$S^\sharp = \frac{1}{f'(\alpha)} S \qquad \textit{and hence} \qquad \mathcal{D}_{S/R} = (f'(\alpha)) \subseteq S$$

In particular the discriminant of the power basis satisfies

$$\operatorname{disc}(1, \alpha, \dots, \alpha^{n-1}) = (-1)^{\frac{n(n-1)}{2}} \ \mathrm{N}_{L/K}(f'(\alpha))$$

where the left-hand side is the determinant $\det(\mathrm{Tr}_{L/K}(\alpha^{i+j-2}))$.

Proof. Write $f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$ and define $q_j(T)$ by

$$\frac{f(X)}{X - \alpha} = \sum_{j=0}^{n-1} q_j(\alpha) X^j$$

Here $q_j(T) = T^{n-1-j} + a_{n-1}T^{n-2-j} + \cdots + a_{j+1}$, with $q_{n-1} = 1$. Lagrange interpolation at the distinct roots α_i of f gives, for $0 \leq k < n$,

$$X^k = \sum_i \alpha_i^k \frac{f(X)}{(X - \alpha_i) f'(\alpha_i)}$$

Comparing the coefficient of X^j yields

$$\mathrm{Tr}_{L/K}\left(\frac{\alpha^k q_j(\alpha)}{f'(\alpha)}\right) = \delta_{kj}$$

Thus the trace-dual basis is $q_j(\alpha)/f'(\alpha)$. The q_j are monic of successive degrees $n-1,\dots,0$, so the $q_j(\alpha)$ form an R-basis of S. This proves $S^\sharp = f'(\alpha)^{-1}S$ and the formula for the different. The signed discriminant formula follows from Theorems 2.26 and 2.29. □

By Theorem 3.23 a prime $\mathfrak{p} \subset R$ divides $\mathrm{disc}(S/R)$ if and only if some $\mathfrak{q} \mid \mathfrak{p}$ divides $\mathcal{D}_{S/R}$. This says that the different (resp. discriminant) is supported exactly on the ramified primes of S (resp. of R). In particular, only finitely many primes ramify in a number field extension.

3.4. The Different and Serre duality

Optional read. This subsection is a geometric detour, and nothing later depends on it. The goal is to explain the picture; the language of curves and cohomology can be taken as motivation.

Earlier we thought of a Dedekind domain as the functions on a smooth curve. Now the different lets us push this analogy a little further. Recall that the trace identifies

$$S^\sharp = \mathcal{D}_{S/R}^{-1} \cong \mathrm{Hom}_R(S, R)$$

So the codifferent is the object which turns an element upstairs into a linear functional downstairs. On a curve, differential forms play a closely related role. The connection is that *ramification changes how differential forms pull back.*

A local picture. Consider a finite separable map $\varphi : C \to D$ of smooth curves over an algebraically closed field k. Think of C as several sheets lying above D. At an unramified point, a local coordinate downstairs is still a local coordinate upstairs. At a ramified point, several sheets come together. The simplest model is

$$t = u^e \qquad \text{with } e \text{ nonzero in } k$$

where t is the coordinate downstairs and u is the coordinate upstairs. Differentiating gives

$$\varphi^*(dt) = eu^{e-1}\,du$$

Thus a differential which did not vanish downstairs now vanishes to order $e-1$ upstairs. This is the differential version of the map slowing down at the branch point.

Where the different appears. We can do the same computation using only the algebra we already know. Take $R = k[t] \subset S = k[u]$, with $t = u^e$. The minimal polynomial of u is $F(X) = X^e - t$, so Theorem 3.24 gives

$$\mathcal{D}_{S/R} = (eu^{e-1}) \qquad \text{and} \qquad S^\sharp = \frac{1}{eu^{e-1}}S$$

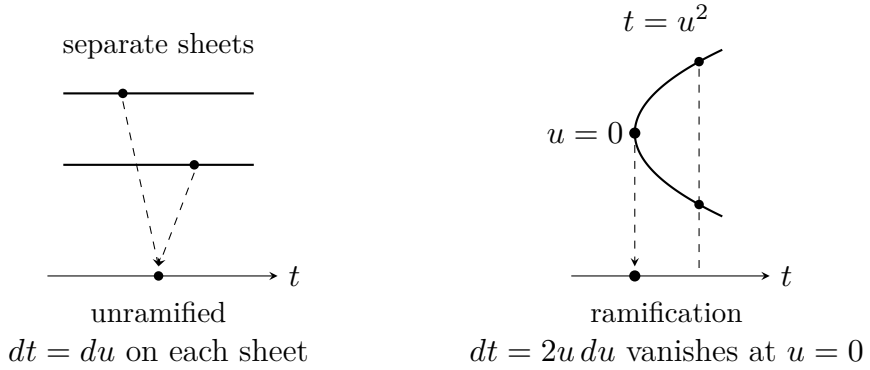


Figure 2: A schematic picture of a double cover, with characteristic different from 2. The right-hand curve is the real slice $t = u^2$: two points above $t > 0$ meet above $t = 0$.

The same derivative appears in both formulas! In particular, inside the rational differentials on C,

$$S^\sharp\, dt = \frac{S}{eu^{e-1}}\, dt = S\, du$$

To recover all regular differentials upstairs from the pulled-back differential dt, we allow exactly the denominators supplied by the codifferent. These denominators cancel the extra vanishing caused by ramification.

The global picture. Write ω_C for the line bundle of regular differential forms on C. Locally its sections look like $h(u)\,du$ with h regular. Let

$$\mathrm{Diff}_{C/D} = \sum_{P \in C} \delta_P [P]$$

be the divisor whose coefficients are the valuations of the different. An ideal cutting out this divisor is $\mathcal{O}_C(-\mathrm{Diff}_{C/D})$; its inverse $\mathcal{O}_C(\mathrm{Diff}_{C/D})$ allows poles of order at most δ_P at P. The local calculation above is a model for the general formula

$$\boxed{\omega_C \cong \varphi^* \omega_D \otimes \mathcal{O}_C(\mathrm{Diff}_{C/D})}$$

Here $\varphi^*\omega_D$ means the line bundle pulled back from D. In the tame case $\delta_P = e_P - 1$, as in the picture. In positive characteristic there can be wild ramification, and then δ_P can be larger. The model $t = u^e$ with $e = 0$ in k is inseparable, so it does not model a separable wild cover.

Why this is related to Serre duality. For a smooth projective curve C, Serre duality says that for every line bundle $\mathcal{L}$ there is a perfect pairing

$$H^1(C, \mathcal{L}) \times H^0(C, \mathcal{L}^{-1} \otimes \omega_C) \longrightarrow k$$

Here H^0 means global sections, while H^1 records gluing obstructions: data on overlaps which cannot be removed by adjusting the choices on each open set. Thus differential forms, with the appropriate twist, give the linear functionals on these gluing obstructions. One can

describe this pairing using residues: multiply local principal parts by differentials and sum the coefficients of $u^{-1}\,du$.

For a finite map, the corresponding duality operation on affine pieces is particularly concrete:

$$\omega_S \cong \mathrm{Hom}_R(S, \omega_R) \cong S^\sharp \otimes_R \omega_R$$

where R and S are the coordinate rings of an affine open downstairs and its inverse image upstairs. This is the finite-map form of duality, and it explains why the very same codifferent occurs in the formula for ω_C. Even its underlying adjunction is familiar linear algebra:

$$\mathrm{Hom}_S(M, \mathrm{Hom}_R(S, R)) \cong \mathrm{Hom}_R(M, R)$$

An R-linear functional ℓ on M corresponds to the S-linear map sending m to the functional $s \mapsto \ell(sm)$.

As a payoff, if C, D are projective, taking degrees in the boxed formula gives Riemann–Hurwitz:

$$2g_C - 2 = \deg(\varphi)(2g_D - 2) + \sum_P \delta_P$$

Here g_C is the genus and $\deg \omega_C = 2g_C - 2$. So the different records exactly the ramification correction to the canonical bundle. For rings of integers, $\mathrm{Hom}_{\mathcal{O}_K}(\mathcal{O}_L, \mathcal{O}_K) = \mathcal{D}_{L/K}^{-1}$ is the arithmetic counterpart of this relative dualizing object. This is a useful reason to remember the codifferent as a duality object, and the different as its vanishing divisor. For the geometric statements, see [3, 4, 5].

3.5. Some Theorems

We will now prove some standard facts about the different and the discriminant.

Theorem 3.25 (Transitivity of the Different)**.** *Let $F \subset K \subset L$ be a tower of finite separable extensions and let $\mathcal{O}_F \subset \mathcal{O}_K \subset \mathcal{O}_L$ be their respective rings of integers. Then the relative different ideals satisfy the transitivity relation:*

$$\mathcal{D}_{L/F} = \mathcal{D}_{L/K} \cdot \mathcal{D}_{K/F}$$

where $\mathcal{D}_{K/F}$ is viewed as the ideal it generates in $\mathcal{O}_L$.

Proof. It is much easier to prove the corresponding identity for the inverse different, also known as the codifferent. Recall that for a finite separable extension E/k, the codifferent $\mathcal{D}_{E/k}^{-1}$ is defined as the fractional ideal:

$$\mathcal{D}_{E/k}^{-1} = \{x \in E \mid \mathrm{Tr}_{E/k}(x\mathcal{O}_E) \subseteq \mathcal{O}_k\}$$

We want to show that $\mathcal{D}_{L/F}^{-1} = \mathcal{D}_{L/K}^{-1} \cdot \mathcal{D}_{K/F}^{-1}$.

Let $x \in L$. By the definition of the codifferent for the extension L/F, we have:

$$\begin{aligned}
x \in \mathcal{D}_{L/F}^{-1} &\iff \mathrm{Tr}_{L/F}(x\mathcal{O}_L) \subseteq \mathcal{O}_F \\
&\iff \mathrm{Tr}_{K/F}(\mathrm{Tr}_{L/K}(x\mathcal{O}_L)) \subseteq \mathcal{O}_F && \text{transitivity of trace} \\
&\iff \mathrm{Tr}_{K/F}(\mathrm{Tr}_{L/K}(x\mathcal{O}_L)\mathcal{O}_K) \subseteq \mathcal{O}_F && \text{linearity of trace on base field} \\
&\iff \mathrm{Tr}_{L/K}(x\mathcal{O}_L) \subseteq \mathcal{D}_{K/F}^{-1} && \text{definition of } \mathcal{D}_{K/F}^{-1} \\
&\iff \mathrm{Tr}_{L/K}(x\mathcal{O}_L\mathcal{D}_{K/F}) \subseteq \mathcal{O}_K && \text{linearity of trace on base field} \\
&\iff x\mathcal{D}_{K/F} \subseteq \mathcal{D}_{L/K}^{-1} && \text{definition of } \mathcal{D}_{L/K}^{-1} \\
&\iff x \in \mathcal{D}_{L/K}^{-1} \cdot \mathcal{D}_{K/F}^{-1}
\end{aligned}$$

Taking the inverse of both sides immediately yields the transitivity formula for the different:

$$\mathcal{D}_{L/F} = \mathcal{D}_{L/K} \cdot \mathcal{D}_{K/F}$$

□

Theorem 3.26 (Tower Law for Discriminants)**.** *For a tower of number fields $L/K/F$, the relative discriminant ideals satisfy:*

$$\Delta_{L/F} = \mathrm{N}_{K/F}(\Delta_{L/K}) \cdot \Delta_{K/F}^{[L:K]}$$

Proof. Recall the transitivity of the relative different ideal:

$$\mathcal{D}_{L/F} = \mathcal{D}_{L/K} \cdot \mathcal{D}_{K/F}$$

The discriminant is the norm of the different, $\Delta_{L/F} = \mathrm{N}_{L/F}(\mathcal{D}_{L/F})$. Applying $\mathrm{N}_{L/F}$ to the transitivity relation and using the transitivity of the norm ($\mathrm{N}_{L/F} = \mathrm{N}_{K/F} \circ \mathrm{N}_{L/K}$), we have:

$$\begin{aligned}
\Delta_{L/F} &= \mathrm{N}_{L/F}(\mathcal{D}_{L/K}) \cdot \mathrm{N}_{L/F}(\mathcal{D}_{K/F}) \\
&= \mathrm{N}_{K/F}\left(\mathrm{N}_{L/K}(\mathcal{D}_{L/K})\right) \cdot \mathrm{N}_{K/F}\left(\mathrm{N}_{L/K}(\mathcal{D}_{K/F})\right)
\end{aligned}$$

By definition, $\mathrm{N}_{L/K}(\mathcal{D}_{L/K}) = \Delta_{L/K}$. Furthermore, since $\mathcal{D}_{K/F}$ is an ideal of $\mathcal{O}_K$, its norm from L to K is simply the ideal raised to the degree of the extension: $\mathrm{N}_{L/K}(\mathcal{D}_{K/F}) = \mathcal{D}_{K/F}^{[L:K]}$. Substituting these in:

$$\Delta_{L/F} = \mathrm{N}_{K/F}(\Delta_{L/K}) \cdot \mathrm{N}_{K/F}(\mathcal{D}_{K/F})^{[L:K]}$$

Recognizing $\mathrm{N}_{K/F}(\mathcal{D}_{K/F})$ as $\Delta_{K/F}$ completes the argument:

$$\Delta_{L/F} = \mathrm{N}_{K/F}(\Delta_{L/K}) \cdot \Delta_{K/F}^{[L:K]}$$

□

Corollary 3.27. *If two number fields have coprime discriminants then they are linearly disjoint over $\mathbb{Q}$. In particular they intersect trivially.*

Proof. Let $\widetilde{K}$ and $\widetilde{L}$ be the Galois closures of the two fields. By Corollary 9.11, their ramified primes are the same as those of K and L respectively. The tower law for discriminants therefore shows that $\widetilde{K} \cap \widetilde{L}$ is unramified at every finite prime. By Corollary 4.9, this intersection is $\mathbb{Q}$. Galois extensions with trivial intersection are linearly disjoint, and the same is then true of their subfields K and L. □

Theorem 3.28. *Let K and L be number fields with rings of integers $\mathcal{O}_K$ and $\mathcal{O}_L$, and discriminants d_K and d_L. If $\gcd(d_K, d_L) = 1$, then*

$$\Delta_{KL} = d_K^{[L:\mathbb{Q}]} \cdot d_L^{[K:\mathbb{Q}]}$$

Proof. By Theorem 3.29 we have $\mathcal{O}_{KL} = \mathcal{O}_K \cdot \mathcal{O}_L$. Moreover by Corollary 3.27 they are linearly disjoint so $\mathcal{O}_{KL} \cong \mathcal{O}_K \otimes \mathcal{O}_L$ and the trace pairing is exactly the one inherited on the tensor product. This gives the result. □

Theorem 3.29. *Let K and L be number fields with rings of integers $\mathcal{O}_K$ and $\mathcal{O}_L$, and discriminants d_K and d_L. If $\gcd(d_K, d_L) = 1$, then the ring of integers of the compositum KL is $\mathcal{O}_{KL} = \mathcal{O}_K\mathcal{O}_L$.*

Proof. The inclusion $\mathcal{O}_K\mathcal{O}_L \subseteq \mathcal{O}_{KL}$ holds trivially. To prove the reverse inclusion, let $\{\alpha_1, \dots, \alpha_n\}$ be an integral basis for $\mathcal{O}_K$. Because $\gcd(d_K, d_L) = 1$, the fields K and L are linearly disjoint over $\mathbb{Q}$, which implies that $\{\alpha_1, \dots, \alpha_n\}$ is also a basis for KL over L. Let $x \in \mathcal{O}_{KL}$ be an arbitrary algebraic integer in the compositum. We can express it uniquely as:

$$x = \sum_{i=1}^{n} c_i \alpha_i \qquad c_i \in L$$

We will show that $c_i \in \mathcal{O}_L$ using the trace pairing. Considering the relative trace of $x\alpha_j$ down to L yields:

$$\mathrm{Tr}_{KL/L}(x\alpha_j) = \sum_{i=1}^{n} c_i \,\mathrm{Tr}_{KL/L}(\alpha_i\alpha_j)$$

Since x and α_j are both algebraic integers, their product $x\alpha_j$ is an algebraic integer. The trace of an algebraic integer is an algebraic integer, so $\mathrm{Tr}_{KL/L}(x\alpha_j) \in \mathcal{O}_L$. Furthermore, because the α_i reside in K, the relative trace of their products restricts to the absolute trace by disjointness of K and L:

$$\mathrm{Tr}_{KL/L}(\alpha_i\alpha_j) = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha_i\alpha_j) \in \mathbb{Z}$$

We now have a linear system of equations $M\vec{c} = \vec{t}$, where:

- M is an $n \times n$ matrix with integer entries $M_{j,i} = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha_i\alpha_j)$.
- $\vec{c} = (c_1, \dots, c_n)^T$ is the column vector of the c_i.
- $\vec{t} = (\mathrm{Tr}(x\alpha_1), \dots, \mathrm{Tr}(x\alpha_n))^T$ is a column vector with entries in $\mathcal{O}_L$.

By Cramer's rule, we can solve for the coefficients as

$$c_i = \frac{\det(M_{[i]})}{\det(M)}$$

where $M_{[i]}$ is the matrix formed by replacing the i-th column of M with $\vec{t}$. By definition the determinant of M is the discriminant d_K. Since $M_{[i]}$ has entries entirely in $\mathcal{O}_L$, its determinant must also lie in $\mathcal{O}_L$. Therefore, $d_K c_i \in \mathcal{O}_L$.

Because $d_K c_i \in \mathcal{O}_L$ for all i we have $d_K x \in \mathcal{O}_K \mathcal{O}_L$. By a symmetric argument, we obtain $d_L x \in \mathcal{O}_K \mathcal{O}_L$. Since $\gcd(d_K, d_L) = 1$, by Bézout's identity there exist integers $u, v \in \mathbb{Z}$ such that $ud_K + vd_L = 1$. We can therefore write x as:

$$x = 1 \cdot x = (ud_K + vd_L)x = u(d_K x) + v(d_L x)$$

Since both $d_K x$ and $d_L x$ belong to the ring $\mathcal{O}_K \mathcal{O}_L$, it follows that $x \in \mathcal{O}_K \mathcal{O}_L$. Thus, $\mathcal{O}_{KL} \subseteq \mathcal{O}_K \mathcal{O}_L$, completing the proof. □

4. The Additive Lattice

Note that $\mathbb{Z}$ is the prototypical example of a Dedekind domain. Given a finite separable field extension $L/\mathbb{Q}$ (in this case L is called a *number field*), Theorem 3.1 tells us that the integral closure of $\mathbb{Z}$ in L, called the *ring of integers* and denoted $\mathcal{O}_L$, is always a Dedekind domain.

4.1. The Minkowski embedding

We have become very familiar with the concept that the integral closure of a PID in a finite separable extension is a lattice. And moreover that the discriminant records the determinant of the trace pairing on this lattice. In the case of a number field, our intuition for lattices becomes much more precise.

From now on we fix a number field $K/\mathbb{Q}$ of degree n and write $\mathcal{O} := \mathcal{O}_K$. Let r_1 be the number of real embeddings and r_2 the number of *pairs* of complex conjugate embeddings, so that

$$n = r_1 + 2r_2$$

Choose embeddings

$$\sigma_1, \dots, \sigma_{r_1} : K \hookrightarrow \mathbb{R} \qquad \text{and} \qquad \tau_1, \dots, \tau_{r_2} : K \hookrightarrow \mathbb{C}$$

where each τ_j is one representative from a conjugate pair (so the full list of embeddings is $\sigma_1, \dots, \sigma_{r_1}, \tau_1, \bar{\tau}_1, \dots, \tau_{r_2}, \bar{\tau}_{r_2}$). We will identify $\mathbb{C} \cong \mathbb{R}^2$ by $x + iy \mapsto (x, y)$.

Construction 4.1 (The additive embedding)**.** Define the map

$$\sigma : K \longrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^n, \qquad x \longmapsto (\sigma_1(x), \dots, \sigma_{r_1}(x), \tau_1(x), \dots, \tau_{r_2}(x))$$

where we view each $\tau_j(x) \in \mathbb{C}$ as a point in $\mathbb{R}^2$ via real and imaginary parts.

Remark 4.2. The Minkowski embedding is an injective homomorphism of additive groups, and it extends $\mathbb{R}$-linearly to an isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$. Furthermore

$$\mathrm{N}(x) = \left(\prod_{i=1}^{r_1} \sigma_i(x)\right) \cdot \left(\prod_{i=1}^{r_2} ((\Re\tau_i(x))^2 + (\Im\tau_i(x))^2)\right)$$

$$\mathrm{Tr}(xy) = \sum_{i=1}^{r_1} \sigma_i(x)\sigma_i(y) + 2\sum_{i=1}^{r_2} (\Re\tau_i(x) \cdot \Re\tau_i(y) - \Im\tau_i(x) \cdot \Im\tau_i(y))$$

which tells us that we can compute the field norm and trace pairing within the embedding. The trace pairing has a negative direction for each complex pair; the usual Euclidean inner product instead uses a plus sign between the real and imaginary terms.

Proposition 4.3. *The subgroup $\sigma(\mathcal{O}) \subseteq \mathbb{R}^n$ is a lattice. Moreover if $d_K \in \mathbb{Z}$ is the discriminant of an integral basis, then*

$$\mathrm{covol}(\sigma(\mathcal{O})) \;=\; 2^{-r_2}\sqrt{|d_K|}$$

Proof.

$\sigma(\mathcal{O})$ is a lattice: By Theorem 3.2 (with $R = \mathbb{Z}$) we have $\mathcal{O} \cong \mathbb{Z}^n$ as an abelian group. Choose a $\mathbb{Z}$-basis $\omega_1, \ldots, \omega_n$ of $\mathcal{O}$. Since σ extends to an $\mathbb{R}$-linear isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^n$, the vectors $\sigma(\omega_1), \ldots, \sigma(\omega_n)$ are $\mathbb{R}$-linearly independent. Therefore

$$\sigma(\mathcal{O}) = \mathbb{Z}\sigma(\omega_1) \oplus \cdots \oplus \mathbb{Z}\sigma(\omega_n)$$

is a lattice.

Computing the covolume: Let V be the usual *embedding matrix* (as in Proposition 2.10) for the ordered list of all n embeddings $\sigma_1, \ldots, \sigma_{r_1}, \tau_1, \bar{\tau}_1, \ldots, \tau_{r_2}, \bar{\tau}_{r_2}$:

$$V_{ij} := \sigma_i(\omega_j) \in \mathbb{C}$$

By Proposition 2.10 we have

$$d_K \;=\; \mathrm{disc}(\omega_1, \ldots, \omega_n) \;=\; (\det V)^2$$

Now let M be the real $n \times n$ matrix whose rows are obtained from V by replacing each conjugate pair of rows

$$(\tau_j(\omega_1), \ldots, \tau_j(\omega_n)) \qquad (\bar{\tau}_j(\omega_1), \ldots, \bar{\tau}_j(\omega_n))$$

with the two real rows

$$(\Re\tau_j(\omega_1), \ldots, \Re\tau_j(\omega_n)) \qquad (\Im\tau_j(\omega_1), \ldots, \Im\tau_j(\omega_n))$$

Concretely, for each j the relation between these pairs of rows is given by

$$\begin{pmatrix} \Re z \\ \Im z \end{pmatrix} \longleftrightarrow \begin{pmatrix} z \\ \bar{z} \end{pmatrix}, \qquad \begin{pmatrix} z \\ \bar{z} \end{pmatrix} = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \begin{pmatrix} \Re z \\ \Im z \end{pmatrix}$$

The determinant of $\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ is $-2i$, so taking all r_2 conjugate pairs we obtain

$$\det V = (\pm(2i)^{r_2}) \det M \qquad \Rightarrow \qquad |\det M| = 2^{-r_2} |\det V|$$

Finally the covolume of $\sigma(\mathcal{O})$ is exactly $|\det M|$, so

$$\operatorname{covol}(\sigma(\mathcal{O})) = |\det M| = 2^{-r_2} |\det V| = 2^{-r_2} \sqrt{|(\det V)^2|} = 2^{-r_2} \sqrt{|d_K|}$$

□

Lemma 4.4. *If $I \subseteq \mathcal{O}$ is a nonzero ideal then $\sigma(I) \subseteq \mathbb{R}^n$ is a sub-lattice of $\sigma(\mathcal{O})$ and*

$$\operatorname{covol}(\sigma(I)) = \operatorname{covol}(\sigma(\mathcal{O})) \cdot [\mathcal{O} : I] = 2^{-r_2} \sqrt{|d_K|} \cdot \mathrm{N}(I)$$

Proof. Since I is a $\mathbb{Z}$-submodule of $\mathcal{O}$ of finite index, its image $\sigma(I)$ is a finite index subgroup of the lattice $\sigma(\mathcal{O})$, hence a sub-lattice. The index is preserved because σ is injective:

$$[\sigma(\mathcal{O}) : \sigma(I)] = [\mathcal{O} : I]$$

Finally for lattices, finite index is the ratio of covolumes:

$$\operatorname{covol}(\sigma(I)) = \operatorname{covol}(\sigma(\mathcal{O})) \cdot [\sigma(\mathcal{O}) : \sigma(I)]$$

and $[\mathcal{O} : I] = \mathrm{N}(I)$ by the discussion in the ideal norm section (for $R = \mathbb{Z}$). □

4.2. Minkowski's convex body theorem

Theorem 4.5 (Minkowski)**.** *Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and let $C \subseteq \mathbb{R}^n$ be convex, centrally symmetric ($C = -C$), and measurable. If*

$$\operatorname{vol}(C) > 2^n \operatorname{covol}(\Lambda)$$

then C contains a nonzero lattice point.

Proof. Let $\pi : \mathbb{R}^n \to \mathbb{R}^n/\Lambda$ be the quotient map. The quotient is a real torus of volume $\operatorname{covol}(\Lambda)$, and π is locally volume preserving. Consider $\frac{1}{2}C = \{x/2 : x \in C\}$. Scaling gives $\operatorname{vol}(\frac{1}{2}C) = \operatorname{vol}(C)/2^n$, so the hypothesis implies

$$\operatorname{vol}\Big(\frac{1}{2}C\Big) > \operatorname{covol}(\Lambda) = \operatorname{vol}(\mathbb{R}^n/\Lambda)$$

Therefore the restriction of π to $\frac{1}{2}C$ cannot be injective (pigeonhole for volume), so we can find distinct $x, y \in \frac{1}{2}C$ with $\pi(x) = \pi(y)$. This means $x - y \in \Lambda$ and $x - y \neq 0$.

Finally, since C is convex and symmetric, we have

$$x, y \in \tfrac{1}{2}C \quad \Rightarrow \quad x - y \in C$$

Indeed, $x, y \in \frac{1}{2}C$ implies $2x, 2y \in C$, so by symmetry $-2y \in C$ and by convexity

$$(2x) + (-2y) \in C + C \subseteq 2C \qquad \Rightarrow \qquad x - y \in C$$

Thus C contains the nonzero lattice point $x - y$. □

4.3. The Minkowski bound

We now feed Theorem 4.5 the specific convex body that remembers the field norm.

Lemma 4.6. *For $T > 0$ define*

$$B_T := \left\{ (x_1, \dots, x_{r_1}, z_1, \dots, z_{r_2}) \in \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \ : \ \sum_{i=1}^{r_1} |x_i| + 2 \sum_{j=1}^{r_2} |z_j| \le T \right\}$$

Then B_T is convex, symmetric, and

$$\operatorname{vol}(B_T) = \frac{2^{r_1 - r_2} \pi^{r_2}}{n!} \, T^n$$

Proof.

(a) Convexity and symmetry are immediate from the defining inequality.

(b) For the volume, by symmetry in the real coordinates we may restrict to $x_i \ge 0$ and multiply by 2^{r_1}. For the complex coordinates we use polar coordinates $z_j = \rho_j e^{i\theta_j}$ so $d(\Re z_j)\, d(\Im z_j) = \rho_j \, d\rho_j \, d\theta_j$ and $\int_0^{2\pi} d\theta_j = 2\pi$. Thus

$$\operatorname{vol}(B_T) = 2^{r_1} (2\pi)^{r_2} \int_{\substack{x_i \ge 0,\ \rho_j \ge 0 \\ \sum x_i + 2\sum \rho_j \le T}} \Big(\prod_{j=1}^{r_2} \rho_j \Big) \, dx \, d\rho$$

Substitute $s_j := 2\rho_j$ so $\rho_j \, d\rho_j = \frac{1}{4} s_j \, ds_j$ and hence $\prod \rho_j \, d\rho_j = 4^{-r_2} \prod s_j \, ds_j$. We get

$$\operatorname{vol}(B_T) = 2^{r_1} (2\pi)^{r_2} 4^{-r_2} \int_{\substack{x_i \ge 0,\ s_j \ge 0 \\ \sum x_i + \sum s_j \le T}} \Big(\prod_{j=1}^{r_2} s_j \Big) \, dx \, ds$$

Now we integrate out the x_i first. For fixed $s = (s_1, \dots, s_{r_2})$ with $\sum s_j \le T$, the region in the x-variables is the standard simplex

$$\{ x_i \ge 0 : x_1 + \cdots + x_{r_1} \le T - \sum s_j \}$$

which has volume $\frac{(T - \sum s_j)^{r_1}}{r_1!}$. Therefore the integral becomes

$$\int_{\substack{s_j \ge 0 \\ \sum s_j \le T}} \frac{(T - \sum s_j)^{r_1}}{r_1!} \Big(\prod_{j=1}^{r_2} s_j \Big) \, ds$$

This is a standard simplex integral (one can do it by induction using $\int_0^A (A-t)^a t^b dt = \frac{a!\, b!}{(a+b+1)!} A^{a+b+1}$), and it evaluates to $\frac{T^n}{n!}$. Putting everything together:

$$\operatorname{vol}(B_T) = 2^{r_1} (2\pi)^{r_2} 4^{-r_2} \cdot \frac{T^n}{n!} = \frac{2^{r_1 - r_2} \pi^{r_2}}{n!} \, T^n$$

□

Theorem 4.7 (Minkowski lemma)**.** *Let $I \subseteq \mathcal{O}$ be a nonzero ideal. Then there exists $0 \neq \alpha \in I$ such that*

$$|\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n}\sqrt{|d_K|}\cdot \mathrm{N}(I)$$

Proof. We apply Theorem 4.5 to the lattice $\Lambda = \sigma(I) \subseteq \mathbb{R}^n$ and the convex body B_T from Lemma 4.6. By Lemma 4.4 we have

$$\mathrm{covol}(\Lambda) = 2^{-r_2}\sqrt{|d_K|}\cdot \mathrm{N}(I)$$

Using Lemma 4.6, the inequality $\mathrm{vol}(B_T) > 2^n \,\mathrm{covol}(\Lambda)$ is

$$\frac{2^{r_1-r_2}\pi^{r_2}}{n!}T^n > 2^n \cdot 2^{-r_2}\sqrt{|d_K|}\cdot \mathrm{N}(I)$$

Since $n = r_1 + 2r_2$, this rearranges to

$$T^n > \left(\frac{4}{\pi}\right)^{r_2} n!\,\sqrt{|d_K|}\cdot \mathrm{N}(I)$$

Choose such a T. Then Minkowski gives a nonzero lattice point $\sigma(\alpha) \in \sigma(I) \cap B_T$, i.e. a nonzero $\alpha \in I$ with

$$\sum_{i=1}^{r_1} |\sigma_i(\alpha)| + 2\sum_{j=1}^{r_2} |\tau_j(\alpha)| \leq T$$

Now apply AM–GM to the n nonnegative numbers

$$|\sigma_1(\alpha)|, \dots, |\sigma_{r_1}(\alpha)|,\ |\tau_1(\alpha)|, |\tau_1(\alpha)|, \dots, |\tau_{r_2}(\alpha)|, |\tau_{r_2}(\alpha)|$$

Their product is $|\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)|$ and their average is $\leq T/n$, so

$$|\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq \left(\frac{T}{n}\right)^n$$

Let T decrease to the positive value for which equality holds in the bound on T^n. All the resulting lattice points lie in one fixed bounded body, so some nonzero α occurs infinitely often. Taking the limit for this α gives

$$|\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n}\sqrt{|d_K|}\cdot \mathrm{N}(I)$$

as desired. □

Theorem 4.8 (Minkowski bound)**.** *Every ideal class in Cl_K contains an integral ideal $\mathfrak{a} \subseteq \mathcal{O}$ with*

$$\mathrm{N}(\mathfrak{a}) \leq M_K \qquad \textit{where} \qquad M_K := \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n}\sqrt{|d_K|}$$

Proof. Let $c \in \mathrm{Cl}_K$ and choose an integral ideal $I \subseteq \mathcal{O}$ representing c^{-1}. Apply Theorem 4.7 to I to obtain $0 \neq \alpha \in I$ with

$$|\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)| \leq M_K \cdot \mathrm{N}(I)$$

Since $\alpha \in I$ we have $(\alpha) \subseteq I$, hence

$$\mathfrak{a} := (\alpha) I^{-1} \subseteq \mathcal{O}$$

is an integral ideal. It satisfies $[\mathfrak{a}] = [I^{-1}] = c$, because (α) is principal. Finally by multiplicativity of ideal norm and Remark 3.10 we get

$$\mathrm{N}(\mathfrak{a}) = \mathrm{N}((\alpha))\,\mathrm{N}(I^{-1}) = \frac{|\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)|}{\mathrm{N}(I)} \leq M_K$$

□

Corollary 4.9. *Every nontrivial finite extension of $\mathbb{Q}$ is ramified on at least one prime.*

Proof. Since there is at least one ideal class, the Minkowski bound must be at least 1. This yields

$$\sqrt{|d_K|} \geq \frac{n^n}{n!}\left(\frac{\pi}{4}\right)^{r_2} \geq \frac{n^n}{n!}\left(\frac{\pi}{4}\right)^{n/2}$$

which turns out to be greater than 1 when $n > 1$. □

Remark 4.10. It turns out that if we assume the *Generalized Riemann Hypothesis* then we get a much smaller bound for generators of the class group. Bach's theorem says that Cl_K is generated by the classes of prime ideals $\mathfrak{p}$ satisfying

$$\mathrm{N}(\mathfrak{p}) \leq 12(\ln |d_K|)^2$$

Thus the search for generating primes only needs norms bounded by a quadratic function of $\ln |d_K|$. This is very useful in computational algebraic number theory.

4.4. Finiteness of the class number

Now the class number finiteness is essentially a counting statement about sub-lattices of $\mathbb{Z}^n$.

Lemma 4.11. *For every real number $B > 0$ there are only finitely many integral ideals $\mathfrak{a} \subseteq \mathcal{O}$ with $\mathrm{N}(\mathfrak{a}) \leq B$.*

Proof. Fix a $\mathbb{Z}$-basis $\mathcal{O} \cong \mathbb{Z}^n$. If $\mathfrak{a} \subseteq \mathcal{O}$ is an ideal then it is in particular a $\mathbb{Z}$-submodule, and

$$\mathrm{N}(\mathfrak{a}) = [\mathcal{O} : \mathfrak{a}]$$

For a fixed integer $m \geq 1$, any subgroup $H \subseteq \mathbb{Z}^n$ of index m contains $m\mathbb{Z}^n$ by Lemma 3.12. Therefore such H correspond to subgroups of the finite abelian group

$$\mathbb{Z}^n / m\mathbb{Z}^n \cong (\mathbb{Z}/m\mathbb{Z})^n$$

A finite group has only finitely many subgroups, so there are only finitely many $\mathbb{Z}$-submodules of index m in $\mathbb{Z}^n$. Finally there are only finitely many integers $m \leq B$, so there are only finitely many ideals with $\mathrm{N}(\mathfrak{a}) \leq B$. □

Theorem 4.12. *The class group* Cl_K *is finite.*

Proof. By Theorem 4.8, every class contains an ideal $\mathfrak{a}$ with $\mathrm{N}(\mathfrak{a}) \leq M_K$. By Lemma 4.11 there are only finitely many such ideals. Therefore there are only finitely many ideal classes. □

5. The Units

The other term in the main exact sequence is $\mathcal{O}^\times$. Again the slogan is linear algebra: multiplication becomes addition after taking logs.

Construction 5.1 (The logarithmic embedding)**.** Define

$$\iota : \mathcal{O}^\times \longrightarrow H := \left\{ x \in \mathbb{R}^{r_1+r_2} : \sum_{i=1}^{r_1+r_2} x_i = 0 \right\}$$

by

$$u \longmapsto (\log|\sigma_1(u)|, \dots, \log|\sigma_{r_1}(u)|, 2\log|\tau_1(u)|, \dots, 2\log|\tau_{r_2}(u)|)$$

This lands in H because for every unit u,

$$\sum_{i=1}^{r_1} \log|\sigma_i(u)| + 2\sum_{j=1}^{r_2} \log|\tau_j(u)| = \log|N_{K/\mathbb{Q}}(u)| = 0$$

The image is called the *multiplicative lattice*. The same formula defines $\iota : K^\times \to \mathbb{R}^{r_1+r_2}$.

Remark 5.2. We have the commutative diagram.

$$\begin{array}{ccccc} & & \overset{\iota}{\frown} & & \\ K^\times & \xrightarrow{\sigma} & (\mathbb{R}^\times)^{r_1} \times (\mathbb{C}^\times)^{r_2} & \xrightarrow{\iota_\infty} & \mathbb{R}^{r_1+r_2} \\ \downarrow{\scriptstyle N_{K/\mathbb{Q}}} & & \downarrow{\scriptstyle \mathrm{N}} & & \downarrow{\scriptstyle \mathrm{Tr}} \\ \mathbb{Q}^\times & \longrightarrow & \mathbb{R}^\times & \xrightarrow{\log|-|} & \mathbb{R} \end{array}$$

The map ι_∞ takes logarithms of absolute values, with weight 2 on complex coordinates. The middle downward map is given by Remark 4.2 and the right downward map is the standard sum of coordinates on $\mathbb{R}^{r_1+r_2}$. The image of $\mathcal{O}_K$ under σ is the additive lattice and the image of $\mathcal{O}_K^\times$ under ι is the multiplicative lattice. We can consider the image of $\mathcal{O}_K \setminus \{0\}$ under ι, which is an additive submonoid. However its intersection with H corresponds to those elements with absolute norm 1, i.e. the image of $\mathcal{O}_K^\times$ by Theorem 3.7.

Theorem 5.3 (Dirichlet unit theorem)**.** *The kernel of* ι *is the finite group of roots of unity*

$$\mu_K := \{u \in \mathcal{O}^\times : u^m = 1 \text{ for some } m \geq 1\}$$

and the image $\iota(\mathcal{O}^\times)$ *is a lattice in* H*. In particular*

$$\operatorname{rank} \mathcal{O}^\times = r_1 + r_2 - 1$$

Proof. We separate the argument into three pieces. Extracting the desired properties from the additive lattice.

$\ker \iota = \mu_K$**:** If $u \in \ker \iota$ then every archimedean absolute value of u is 1. Therefore under the additive embedding from Construction 4.1 we have

$$\sigma(u^m) \in [-1, 1]^n$$

for every $m \geq 1$. Since $\sigma(\mathcal{O}_K)$ is the additive lattice, this box contains only finitely many lattice points. Hence two powers coincide, so u is torsion. The whole kernel lies in this bounded box, so it is finite. The converse is immediate.

$\iota(\mathcal{O}^\times)$ **is discrete:** Fix $B > 0$. If $\|\iota(u)\|_\infty \leq B$ then every embedding of u has absolute value at most e^B. Thus $\sigma(u)$ lies in a bounded box in $\mathbb{R}^n$, and the additive lattice meets a bounded box in only finitely many points. Therefore bounded subsets of H meet $\iota(\mathcal{O}^\times)$ in only finitely many points.

$\iota(\mathcal{O}^\times)$ **has full rank:** If $r_1 + r_2 = 1$, then $H = \{0\}$ and there is nothing to prove. Otherwise write $\iota(u) = (u_1, \dots, u_{r_1+r_2})$. Fix one of the chosen embeddings τ_k. We would like to construct some unit u with

$$u_k > 0 \qquad u_i < 0 \text{ for } i \neq k$$

We construct an infinite sequence of non-zero algebraic integers $\alpha_1, \alpha_2, \cdots \in \mathcal{O}_K$ by iteratively applying Minkowski's Convex Body Theorem to products of real intervals and complex disks. For the first step, choose positive real numbers $c_{1,\tau}$ for each real embedding and each chosen complex embedding τ such that $\prod_\tau c_{1,\tau}^{N_\tau} = C' > (2/\pi)^{r_2}\sqrt{|d_K|}$ (where $N_\tau = 1$ for real embeddings and 2 for complex embeddings). By Minkowski's Theorem, there exists a non-zero $\alpha_1 \in \mathcal{O}_K$ such that $|\tau(\alpha_1)| \leq c_{1,\tau}$ for all τ.

For the m-th step (where $m > 1$), choose new positive real bounds $c_{m,\tau}$ such that for all $\tau \neq \tau_k$, we strictly shrink the box:

$$c_{m,\tau} < |\tau(\alpha_{m-1})|$$

To maintain the product C', we compensate by making the bound c_{m,τ_k} sufficiently large. Minkowski's Theorem again guarantees the existence of a non-zero $\alpha_m \in \mathcal{O}_K$ inside this new box, satisfying $|\tau(\alpha_m)| \leq c_{m,\tau}$ for all τ. This construction yields an infinite sequence $(\alpha_m)_{m=1}^\infty$ in $\mathcal{O}_K$. By the definition of our bounding boxes, the absolute values in every direction $\tau \neq \tau_k$ are strictly decreasing:

$$|\tau(\alpha_1)| > |\tau(\alpha_2)| > |\tau(\alpha_3)| > \dots$$

However, the absolute norm of every α_m is bounded by the same constant:

$$|N_{K/\mathbb{Q}}(\alpha_m)| = \prod_\tau |\tau(\alpha_m)|^{N_\tau} \leq C'$$

Since there are only finitely many principal ideals in $\mathcal{O}_K$ with bounded norm there must exist indices $m < n$ such that α_m and α_n generate the exact same principal ideal $(\alpha_m) = (\alpha_n)$. Because they generate the same ideal, they are associates. Thus, there exists a unit $u \in \mathcal{O}_K^*$ such that:

$$\alpha_n = \alpha_m \cdot u \implies u = \frac{\alpha_n}{\alpha_m}$$

We now verify the logarithmic coordinates of u. For any embedding $\tau \neq \tau_k$, since $m < n$, our strictly decreasing condition ensures that $|\tau(\alpha_n)| < |\tau(\alpha_m)|$. Therefore:

$$|\tau(u)| = \frac{|\tau(\alpha_n)|}{|\tau(\alpha_m)|} < 1$$

Taking the real logarithm, we obtain:

$$\log|\tau(u)| < 0 \quad \text{for all } \tau \neq \tau_k$$

Since u is a unit the sum of all its coordinates is 0. Therefore the remaining k-th term must be strictly positive to balance the equation. This completes the construction of the desired unit. Now construct units $u_1, \dots, u_{r_1+r_2}$ of this form for every index. We claim that they span a subspace of rank at least $r_1 + r_2 - 1$. To see this, assemble them as the rows of a matrix.

$$A_{ij} = \iota(u_i)_j \qquad A = \begin{bmatrix} + & - & \cdots & - \\ - & + & \cdots & - \\ \vdots & \vdots & \ddots & \vdots \\ - & - & \cdots & + \end{bmatrix}$$

Assume for contradiction that the first $s := r_1 + r_2 - 1$ columns of A are linearly dependent. Then there exist real numbers $c_1, \dots, c_s$, not all zero, such that for every row i:

$$\sum_{j=1}^{s} A_{ij} c_j = 0$$

Let k be the index of the coefficient with the maximum absolute value, so $|c_k| \geq |c_j|$ for all j, and strictly $|c_k| > 0$. Evaluating the k-th row and isolating the diagonal term yields:

$$\begin{aligned} A_{kk} c_k &= -\sum_{j \neq k}^{s} A_{kj} c_j \\ A_{kk}|c_k| &\leq \sum_{j \neq k}^{s} |A_{kj}||c_j| && \text{triangle inequality} \\ &\leq \sum_{j \neq k}^{s} (-A_{kj})|c_k| && \text{maximality and } |A_{kj}| = -A_{kj} \\ A_{kk} &\leq -\sum_{j \neq k}^{s} A_{kj} \end{aligned}$$

Subtracting the RHS gives $\sum_{j=1}^{s} A_{kj} \leq 0$. However the sum of the indices along the entire row must be 0, so this sum must be positive. This is a contradiction and we are done.

□

So up to torsion, the unit group is literally a lattice.

Definition 5.4 (Fundamental units)**.** Write $r := r_1 + r_2 - 1$. A collection of units $\epsilon_1, \dots, \epsilon_r \in \mathcal{O}^\times$ is called a system of *fundamental units* if

$$\iota(\mathcal{O}^\times) = \mathbb{Z}\iota(\epsilon_1) \oplus \cdots \oplus \mathbb{Z}\iota(\epsilon_r)$$

Corollary 5.5. *If $\epsilon_1, \dots, \epsilon_r$ are fundamental units then every unit $u \in \mathcal{O}^\times$ may be written uniquely as*

$$u = \zeta \epsilon_1^{n_1} \cdots \epsilon_r^{n_r} \qquad (\zeta \in \mu_K,\ n_i \in \mathbb{Z})$$

Equivalently,

$$\mathcal{O}^\times \cong \mu_K \times \mathbb{Z}^r$$

Proof. Since $\iota(\mathcal{O}^\times)$ is a free abelian group of rank r, choosing a $\mathbb{Z}$-basis of it and lifting gives the ϵ_i. The only ambiguity in lifting is the kernel of ι, which is exactly μ_K by Theorem 5.3. □

Remark 5.6. This is the multiplicative analogue of the additive lattice. The additive embedding put $\mathcal{O}$ inside $\mathbb{R}^n$ as a lattice of covolume $2^{-r_2}\sqrt{|d_K|}$. The logarithmic embedding puts $\mathcal{O}^\times$ inside the sum-0 hyperplane H as a lattice of rank $r_1 + r_2 - 1$. Deleting any one coordinate identifies H with $\mathbb{R}^{r_1+r_2-1}$. The covolume of the resulting lattice is called the *regulator* R_K. With the Euclidean measure induced on H itself, the covolume is $\sqrt{r_1 + r_2} R_K$.

Example 5.7.

- If K is imaginary quadratic then $r_1 = 0$ and $r_2 = 1$, so $r = 0$. Thus $\mathcal{O}^\times = \mu_K$ is finite.
- If K is real quadratic then $r_1 = 2$ and $r_2 = 0$, so $r = 1$. Thus there is a single fundamental unit $\epsilon > 1$ and
$$\mathcal{O}^\times = \{\pm\epsilon^n : n \in \mathbb{Z}\}$$

So, up to roots of unity, computing the unit group is just finding a basis of the multiplicative lattice.

6. Local Fields

In this section we will discuss local fields.

6.1. Completions

Definition 6.1 (Analytical I-adic Completion)**.** Let R be a commutative ring and I an ideal of R. We define the I-adic order $v_I(x) = \sup\{n \geq 0 \mid x \in I^n\}$, which induces a pseudometric $d(x, y) = c^{v_I(x-y)}$ for some fixed constant $c \in (0, 1)$.

Let $\mathcal{C}$ be the ring of all Cauchy sequences in R with respect to this metric, and let $\mathcal{N}$ be the ideal of all null sequences (sequences converging to 0). The *analytical I-adic completion* of R is defined as the quotient ring:

$$\hat{R}_{\text{an}} = \mathcal{C}/\mathcal{N}$$

It is worth noting that the "metric" we have defined is not necessarily a true metric unless $\cap_{n=1}^{\infty} I^n = 0$. This is because $d(x, y) = 0$ does not necessarily imply that $x = y$. The order satisfies $v_I(xy) \geq v_I(x) + v_I(y)$ and $v_I(x + y) \geq \min(v_I(x), v_I(y))$, which give the needed continuity of the ring operations. In the cases we care about the intersection is zero.

Definition 6.2 (Algebraic I-adic Completion)**.** Let R be a ring and I an ideal of R. The natural projection maps $\pi_n : R/I^{n+1} \to R/I^n$ form an inverse system of rings. The *algebraic I-adic completion* of R is defined as the inverse limit of this system:

$$\hat{R}_{\text{alg}} = \varprojlim_n R/I^n$$

This ring consists of all coherent sequences $(a_n)_{n\geq 1}$, where $a_n \in R/I^n$ and $\pi_n(a_{n+1}) = a_n$ for all $n \geq 1$.

Theorem 6.3. *Let R be a ring and I an ideal of R. The analytical I-adic completion is naturally isomorphic to the algebraic I-adic completion. That is, there exists a ring isomorphism:*

$$\hat{R}_{an} \cong \hat{R}_{alg}$$

Proof. Let $[(x_k)] \in \hat{R}_{\text{an}}$ be the equivalence class of a Cauchy sequence $(x_k)_{k\geq 1}$. Because (x_k) is Cauchy, for every integer $n \geq 1$, the sequence of projections $x_k \pmod{I^n}$ is eventually constant. Let $a_n \in R/I^n$ denote this stabilized limit.

By construction, $a_{n+1} \equiv a_n \pmod{I^n}$, so the sequence $(a_n)_{n\geq 1}$ is a coherent sequence in $\varprojlim R/I^n$. The mapping $[(x_k)] \mapsto (a_n)_{n\geq 1}$ is a ring homomorphism. Its kernel consists of null sequences. Conversely, choosing representatives $x_n \in R$ of a coherent sequence a_n gives a Cauchy sequence with these stabilized residues. This proves surjectivity and the desired isomorphism. □

Intuitively, taking a completion is a stronger form of looking at a ring "locally". In the algebraic geometry perspective, consider localizing at a point (maximal prime ideal):

- Localization is what happens when you look at all rational functions, and only enforce they are defined at that point.
- Completion is what happens when you consider each function by its "derivatives" or power series at that point. And enforce that every power series should define a function.

To formally state that completion is some kind of local transformation, we consider the following theorems:

Lemma 6.4 (Isomorphism of Quotients)**.** *Let R be a ring, I an ideal of R, and S a multiplicatively closed subset of R. Assume that for every $s \in S$, the image of s in the quotient ring R/I is a unit. Then $S^{-1}\widehat{R}_I = \widehat{R}_I$.*

Proof. The statement that s is a unit modulo I is that

$$sx = 1 \pmod{I}$$

for some $x \in R$. Write $sx = 1 - t$ with $t \in I$. The geometric series converges in $\widehat{R}_I$ and gives

$$s^{-1} = x \sum_{j=0}^{\infty} t^j$$

Its truncations also give inverses modulo every I^n. Thus $\widehat{R}_I$ already satisfies the universal property of localization at S. □

Theorem 6.5 (Completion Commutes with Localization)**.** *Let R be a ring, I an ideal, and S a multiplicatively closed subset whose elements map to units in R/I. Then there is a natural isomorphism:*

$$\widehat{R}_I \cong (\widehat{S^{-1}R})_{IS^{-1}R}$$

Proof. By definition, the algebraic completions are defined as inverse limits:

$$\widehat{R}_I = \varprojlim_n R/I^n \quad \text{and} \quad (\widehat{S^{-1}R})_{IS^{-1}R} = \varprojlim_n S^{-1}R/I^n S^{-1}R$$

By the assumption $S^{-1}(R/I^n) = R/I^n$ therefore by exactness of localization (commuting with quotients) we have $\phi_n : R/I^n \to S^{-1}R/I^n S^{-1}R$ are isomorphisms for all $n \geq 1$. Furthermore, these isomorphisms commute with the projection maps of the inverse systems. Because the inverse limits of isomorphic inverse systems are themselves isomorphic, the completions are isomorphic. □

6.2. Lifting Roots

In this section we will derive some results about lifting roots. We will prove an extremely general form of Hensel's lemma, for which we need to use the Resultant. For the remainder we let R be a discrete valuation ring with valuation v and $\mathfrak{p} := \{x \in R \mid v(x) > 0\}$, such that R is complete with respect to its valuation. We let K be the fraction field of R, which is a complete discretely valued field. When the residue field is finite, K is a local field. Normalize v so that a uniformizer has valuation 1, and set $|x| = c^{v(x)}$ for a fixed $0 < c < 1$. For a polynomial $f = \sum a_i x^i$, write $v(f) = \min_i v(a_i)$, with $v(0) = \infty$.

Theorem 6.6 (Strong Hensel's Lemma)**.** *Let $h, f, g \in R[x]$ be monic, with f, g of positive degree, $\deg h = \deg f + \deg g$, and let $e = \mathrm{Res}(f, g) \neq 0$. If*

$$v(h - fg) > 2v(e)$$

then there exist unique monic polynomials $F, G \in R[x]$ such that

- $h = FG$
- $\deg F = \deg f \qquad \deg G = \deg g$
- $v(F - f) \geq v(h - fg) - v(e) > v(e) \qquad v(G - g) \geq v(h - fg) - v(e) > v(e)$

Proof. Write $t = v(e)$. If $h = fg$, the existence statement is immediate. Otherwise write $h - fg = e^2\delta$ with $v(\delta) > 0$. Since all three polynomials are monic, $\deg \delta < \deg f + \deg g$. By Lemma 2.22 there are unique $U, V \in R[x]$ with

$$fV + gU = e\delta \qquad \deg U < \deg f \qquad \deg V < \deg g$$

Moreover $v(U), v(V) \geq v(\delta)$ by the coefficient statement in that lemma. Set $f_1 = f + eU$ and $g_1 = g + eV$. Then

$$h - f_1 g_1 = e^2\delta - e(fV + gU) - e^2UV = -e^2UV$$

so

$$v(h - f_1 g_1) \geq 2t + 2v(\delta) \qquad v(f_1 - f), v(g_1 - g) \geq t + v(\delta) > t$$

The resultant is a polynomial in the coefficients, hence

$$v(\mathrm{Res}(f_1, g_1) - e) > t \qquad v(\mathrm{Res}(f_1, g_1)) = t$$

Thus we can repeat the construction using the new resultant $e_1 = \mathrm{Res}(f_1, g_1)$. The new error factor is $\delta_1 = -e^2UV/e_1^2$, whose valuation is at least $2v(\delta)$ because e/e_1 is a unit. At every step the excess $v(h - f_k g_k) - 2t$ at least doubles, and the corrections tend to zero. Completeness gives limits F, G with $h = FG$ and the stated degree and valuation bounds.

For uniqueness, suppose $F + A, G + B$ is another such pair, with $\deg A < \deg F$ and $\deg B < \deg G$. If either difference is nonzero, put $w = \min(v(A), v(B)) > t$. Then

$$FB + GA = -AB$$

The resultant of F, G has valuation t. Applying its adjugate matrix to this equation shows that every coefficient of $\mathrm{Res}(F, G)A$ and $\mathrm{Res}(F, G)B$ has valuation at least $2w$. Thus $t + w \geq 2w$, contradicting $w > t$. Therefore $A = B = 0$. □

When the resultant is a unit, the same correction argument works in a ring complete and separated for the I-adic topology, starting with an error in $I[x]$. We have many immediate corollaries of this result, which are normally proven on their own.

Corollary 6.7. *If a polynomial $f \in R[X]$ has a simple root a_0 modulo $\mathfrak{p}$ then it has a unique root a with $a \equiv a_0 \pmod{\mathfrak{p}}$.*

Corollary 6.8. *If $h, f, g \in R[x]$ are monic polynomials with $h = fg \pmod{\mathfrak{p}}$ and f, g are coprime modulo $\mathfrak{p}$ then they lift to unique F, G with $h = FG$ and $F = f \pmod{\mathfrak{p}}$, $G = g \pmod{\mathfrak{p}}$.*

In the cases that we care about, R will be a DVR, complete with respect to the valuation on its maximal ideal. In this case, we can strengthen the previous result.

Theorem 6.9 (Newton's Lemma)**.** *Let $f \in R[X]$ and suppose $a_0 \in R$ satisfies*

$$|f(a_0)| < |f'(a_0)|^2$$

Then there is a unique root a with $|a - a_0| < |f'(a_0)|$, and it satisfies

$$|a - a_0| \leq \left| \frac{f(a_0)}{f'(a_0)} \right|$$

The sequence

$$a_{k+1} = a_k - \frac{f(a_k)}{f'(a_k)}$$

converges to a.

Proof. Set $t = v(f'(a_0))$. If $f(a_0) = 0$, the sequence is constant. Otherwise suppose $s_k = v(f(a_k)) > 2t$ and $v(f'(a_k)) = t$. The Newton correction $b_k = -f(a_k)/f'(a_k)$ has valuation $s_k - t > t$, so $a_{k+1} \in R$. Taylor expansion gives

$$f(a_k + b_k) = f(a_k) + f'(a_k)b_k + b_k^2 c_k = b_k^2 c_k \qquad c_k \in R$$

and $f'(a_k + b_k) - f'(a_k)$ is divisible by b_k. Consequently

$$s_{k+1} \geq 2(s_k - t) \qquad v(f'(a_{k+1})) = t$$

The excess $s_k - 2t$ at least doubles, so the corrections tend to zero and the sequence converges to a root a. The first correction bounds the total change, giving the claimed inequality. If a, b are two roots in the stated ball, Taylor expansion gives

$$0 = f(b) - f(a) = (b - a)(f'(a) + (b - a)c) \qquad c \in R$$

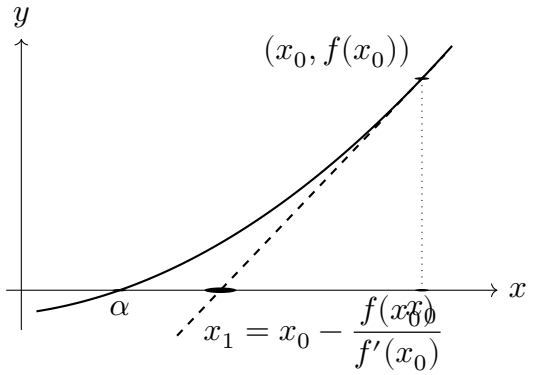
The expression in parentheses has valuation t, so $a = b$.

The connection with the resultant is given by

$$f(x) - f(a_0) = (x - a_0)q(x) \qquad q(x) = \frac{f(x) - f(a_0)}{x - a_0} \qquad \mathrm{Res}(x - a_0, q) = q(a_0) = f'(a_0) \tag{1}$$

where the last equality follows by differentiating and evaluating at a_0. □

The process of finding the root is similar to the usual Newton method, in which one approximates the root of a polynomial by going in the direction of the derivative. If the derivative is "big enough" relative to the value of the polynomial, you will be converging to a root.



Hensel's lemma turns sufficiently good approximate roots into actual roots. Now we want to understand where roots live when they do not lie in K itself. There are two useful pictures: a Newton polygon measures their valuations, and the structure of a local extension separates the change in residue field from the change in uniformizer.

6.2.1 Extending the valuation

For the moment R is still an arbitrary complete DVR, with fraction field K and residue field k.

Theorem 6.10 (Finite extensions of a complete DVR)**.** *Let L/K be a finite separable extension of degree n, and let S be the integral closure of R in L. Then S is a complete DVR with a unique maximal ideal $\mathfrak{m}_L$. If*

$$\mathfrak{m}_K S = \mathfrak{m}_L^e \qquad \text{and} \qquad f = [S/\mathfrak{m}_L : R/\mathfrak{m}_K]$$

then $[L : K] = ef$. Writing v_K, v_L for the valuations normalized to have image $\mathbb{Z}$, we have

$$v_L|_K = e v_K \qquad \text{and} \qquad v_K(\mathrm{N}_{L/K}(x)) = f v_L(x)$$

The unique valuation on L extending v_K is therefore

$$w_L = \frac{1}{e} v_L = \frac{1}{n} v_K \circ \mathrm{N}_{L/K}$$

and L is complete for this valuation.

Proof. By Theorems 3.1 and 3.2, S is Dedekind and finite free over R. In particular it is complete for the $\mathfrak{m}_K$-adic topology. If $\mathfrak{m}_K S = \prod_{i=1}^{g} \mathfrak{q}_i^{e_i}$, then CRT at every power gives

$$S \cong \varprojlim_r S/\mathfrak{m}_K^r S \cong \prod_{i=1}^{g} \varprojlim_r S/\mathfrak{q}_i^{re_i}$$

Each factor is nonzero. Since S is a domain, this product can have only one factor. Thus S is local, hence a DVR, and the two ideal-adic topologies agree. The identity $n = ef$ is now Theorem 3.4 with $g = 1$. The formulas for valuations follow from the factorization of $\mathfrak{m}_K S$ and the ideal norm of $\mathfrak{m}_L$.

Any valuation extending v_K has a valuation ring containing S, because elements integral over R have nonnegative valuation. A valuation ring containing a DVR with the same fraction field is either that DVR or the entire field: if it contains an element of negative v_L-valuation, it contains the inverse of a uniformizer. The entire field would give the trivial valuation, so the valuation ring must be S. The restriction to K fixes the scaling, giving uniqueness. Finally a Cauchy sequence in L is bounded, so multiplying it by a fixed power of a uniformizer puts it in S, where it converges. □

The distinction between v_L and w_L is useful to remember. With the first convention the new uniformizer has valuation 1; with the second it has valuation $1/e$ and the old uniformizer still has valuation 1. We use the second convention when drawing Newton polygons over K. These valuations extend compatibly to an algebraic closure $\overline{K}$. For a purely inseparable element the extension is forced by $v(\alpha) = p^{-r} v(\alpha^{p^r})$, so this also covers positive characteristic.

6.2.2 Newton Polygons

Suppose that $F(X) = \sum_{i=0}^{n} a_i X^i$ and that $\alpha \neq 0$ is a root. If $v(\alpha) = r$, then the terms of $F(\alpha)$ have valuations

$$v(a_i) + ir$$

The smallest of these numbers must occur at least twice. Otherwise the term with smallest valuation would dominate the sum, and the sum could not be zero. This turns the problem into a picture in the plane.

Definition 6.11 (Newton polygon)**.** Let $F = \sum_{i=0}^{n} a_i X^i \in K[X]$ with $a_0 a_n \neq 0$. The *Newton polygon* of F is the lower convex boundary of the points

$$(i, v_K(a_i)) \qquad \text{for } a_i \neq 0$$

between horizontal coordinates 0 and n. Equivalently, stretch a string below the coefficient points and pull it upwards until it becomes taut. Zero coefficients have valuation ∞ and are omitted.

A line of slope $-r$ has equation $y + rx = b$. Moving it upwards until it touches the coefficient points finds exactly the minimum of $v(a_i) + ir$. So an edge of slope $-r$ is a place where cancellation between terms of a possible root of valuation r can occur. The following theorem says that the picture also counts the roots.

Theorem 6.12 (Newton polygon theorem)**.** *If an edge of the Newton polygon of F has slope s and horizontal length m, then F has exactly m roots $\alpha \in \overline{K}$ with*

$$v(\alpha) = -s$$

counted with multiplicity, where v extends v_K. Moreover, if F is monic, the roots belonging to each slope give a monic factor $F_s \in K[X]$, and

$$F = \prod_s F_s \qquad \deg F_s = m_s$$

Proof. Dividing by the leading coefficient only translates the polygon vertically, so assume F monic. List its roots so that $r_1 := v(\alpha_1) \leq \cdots \leq r_n := v(\alpha_n)$. The coefficient a_{n-j} is, up to sign, the sum of the products of j distinct indexed roots. Thus

$$v(a_{n-j}) \geq r_1 + \cdots + r_j$$

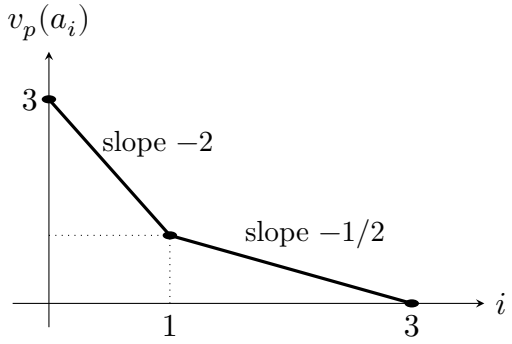
If $r_j < r_{j+1}$, the product of the first j roots is the unique term of smallest valuation, so equality holds. Equality also holds for $j = 0, n$. Consequently the lower boundary passes through the endpoints of each block of equal root valuations, and the intervening coefficient points lie on or above it. Read from left to right, a block of m roots of valuation r gives an edge of length m and slope $-r$.

Uniqueness of the extended valuation implies that all conjugates of a root have the same valuation. Therefore each irreducible factor of F contributes to just one slope. Grouping these irreducible factors, with their multiplicities, gives the polynomials F_s over K. □

Example 6.13. Over $\mathbb{Q}_p$, consider

$$F(X) = X^3 + pX + p^3$$

The nonzero coefficient points are $(0,3), (1,1), (3,0)$.



There is one root of valuation 2 and two roots of valuation $1/2$. The first slope gives a linear factor over $\mathbb{Q}_p$. The other gives an irreducible quadratic factor, since neither of its roots can lie in $\mathbb{Q}_p$, whose nonzero elements have integer valuations. Thus the picture detects a root even though reduction modulo p only gives X^3.

We can actually find that root using Hensel. Substitute $X = p^2Y$ and divide by p^3 to obtain

$$p^3Y^3 + Y + 1$$

which has the simple root $Y = -1$ modulo p. Its lift gives the desired root $\alpha \equiv -p^2 \pmod{p^3}$.

Corollary 6.14 (Slope denominators and irreducibility)**.** *If a separable root α has valuation a/b in lowest terms, then b divides the ramification index of $K(\alpha)/K$. If a monic polynomial of degree n has a single Newton-polygon edge from $(0,m)$ to $(n,0)$ with $\gcd(m,n) = 1$, then it is irreducible over K.*

Proof. The extended value group of $K(\alpha)$ is $\frac{1}{e}\mathbb{Z}$, proving the first statement. For the second, every root has valuation m/n. A monic factor of degree d would have constant coefficient of valuation $dm/n \in \mathbb{Z}$, so n divides d. □

In particular, an Eisenstein polynomial has the single edge from $(0,1)$ to $(n,0)$, which gives another proof of Eisenstein's criterion. The polygon groups roots by valuation, but a group of roots of the same valuation can still split into several factors. For example, when p is odd, $X^2 - 1$ has a single horizontal edge and two linear factors. To go further, one combines the polygon with reduction of suitably rescaled or translated polynomials and Hensel's lemma. See also [1, Section 7] for this viewpoint; its coefficient-order convention reverses the sign of the slopes.

6.2.3 What a local field looks like

From now on assume that $k = R/\mathfrak{m}_K$ is finite, say $k = \mathbb{F}_q$ with q a power of p. All field extensions under discussion will be finite and separable. We also write $\mathcal{O}_K = R$ and choose a uniformizer π_K.

Definition 6.15 (Nonarchimedean local field)**.** A *nonarchimedean local field* is a complete discretely valued field with finite residue field. In this section we abbreviate this to *local field*.

The adjective "local" also has a topological explanation. The ring $\mathcal{O}_K = \varprojlim_r \mathcal{O}_K/\mathfrak{m}_K^r$ is an inverse limit of finite discrete spaces, hence compact, and it is an open neighbourhood of 0 in K. Thus K is locally compact. The archimedean local fields are $\mathbb{R}$ and $\mathbb{C}$; our valuation-ring picture concerns the nonarchimedean ones.

Proposition 6.16 (Digits and Teichmüller representatives)**.** *Each $a \in k$ has a unique lift $[a] \in \mathcal{O}_K$ satisfying*

$$[a]^q = [a] \qquad [a] \equiv a \pmod{\mathfrak{m}_K}$$

These lifts satisfy $[ab] = [a][b]$. Every $x \in \mathcal{O}_K$ has a unique convergent expansion

$$x = [a_0] + [a_1]\pi_K + [a_2]\pi_K^2 + \cdots \qquad a_i \in k$$

Proof. Every element of k is a simple root of $X^q - X$, whose derivative reduces to -1. Hensel gives the unique lifts, and uniqueness applied to the product gives multiplicativity. For the expansion, choose a_0 to be the residue of x, subtract $[a_0]$, divide by π_K, and repeat. The error after r digits lies in $\mathfrak{m}_K^r$, so completeness gives convergence. The first digit where two expansions differ proves uniqueness. □

This is the familiar base-p expansion with a new uniformizer and a larger alphabet of digits. In characteristic zero these representatives are generally not closed under addition: carrying is part of the arithmetic.

Theorem 6.17 (Classification of local fields)**.** *A nonarchimedean local field is either a finite extension of $\mathbb{Q}_p$, or is isomorphic to $\mathbb{F}_q((t))$ for some prime power q. In the second case, choosing a uniformizer gives*

$$\mathcal{O}_K \cong \mathbb{F}_q[[t]] \qquad K \cong \mathbb{F}_q((t))$$

Proof. If $\operatorname{char} K = p$, the roots of $X^q - X$ are closed under addition as well as multiplication, by Freshman's dream. They form a copy of $\mathbb{F}_q$ inside $\mathcal{O}_K$. The expansions in Proposition 6.16, with $t \mapsto \pi_K$, now give the claimed ring and field isomorphisms.

If $\operatorname{char} K = 0$, completing the prime field embeds $\mathbb{Q}_p$ into K. Indeed, integers prime to p are units, and $v_K(p) > 0$. Write $e_0 = v_K(p)$, so $p\mathcal{O}_K = \mathfrak{m}_K^{e_0}$. The quotient $\mathcal{O}_K/p\mathcal{O}_K$ is a finite-dimensional $\mathbb{F}_p$-vector space. Choose lifts $b_1, \ldots, b_d$ of a basis. Successively expanding modulo $p, p^2, \ldots$ expresses every element of $\mathcal{O}_K$ as

$$c_1 b_1 + \cdots + c_d b_d \qquad c_i \in \mathbb{Z}_p$$

Thus $\mathcal{O}_K$ is finite over $\mathbb{Z}_p$, and $K = \mathcal{O}_K[1/p]$ is finite over $\mathbb{Q}_p$. Conversely, Theorem 6.10 shows that finite extensions of $\mathbb{Q}_p$ are local fields, and the Laurent series examples are immediate. □

Proposition 6.18 (The multiplicative structure)**.** *Put $U_j = 1 + \mathfrak{m}_K^j$ for $j \geq 1$. After choosing π_K, there is a decomposition*

$$K^\times \cong \pi_K^{\mathbb{Z}} \times \mu_{q-1}(K) \times U_1$$

where $\mu_{q-1}(K) = \{x \in K^\times : x^{q-1} = 1\}$ maps isomorphically to $k^\times$ by reduction. Moreover

$$U_j/U_{j+1} \cong (k, +) \qquad j \geq 1$$

Proof. Every nonzero element is a unique power of π_K times a unit. Divide the unit by the Teichmüller lift of its residue to leave an element of U_1. Finally

$$(1 + a\pi_K^j)(1 + b\pi_K^j) \equiv 1 + (a + b)\pi_K^j \pmod{\mathfrak{m}_K^{j+1}}$$

so the map $1 + a\pi_K^j \mapsto \bar{a}$ gives the last isomorphism. □

So multiplication is organized by a valuation, a nonzero residue, and then successive additive residue digits. In particular U_1 is an inverse limit of finite p-groups, often called a *pro-p group.*

6.2.4 Unramified extensions

Let L/K be a finite extension of local fields, with residue fields k_L/k_K. Recall that $[L : K] = ef$. The extension is *unramified* if $e = 1$, and *totally ramified* if $f = 1$. Residue fields are finite and hence perfect, so there is no additional residue-separability condition to check here.

Theorem 6.19 (Lifting the residue field)**.** *Finite unramified extensions of K correspond to finite extensions of k_K. More explicitly, if $\bar{F} \in k_K[X]$ is monic irreducible of degree f and $F \in \mathcal{O}_K[X]$ is any monic lift, then*

$$E = K(\alpha), \qquad F(\alpha) = 0$$

is unramified of degree f, with $\mathcal{O}_E = \mathcal{O}_K[\alpha]$ and $k_E \cong k_K[X]/(\bar{F})$. Every unramified extension arises this way, and embeddings of residue fields lift uniquely to embeddings of the corresponding extensions.

Proof. Reduction and Gauss's lemma show that F is irreducible over K. It is separable since $\bar{F}$ is separable. The residue of α generates a degree-f subextension of k_E/k_K, while $[E : K] = f$. The equality $[E : K] = e(E/K)f(E/K)$ therefore forces $e(E/K) = 1$ and $[k_E : k_K] = f$. The powers of α span $\mathcal{O}_E/\mathfrak{m}_K\mathcal{O}_E$ over k_K, so Nakayama applied to $\mathcal{O}_E/\mathcal{O}_K[\alpha]$ gives $\mathcal{O}_E = \mathcal{O}_K[\alpha]$.

Conversely, choose a primitive element of the residue field of an unramified extension and lift its minimal polynomial to F. Hensel produces a root of F in the extension with the chosen residue. That root generates the extension by the degree calculation above. For an embedding of residue fields, Hensel likewise gives a unique root with the required image residue, and hence a unique field embedding. □

The same lifting argument works when the target extension is ramified: an embedding of the residue field of an unramified extension lifts uniquely into any finite extension with that residue field embedding.

Corollary 6.20 (Unramified extensions and Frobenius)**.** *For every $f \geq 1$ there is, up to K-isomorphism, a unique unramified extension K_f/K of degree f. It is Galois, and reduction gives*

$$\mathrm{Gal}(K_f/K) \cong \mathrm{Gal}(\mathbb{F}_{q^f}/\mathbb{F}_q) \cong \mathbb{Z}/f\mathbb{Z}$$

Its distinguished generator, the arithmetic Frobenius*, is characterized by*

$$\mathrm{Frob}_q(x) \equiv x^q \pmod{\mathfrak{m}_{K_f}} \qquad (x \in \mathcal{O}_{K_f})$$

One may also construct K_f as $K(\zeta_{q^f-1})$, where ζ_{q^f-1} is a primitive root of unity of that order.

Proof. Apply Theorem 6.19 to the finite fields and their embeddings. For the last statement, lift a generator of $\mathbb{F}_{q^f}^\times$ by Proposition 6.16. Its lift has order $q^f - 1$, and its residue already generates the degree-f residue extension. □

More generally, adjoining roots of unity of order prime to p gives an unramified extension. Their residues occur in some finite extension of $\mathbb{F}_q$, and Hensel lifts all of them there uniquely.

Theorem 6.21 (The unramified–totally ramified tower)**.** *Every finite extension L/K of local fields contains a unique maximal unramified subextension E/K. It satisfies*

$$[E : K] = f(L/K) \qquad [L : E] = e(L/K) \qquad k_E = k_L$$

In particular L/E is totally ramified.

Proof. Choose a primitive element of k_L/k_K and a monic lift of its minimal polynomial. Hensel lifts the element to $\alpha \in \mathcal{O}_L$. Then $E = K(\alpha)$ is unramified and has residue field k_L by Theorem 6.19. The tower laws for e and f give the stated degrees. Every unramified subextension of L/K embeds into E by its residue field embedding, and uniqueness of the lift shows that this embedding is its original inclusion in L. Thus all such subextensions lie in E. □

$$K \xhookrightarrow[\text{degree } f]{\text{unramified}} E \xhookrightarrow[\text{degree } e]{\text{totally ramified}} L$$

The first step enlarges the residue field while keeping the value group fixed. The second keeps the residue field fixed and changes the extended value group from $\mathbb{Z}$ to $\frac{1}{e}\mathbb{Z}$. Now we want to understand this second step explicitly.

6.2.5 Totally ramified extensions and Eisenstein polynomials

Theorem 6.22 (A uniformizer generates the ramified part)**.** *Let L/E be a totally ramified extension of local fields of degree e, and choose any uniformizer π_L. Then*

$$L = E(\pi_L) \qquad \mathcal{O}_L = \mathcal{O}_E[\pi_L]$$

and the minimal polynomial of π_L over E is Eisenstein. Conversely, a root of a monic separable Eisenstein polynomial of degree e over $\mathcal{O}_E$ generates a totally ramified extension of degree e and is a uniformizer there.

Proof. Normalize $v_L(\pi_L) = 1$, so $v_L(E^\times) = e\mathbb{Z}$. In a linear combination

$$\sum_{i=0}^{e-1} c_i \pi_L^i \qquad (c_i \in E)$$

the nonzero terms have distinct valuations modulo e. There can therefore be no cancellation of the term of smallest valuation. This proves that $1, \pi_L, \dots, \pi_L^{e-1}$ are linearly independent, hence form an E-basis of L. It also proves that the sum is integral exactly when every c_i lies in $\mathcal{O}_E$, giving the assertion about rings of integers.

All conjugates of π_L have extended E-valuation $1/e$. Their elementary symmetric functions show that every nonleading coefficient of the minimal polynomial has positive, hence at least 1, E-valuation. The constant coefficient has valuation exactly $e \cdot (1/e) = 1$. Thus the polynomial is Eisenstein.

Conversely, the Newton polygon of an Eisenstein polynomial of degree e has slope $-1/e$, and Corollary 6.14 proves irreducibility. The ramification index of the extension generated by a root is divisible by e and at most its degree e. It is therefore e, the residue degree is 1, and the root has normalized valuation 1. □

Combining this with Theorem 6.21, every finite extension of local fields can be built by first lifting a residue-field generator and then adjoining a root of an Eisenstein polynomial. In particular, with $E = K(\alpha)$ the unramified part,

$$\mathcal{O}_L = \mathcal{O}_K[\alpha, \pi_L]$$

This is a very concrete description of the local ring of integers.

Example 6.23. The following extensions all start from $\mathbb{Q}_3$.

L	e	f	construction
$\mathbb{Q}_3(i)$	1	2	X^2+1 irreducible modulo 3
$\mathbb{Q}_3(\sqrt{3})$	2	1	X^2-3 Eisenstein
$\mathbb{Q}_3(\sqrt[3]{3})$	3	1	X^3-3 Eisenstein
$\mathbb{Q}_3(i, \sqrt[3]{3})$	3	2	the first and third steps together

For the last row, 3 is still a uniformizer in the unramified quadratic field $\mathbb{Q}_3(i)$. Thus X^3-3 remains Eisenstein there, giving total degree 6 and residue field $\mathbb{F}_9$.

6.2.6 Tame and wild ramification

Definition 6.24. An extension L/K of local fields with residue characteristic p is *tamely ramified* if $p \nmid e(L/K)$, and *wildly ramified* if $p \mid e(L/K)$. In particular unramified extensions are tame.

The reason to separate these cases is again Hensel's lemma. When e is a unit, taking an eth root of a unit close to 1 is easy. This lets us simplify the Eisenstein description substantially.

Proposition 6.25 (The shape of a tame extension)**.** *If L/E is totally tamely ramified of degree e, then for a chosen uniformizer π_E there is a unit $u \in \mathcal{O}_E^\times$ such that*

$$L = E(\theta) \qquad \theta^e = u\pi_E$$

Conversely, adjoining such a root with $p \nmid e$ gives a totally tamely ramified extension of degree e.

Proof. Choose a uniformizer π_L and write $\pi_L^e = c\pi_E$ with $c \in \mathcal{O}_L^\times$. Since the residue fields agree, choose $u \in \mathcal{O}_E^\times$ with the same residue as c. Then $c/u \equiv 1 \pmod{\mathfrak{m}_L}$. The polynomial $X^e - c/u$ has the simple root 1 modulo $\mathfrak{m}_L$, so Hensel gives $b \in 1 + \mathfrak{m}_L$ with $b^e = c/u$. Now $\theta = \pi_L/b$ is a uniformizer and satisfies $\theta^e = u\pi_E$. Apply Theorem 6.22. The converse follows because $X^e - u\pi_E$ is Eisenstein and separable. □

Thus tame ramification really does have the shape "take a root of a local parameter", after allowing a unit factor. The unit factor matters: $\mathbb{Q}_3(\sqrt{3})$ and $\mathbb{Q}_3(\sqrt{6})$ are distinct ramified quadratic extensions, both with $e = 2, f = 1$. Their ratio would give $\sqrt{2}$, which generates the unramified quadratic extension.

The different measures how much more complicated wild ramification is.

Proposition 6.26 (The local different)**.** *For a finite extension L/K of local fields, write*

$$\mathcal{D}_{L/K} = \mathfrak{m}_L^\delta$$

Then

$$\delta \geq e - 1 \qquad \textit{with equality if and only if } L/K \textit{ is tame}$$

If E is the maximal unramified subextension and F is the minimal polynomial of a uniformizer π_L over E, then

$$\mathcal{D}_{L/K} = (F'(\pi_L)) \qquad \textit{and} \qquad \operatorname{disc}(\mathcal{O}_L/\mathcal{O}_K) = \mathfrak{m}_K^{f\delta}$$

Proof. The unramified extension E/K has trivial different by Theorem 3.23. The trace-dual proof of Theorem 3.25 applies equally to complete DVRs, so $\mathcal{D}_{L/K} = \mathcal{D}_{L/E}$. Now use Theorems 3.24 and 6.22.

Write $F(X) = X^e + \sum_{i=0}^{e-1} a_i X^i$. For $1 \leq i < e$, the term $ia_i\pi_L^{i-1}$ in $F'(\pi_L)$ has valuation at least $e + i - 1 \geq e$. If $p \nmid e$, the leading term $e\pi_L^{e-1}$ has valuation exactly $e - 1$, so it determines the valuation of the sum. If $p \mid e$, every derivative term has valuation at least e; zero terms are simply omitted. Separability ensures $F'(\pi_L) \neq 0$. Finally the discriminant is the norm of the different, and $\mathrm{N}_{L/K}(\mathfrak{m}_L) = \mathfrak{m}_K^f$. □

Example 6.27. For $L = \mathbb{Q}_3(\sqrt{3})$, the different is $(2\sqrt{3})$ and has exponent $1 = e - 1$. For $L = \mathbb{Q}_3(\pi)$ with $\pi^3 = 3$, the derivative is $3\pi^2 = \pi^5$. Thus

$$e = 3 \qquad \delta = 5 > e - 1 \qquad \operatorname{disc}(\mathcal{O}_L/\mathbb{Z}_3) = (3^5)$$

The extra divisibility comes from the factor 3 in the derivative.

The Galois picture. If L/K is Galois, its Galois group G acts on the unique prime upstairs, so the whole group is the decomposition group. Reduction gives

$$1 \longrightarrow I \longrightarrow G \longrightarrow \operatorname{Gal}(k_L/k_K) \longrightarrow 1 \qquad |I| = e$$

where I consists of the automorphisms acting trivially on the residue field. Its fixed field is exactly the maximal unramified subextension E. Indeed, Hensel uniqueness shows that an automorphism fixes E exactly when it fixes the residue field, and normality lets every automorphism of E/K extend to L. In particular Frobenius is naturally an element of G/I; when the extension is unramified it is an element of G itself.

We record the usual finer structure without proof. Define the *lower ramification groups* by

$$G_j = \{\sigma \in G : v_L(\sigma(a) - a) \geq j + 1 \text{ for every } a \in \mathcal{O}_L\} \qquad (j \geq 0)$$

Then $G_0 = I$, G_1 is a p-group called *wild inertia*, and G_0/G_1 is cyclic of order prime to p. The groups eventually become trivial, and

$$\delta = \sum_{j \geq 0} (|G_j| - 1)$$

The picture is that G_j consists of symmetries invisible modulo $\mathfrak{m}_L^{j+1}$. The first quotient records multiplication of a uniformizer by a residue-field unit; the higher successive quotients sit in the additive residue field. Thus tame inertia is cyclic, and the additional depth is controlled by p-groups. The formula for δ counts precisely this extra depth. See [1, Section 7] and [2] for further details on local extensions and ramification.

6.2.7 Approximating extensions and returning to number fields

There is one more useful consequence of completeness: a sufficiently accurate approximation to a field generator remembers the field it generates.

Lemma 6.28 (Krasner's lemma)**.** *Let $\alpha, \beta \in \overline{K}$ be separable over K, with $\alpha \notin K$. If*

$$|\beta - \alpha| < \min_{\substack{\alpha' \neq \alpha \\ \alpha' \text{ conjugate to } \alpha}} |\alpha - \alpha'|$$

then $K(\alpha) \subseteq K(\beta)$. In particular, if the two degrees over K agree, then the fields agree.

Proof. Work in a finite Galois extension containing both elements and all their conjugates. If σ fixes β, uniqueness of the extended absolute value gives

$$|\sigma(\alpha) - \alpha| \leq \max\{|\sigma(\alpha) - \beta|, |\beta - \alpha|\} = |\beta - \alpha|$$

The hypothesis forces $\sigma(\alpha) = \alpha$. The Galois correspondence now gives the desired inclusion. □

So Hensel controls the root, while Krasner says when controlling the root also controls its extension field.

Finally, here is why local fields tell us about primes in a number field.

Theorem 6.29 (Completing separates the primes)**.** *Let L/K be an extension of number fields and let $\mathfrak{p}$ be a nonzero prime of $\mathcal{O}_K$. Write $K_\mathfrak{p}$ for the completion of K, and $L_\mathfrak{q}$ for the completion at a prime $\mathfrak{q}$ above $\mathfrak{p}$. Then*

$$L \otimes_K K_\mathfrak{p} \cong \prod_{\mathfrak{q}|\mathfrak{p}} L_\mathfrak{q} \qquad [L_\mathfrak{q} : K_\mathfrak{p}] = e(\mathfrak{q}|\mathfrak{p}) f(\mathfrak{q}|\mathfrak{p})$$

Proof. Localize $\mathcal{O}_K$ at $\mathfrak{p}$ and put $S = (\mathcal{O}_L)_\mathfrak{p}$. For every r, CRT gives

$$S/\mathfrak{p}^r S \cong \prod_{\mathfrak{q}|\mathfrak{p}} S/\mathfrak{q}^{re(\mathfrak{q}|\mathfrak{p})} S$$

Taking inverse limits gives a product of complete DVRs. Since S is finite free over $(\mathcal{O}_K)_\mathfrak{p}$, its completion is obtained by tensoring with the completed base ring. Inverting a base uniformizer therefore gives the displayed product of fields. Completion preserves the uniformizers, their ramification indices, and the residue fields, so the degree formula follows from Theorem 6.10. □

Thus the global identity $[L : K] = \sum_{\mathfrak{q}|\mathfrak{p}} e_\mathfrak{q} f_\mathfrak{q}$ is a sum of local degrees. If $L = K(\alpha)$ with minimal polynomial F, factoring F into irreducibles over $K_\mathfrak{p}$ gives precisely these local fields. Hensel lifts coprime factors from the residue field, and Newton polygons help separate factors whose roots have different valuations even when reduction has repeated factors. This is the bridge back to the computations in the next section.

7. Computing in Number Fields

We've done a lot of theoretical thinking and structural work, however we've not yet figured out how to actually compute stuff. Number theory is one of the most computationally rich fields of mathematics, so it is worth learning that side of it. Therefore in this section we'll learn methods for computing things about number fields.

7.1. Hermite Normal Form

In this section we will learn a tool that allows explicit computation with lattices. While we have previously seen lattices abstractly as finitely generated $\mathbb{Z}$-modules, to write code that does concrete calculations we need a formalism to represent them.

Proposition 7.1. *The following operations do not change the lattice the columns of a matrix A generates:*

(a) Swapping two columns.

(b) Adding or subtracting an integer multiple of one column from another, or multiplying a column by -1.

(c) Pick a row r and let c_i, c_j denote the ith and jth columns of A. Using the Euclidean algorithm compute d, x, y so:

$$d = \gcd(a_{ri}, a_{rj}) = xa_{ri} + ya_{rj}$$

then, if $d \neq 0$, perform the simultaneous operations

$$c_i \leftarrow xc_i + yc_j \qquad c_j \leftarrow \frac{a_{rj}}{d}c_i - \frac{a_{ri}}{d}c_j$$

which sends a_{ri} to d, sends a_{rj} to 0, and corresponds to using (a) and (b) to perform the Euclidean algorithm via column operations.

Definition 7.2 (Hermite Normal Form)**.** Let A be an $n \times k$ integer matrix of rank n, whose columns generate a lattice $\Lambda \subseteq \mathbb{Z}^n$. We call A in *Hermite Normal Form* (HNF) if it is of the form $[H \mid 0]$ (up to column permutation), where H is an $n \times n$ matrix satisfying:

(a) H is lower triangular ($h_{ij} = 0$ for $i < j$).

(b) $h_{ii} > 0$ for all i.

(c) $0 \leq h_{ij} < h_{ii}$ for all $j < i$.

Representing our lattice vectors as columns is standard and algebraically convenient. By enforcing H to be lower triangular, the HNF gives us a canonical choice of basis for any lattice.

Theorem 7.3. *For any integer lattice $\Lambda \subseteq \mathbb{Z}^n$ of rank n, there exists a unique $n \times n$ matrix H in Hermite Normal Form which generates the lattice. Therefore, two matrices generate the same column lattice if and only if they have the same HNF.*

Proof.

Existence: Let A be a matrix whose columns generate the lattice. We apply integer column operations. Working from the top row down, we use Proposition 7.1 (c) on the rows to eliminate entries to the right of the diagonal, making the matrix lower triangular. We then multiply columns by -1 if necessary to ensure $h_{ii} > 0$. Finally, we subtract multiples of column i from column j (for $j < i$) to reduce the off-diagonal entries in the i-th row modulo h_{ii}.

Uniqueness: We proceed by induction on n. Assume uniqueness holds for lattices of rank $n-1$. We establish the uniqueness of H in three parts:

- *Uniqueness of h_{nn}:* Because H is lower triangular, the intersection of Λ with the n-th coordinate axis, $\Lambda \cap (\{0\}^{n-1} \times \mathbb{Z})$, is a 1-dimensional lattice generated by $(0, \ldots, 0, h_{nn})^T$. Since the HNF definition requires $h_{nn} > 0$, h_{nn} is uniquely determined as the positive generator of this sub-lattice.
- *Uniqueness of the top-left block:* Let $\pi(\Lambda) \subset \mathbb{Z}^{n-1}$ be the projection of Λ onto the first $n-1$ coordinates. The first $n-1$ columns of H, truncated to their first $n-1$ rows, form an HNF basis for $\pi(\Lambda)$. By the inductive hypothesis, this $(n-1) \times (n-1)$ top-left block is uniquely determined by $\pi(\Lambda)$.

- *Uniqueness of the last row:* Suppose H and H' are two valid HNF bases for Λ. By steps 1 and 2, they share the same h_{nn} and top-left block. Let c_i and c'_i be the i-th columns of H and H'. Since both bases generate Λ, their difference $c_i - c'_i \in \Lambda$. Because their first $n-1$ coordinates are identical, we have:

$$c_i - c'_i = (0, \dots, 0, h_{ni} - h'_{ni})^T$$

  By step 1, this difference must be an integer multiple of $(0, \dots, 0, h_{nn})$. Thus, $h_{ni} \equiv h'_{ni} \pmod{h_{nn}}$. The HNF definition requires that $0 \leq h_{ni}, h'_{ni} < h_{nn}$. The only multiple of h_{nn} that fits the bounded difference is 0. Therefore, $h_{ni} = h'_{ni}$.

Since all components of the matrix are uniquely fixed by the geometry of Λ, the HNF is unique.

□

Remark 7.4. Given a $n \times r$ integer matrix A with full rank, we let $\mathrm{HNF}(A)$ denote the unique $n \times n$ Hermite Normal Form matrix generating the same lattice. The above proof (existence) gives an algorithm for computing it.

In algebraic number theory, we often choose a convenient canonical basis, such as a power basis in the next section, and would like to represent our lattice vectors in terms of that. However, the lattice we compute often contains this basis strictly. Therefore to represent our basis vectors we would need to have rational numbers in the matrix. This is a computational disaster, so we use the global denominator.

Definition 7.5 (Global Denominator)**.** Let $\Lambda \subset \mathbb{Q}^n$ be a lattice. We call an integer $d \in \mathbb{Z}_{>0}$ a *global denominator* if $d\Lambda \subseteq \mathbb{Z}^n$.

This lets us represent any lattice basis as $\frac{1}{d}H$, where H is an integer matrix in HNF. Even better, to avoid intermediate coefficient explosion when computing the HNF of a full-rank sub-lattice, we can work entirely modulo a known integer.

Remark 7.6. Let $\Lambda \subseteq \mathbb{Z}^n$ be a full-rank sub-lattice spanned by the columns of an $n \times n$ matrix A and let D be any positive multiple of the index $[\mathbb{Z}^n : \Lambda]$. We can compute $\mathrm{HNF}(A)$ by working entirely modulo D.

Proof. Note $D\mathbb{Z}^n \subseteq \Lambda$ by Lemma 3.12, the lattice is generated by the columns of A together with the columns of DI_n. Therefore, any column operations used to compute the HNF can be reduced modulo D by adding or subtracting the columns of DI_n. □

Remark 7.7. Writing the basis of a lattice in HNF allows quick computation of how to write any vector in the basis. This is because representing v corresponds to solving

$$Mx = v$$

and we can determine x_i inductively from the ith row, starting from the top, since at each step the equation only has one unknown.

The Hermite Normal Form is the standard way to crush a bunch of generators into a nice basis, and moreover it is very good for many computations. Therefore in the next sections, when I say choose a basis, unless otherwise implied, it is understood that the basis is the Hermite Normal basis.

7.2. Computing the Discriminant

In general, the first step to understanding a number field is trying to learn something about its discriminant.

Example 7.8 (The Discriminant of a Polynomial and its Field)**.** Let R be an integral domain of characteristic 0, and let $f(x) \in R[x]$ be a monic irreducible polynomial of degree n with roots $\alpha_1, \dots, \alpha_n$ in a splitting field L over $K = \operatorname{Frac}(R)$. There are three ways we may define Δ_f (the discriminant of f):

Root Differences: Defined essentially as the product of squared differences of the roots. This vanishes if and only if roots collide (i.e., f is inseparable).

$$\Delta_f = \prod_{1 \le i < j \le n} (\alpha_i - \alpha_j)^2$$

The Resultant: Calculable without knowing the roots, using the resultant of f and its derivative f':

$$\Delta_f = (-1)^{\frac{n(n-1)}{2}} \operatorname{Res}(f, f')$$

Derivative Values: Expresses Δ_f via the values of the derivative at the roots.

$$\Delta_f = (-1)^{\frac{n(n-1)}{2}} \prod_{i=1}^{n} f'(\alpha_i) = (-1)^{\frac{n(n-1)}{2}} n^n \prod_{i=1}^{n-1} f(\beta_i)$$

where the β_i are the roots of f'.

Proof. These equivalences follow from Theorem 2.29 and Corollary 2.16. Intuitively, they all are different ways you might try to test whether f and f' share roots.

- The product $(\alpha_i - \alpha_j)^2$ computes whether f has a double root, which is true if and only if f and f' share a root.
- If f and f' share roots then dividing by their greatest common divisor gives nonzero polynomials a and b with:

  $$af + bf' = 0 \qquad \deg a < \deg f' \qquad \deg b < \deg f$$

  This forces the determinant of the Sylvester matrix to be 0 since it corresponds exactly to a linear dependence among the columns.
- If f and f' share roots then evaluating one at the roots of the other will give 0.

□

Theorem 7.9. *If $f \in R[x]$ is a monic irreducible polynomial and α is one of its roots then* $\operatorname{disc} R[\alpha]/R$ *(with respect to the trace pairing on $K(\alpha)/K$) is exactly Δ_f.*

Proof. Since f is irreducible it is the minimal polynomial of α. We may use the power basis $\{1, \alpha, \dots, \alpha^{n-1}\}$ to compute $\operatorname{disc} R[\alpha]/R$. By Proposition 2.10 this is the square of the determinant of the embedding matrix $V_{ij} = \sigma_i(\alpha^{j-1})$. This matrix is precisely the Vandermonde matrix of the roots $\alpha_1, \dots, \alpha_n$ of f. Thus:

$$\Delta(1, \dots, \alpha^{n-1}) = (\det V)^2 = \prod_{i<j} (\alpha_i - \alpha_j)^2 = \Delta_f$$

□

This tells us that if $\mathcal{O}_L/\mathcal{O}_K$ is *monogenic* (so $\mathcal{O}_L = \mathcal{O}_K[\alpha]$), then the discriminant $\Delta_{L/K}$ is exactly Δ_f where f is the minimal polynomial of α.

Algorithm 7.10. Computing huge resultants is generally hard, however there is an algorithm to do it quickly over $\mathbb{Z}$. We will mainly focus on the Sylvester matrix definition.

(a) Use the Hadamard bound to see that the resultant's value is bounded by the product of the Euclidean norms of the rows.

(b) Pick distinct small primes p_i not dividing the leading coefficients of f and g, whose product is greater than twice the bound in (a).

(c) Use (d) to compute the resultant mod p_i by working in $\mathbb{F}_{p_i}$.

(d) Note the formula of Remark 2.19

$$\operatorname{Res}(f, g) = (-1)^{\deg f \cdot \deg g} \operatorname{lc}(g)^{\deg f - \deg r} \cdot \operatorname{Res}(g, r)$$

where $\operatorname{lc}(g)$ is the leading coefficient of g and $r \neq 0$ is the remainder of f after division by g. Use the stopping rules in Remark 2.19. Which lets us compute using the Euclidean algorithm, which is not very expensive in finite fields.

(e) Use the Chinese Remainder Theorem and choose the unique representative in the interval given by the bound in (a).

For a human, you should probably ignore the parts for working in mod p and just use the Euclidean algorithm.

Proposition 7.11 (Discriminant and index)**.** *If $A \subseteq B$ are orders in K, then*

$$\operatorname{disc}(A/\mathbb{Z}) = [B : A]^2 \operatorname{disc}(B/\mathbb{Z})$$

In particular,

$$\operatorname{disc}(A/\mathbb{Z}) = [\mathcal{O} : A]^2 \operatorname{disc}(\mathcal{O}/\mathbb{Z})$$

Proof. Recall Lemma 2.2 that changing the basis scales the discriminant by the square of the determinant of the change of basis matrix. □

Algorithm 7.12. This gives the following general algorithm to compute the discriminant:

(a) Find an integral primitive element α of the field extension. The discriminant of the power basis is equal to Δ_f where f is the minimal polynomial of α.

(b) While computing the ring of integers, we enlarge this order in iterative steps. At each step, use Proposition 7.11 to keep track of the running discriminant.

As a human, it is sometimes worth trying to use the theorems in Section 3.5 to get a shortcut.

7.3. Zassenhaus Round 2

In this section I will give the machinery necessary to perform the "Round 2" algorithm for finding the ring of integers. The nice thing about this is that it is a relatively practical algorithm for humans to use, and also has pretty solid asymptotics for computers (better algorithms do exist).

The computation of $\mathcal{O}$ is local at the primes dividing the index of a convenient starting order. In practice one usually begins with a monogenic order

$$A = \mathbb{Z}[\alpha] \subseteq \mathcal{O}$$

coming from an integral primitive element α. There are then two tools:

- Dedekind's criterion, which often proves immediately that a given prime does *not* divide $[\mathcal{O} : A]$;
- the *Round 2* step, which enlarges A at the remaining bad primes until it becomes maximal.

Intuitively, if we don't have enough integers, our discriminant will be too large since our lattice is too small. This will give "fake ramification", which we can check on each prime.

Corollary 7.13 (The first discriminant test)**.** *If a prime p divides $[\mathcal{O} : A]$, then p^2 divides* $\operatorname{disc}(A/\mathbb{Z})$.

Proof. This is immediate from Proposition 7.11. □

So only the primes p with $p^2 \mid \operatorname{disc}(A/\mathbb{Z})$ can possibly be bad.

Proposition 7.14 (Local description of p-maximality)**.** *Let $A \subseteq \mathcal{O}$ be an order and let p be a rational prime. We say that A is* p-maximal *if it satisfies one of the equivalent conditions:*

(a) $p \nmid [\mathcal{O} : A]$;

(b) for every prime $\mathfrak{p} \subseteq A$ with $\mathfrak{p} \cap \mathbb{Z} = (p)$, one has

$$A_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$$

(c) for every such $\mathfrak{p}$, the local ring $A_{\mathfrak{p}}$ is a DVR.

Proof. Let $M := \mathcal{O}/A$, a finite abelian group of order $[\mathcal{O} : A]$.

$(a) \Leftrightarrow (b)$**:** The condition $p \nmid [\mathcal{O} : A]$ is equivalent to saying that the p-primary subgroup of M is zero. If $\mathfrak{p} \cap \mathbb{Z} = (p)$, then every prime $q \neq p$ is a unit in $A_{\mathfrak{p}}$, so localizing at $\mathfrak{p}$ kills all q-primary torsion. Thus $M_{\mathfrak{p}} = 0$ for every $\mathfrak{p} \mid p$ if and only if the p-primary subgroup of M is zero. But

$$M_{\mathfrak{p}} \cong \mathcal{O}_{\mathfrak{p}}/A_{\mathfrak{p}}$$

so $M_{\mathfrak{p}} = 0$ is exactly the equality $A_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$.

$(b) \Rightarrow (c)$**:** Under (b), $\mathcal{O}_{\mathfrak{p}} = A_{\mathfrak{p}}$ is local. It is a localization of a Dedekind domain, hence a DVR.

$(c) \Rightarrow (b)$**:** The integral closure of $A_{\mathfrak{p}}$ in K is $\mathcal{O}_{\mathfrak{p}}$. If $A_{\mathfrak{p}}$ is a DVR, then it is integrally closed, so it already equals its integral closure.

□

Corollary 7.15. *$A = \mathcal{O}$ if and only if A is p-maximal for all primes p.*

Proof. We can check whether two rings are equal locally. □

Lemma 7.16 (The local idealizer test)**.** *Let R be a one-dimensional Noetherian local domain with maximal ideal $\mathfrak{m}$. Then*

$$(\mathfrak{m} : \mathfrak{m}) = R \qquad \Longleftrightarrow \qquad R \text{ is a DVR}$$

Proof. If R is a DVR and $\mathfrak{m} = (\pi)$, then

$$(\mathfrak{m} : \mathfrak{m}) = (\pi R : \pi R) = (R : R) = R$$

because R is an integral domain.

Conversely, suppose R is not a DVR. Then $\mathfrak{m}$ must not be principal and is therefore not invertible by Theorem 1.4. In this case we must have $\mathfrak{m}\mathfrak{m}^{-1} = \mathfrak{m}$ (it contains $\mathfrak{m}$ and is contained in R) so we have $(\mathfrak{m} : \mathfrak{m}) = \mathfrak{m}^{-1}$. It suffices to show $R \subsetneq \mathfrak{m}^{-1}$. This is due to Lemma 1.5. □

Theorem 7.17 (One step of Round 2)**.** *Let $A \subseteq K$ be an order, let p be a rational prime, and set:*

$$I := \sqrt{pA} \qquad A' := (I : I)$$

Then A' is an order satisfying:

$$A \subseteq A' \subseteq \frac{1}{p}A$$

Moreover $[A' : A]$ is a power of p, and

$$A' = A \qquad \Longleftrightarrow \qquad A \text{ is } p\text{-maximal.}$$

Proof. Note $pA \subseteq I \subseteq A$ so

$$x \in (I : I) \implies x \in (A : pA) = \frac{1}{p}A$$

and $(I : I) \subseteq \frac{1}{p}A$. The idealizer is a subring containing A. It is a finitely generated torsion-free $\mathbb{Z}$-module of full rank, so it is an order. Now $[A' : A]$ must divide $[\frac{1}{p}A : A]$ and is therefore a power of p. Finally, let $\mathfrak{p} \subseteq A$ be a prime with $\mathfrak{p} \cap \mathbb{Z} = (p)$. Localizing gives

$$I_{\mathfrak{p}} = \sqrt{pA_{\mathfrak{p}}} = \mathfrak{p}A_{\mathfrak{p}}$$

because $A_{\mathfrak{p}}$ is local and $\mathfrak{p}A_{\mathfrak{p}}$ is its unique prime above p. Hence

$$A'_{\mathfrak{p}} = (I_{\mathfrak{p}} : I_{\mathfrak{p}}) = (\mathfrak{p}A_{\mathfrak{p}} : \mathfrak{p}A_{\mathfrak{p}})$$

By Lemma 7.16, this equals $A_{\mathfrak{p}}$ if and only if $A_{\mathfrak{p}}$ is a DVR. By Proposition 7.14, the latter is equivalent to A being p-maximal. □

This is very mathematically elegant and gives a good way of computing the ring of integers.

Remark 7.18. The idealizer is computed by linear algebra over $\mathbb{F}_p$. The first step is to compute $I = \sqrt{pA}$. Since $pA \subseteq I$ we work entirely in A/pA.

(a) Pick a basis $\beta_1, \dots, \beta_n$ for A. Compute the integer matrix M^r corresponding to the action of each β_r:

$$\beta_r \beta_j = \sum_i M^r_{ij} \beta_i$$

(b) Using (a) compute $a_{ij} \in \mathbb{F}_p$ with:

$$\beta_j^p \equiv \sum_i a_{ij} \beta_i \pmod{p}$$

and note by Freshman's dream the pth power map is exactly the matrix P with coefficients a_{ij}.

(c) Choose m minimal so $p^m > n$. By Freshman's dream the p^m power map is exactly P^m. Now I/pA is exactly the null space of this matrix. Use RREF to find a basis of this null space, lift its vectors to A, and adjoin $p\beta_1, \dots, p\beta_n$. Taking HNF gives a $\mathbb{Z}$-basis $\gamma_1, \dots, \gamma_n$ for I.

Now we will compute the full idealizer $A' = (I : I)$. We use the fact that $A \subseteq A' \subseteq \frac{1}{p}A$ to work in $(\frac{1}{p}A)/A \cong A/pA$.

(d) Compute matrices N^r corresponding to the action of β_i on I:

$$\beta_r \gamma_j = \sum_i N^r_{ij} \gamma_i \qquad N^r = B^{-1} M^r B$$

Here the columns of B are the γ_i written in the β_i-basis. Compute N^r over $\mathbb{Q}$; its entries are integers since I is an A-ideal. We can then reduce modulo p.

(e) For $x = \frac{1}{p}(c_1\beta_1 + \cdots + c_n\beta_n)$ we have

$$x \in (I : I) \iff c_1 N^1 + \cdots + c_n N^n \equiv 0 \pmod{p}$$

which is a system of n^2 congruences modulo p. RREF of an $n^2 \times n$ matrix gives a basis of A'/A. Lift these vectors to $\frac{1}{p}A$, adjoin a basis of A, and take HNF to get a basis of A'.

Algorithm 7.19 (Round 2 algorithm)**.** Start with any order $A_0 \subseteq \mathcal{O}$. For each prime p with $p^2 \mid \operatorname{disc}(A_0/\mathbb{Z})$, repeat the following step (using Remark 7.18):

$$A \leftarrow (\sqrt{pA} : \sqrt{pA})$$

until it stabilizes. After doing this for every such prime p, the resulting order is $\mathcal{O}$.

Proof. By Theorem 7.17, the iteration at a fixed prime p stabilizes exactly when the current order is p-maximal. Each strict enlargement

$$A \subsetneq A' := (\sqrt{pA} : \sqrt{pA})$$

has index a positive power of p. Therefore Proposition 7.11 shows that

$$v_p(\operatorname{disc}(A'/\mathbb{Z})) < v_p(\operatorname{disc}(A/\mathbb{Z}))$$

Since this valuation is a nonnegative integer, the iteration must terminate.

When the process is finished for every prime p with $p^2 \mid \operatorname{disc}(A_0/\mathbb{Z})$, the final order is p-maximal for every prime p. Hence the final order is $\mathcal{O}$ itself. □

The last bit of machinery is a special case, it turns out that when the order is monogenic the computation is much easier.

Theorem 7.20 (Dedekind's Criterion)**.** *Let $K = \mathbb{Q}(\alpha)$ be a number field where α is an algebraic integer with minimal monic polynomial $f(x) \in \mathbb{Z}[x]$ of degree n. Let $A = \mathbb{Z}[\alpha]$ and p be a prime. Let the factorization of $f(x)$ modulo p be $\bar{f}(x) = \prod_i \bar{g}_i(x)^{e_i}$.*

Let $g_i(x)$ be monic lifts of the $\bar{g}_i(x)$ and define the following polynomials in $\mathbb{Z}[x]$:

- $h(x) = \prod_i g_i(x)^{e_i - 1}$
- $g(x) = \prod_i g_i(x)$
- $k(x) = \frac{f(x) - g(x)h(x)}{p}$

Then

- *The order A is p-maximal if and only if $\gcd(\bar{h}(x), \bar{k}(x)) = 1$ in $\mathbb{F}_p[x]$.*
- *If A is not p-maximal, let $d(x) = \gcd(\bar{g}(x), \bar{h}(x), \bar{k}(x))$ in $\mathbb{F}_p[x]$ and let $U(x) \in \mathbb{Z}[x]$ be any monic lift of $\frac{\bar{f}(x)}{d(x)}$. A strictly larger order is given by*

$$A' = (\sqrt{pA} : \sqrt{pA}) = A\left[\frac{U(\alpha)}{p}\right]$$

- *A' has basis*

$$1, \alpha, \alpha^2, \dots, \alpha^{\deg U - 1}, \frac{U(\alpha)}{p}, \frac{\alpha U(\alpha)}{p}, \dots, \frac{\alpha^{\deg(d)-1} U(\alpha)}{p}$$

so the index $[A' : A]$ is $p^{\deg(d)}$.

Proof. We utilize the Chinese Remainder Theorem:

$$\frac{A}{pA} \cong \frac{\mathbb{F}_p[x]}{\bar{f}(x)} \cong \prod_i \frac{\mathbb{F}_p[x]}{\bar{g}_i(x)^{e_i}}$$

Note the p-radical ideal is $I = \sqrt{pA} = pA + g(\alpha)A$. We will compute the multiplier ring $(I : I) = \{x \in K \mid xI \subseteq I\}$. Suppose $x \in (I : I)$. By Theorem 7.17 we can let $x = v(\alpha)/p$ for $v \in \mathbb{Z}[x]$. Now

$$x \in (A : I) \implies \bar{v}(\alpha)\bar{g}(\alpha) = 0 \pmod{p}$$

So, after subtracting an element of A, we can write $x = \frac{m(\alpha)h(\alpha)}{p}$ for $m(x) \in \mathbb{Z}[x]$. For $x \in (I : I)$, we require:

- $x \cdot p \in I \implies \bar{g} \mid \bar{m}\bar{h}$ in $\mathbb{F}_p[x]$.
- $x \cdot g(\alpha) \in I \implies \frac{m(\alpha)h(\alpha)g(\alpha)}{p} \equiv -m(\alpha)k(\alpha) \in I \implies \bar{g} \mid \bar{m}\bar{k}$ in $\mathbb{F}_p[x]$.

These hold simultaneously if and only if $\bar{g}$ divides $\bar{m} \cdot \gcd(\bar{h}, \bar{k})$. Let $d = \gcd(\bar{g}, \bar{h}, \bar{k})$. The minimal degree multiplier is $\bar{m} = \frac{\bar{g}}{d}$. Therefore the allowed classes of x modulo A are precisely the multiples of $U(\alpha)/p$, where $\bar{U} = \bar{f}/d$. Thus

$$A' = (I : I) = A + A\frac{U(\alpha)}{p}$$

So adjoining $\frac{U(\alpha)}{p}$ yields the overorder A'. Moreover $A' = A$ if and only if:

$$U(\alpha) = 0 \pmod{p} \iff d = 1$$

To compute the index $[A' : A]$, consider the standard $\mathbb{Z}$-basis of A: $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$. Since $U(x)$ is monic of degree $n - \deg(d)$ and $\bar{U}(x)$ is a factor of $\bar{f}(x)$, the elements $\frac{U(\alpha)}{p}, \frac{\alpha U(\alpha)}{p}, \dots, \frac{\alpha^{\deg(d)-1}U(\alpha)}{p}$ can replace the basis elements $\alpha^{n-\deg(d)}, \dots, \alpha^{n-1}$ to form a $\mathbb{Z}$-basis for A'.

The transition matrix to this new basis is upper-triangular. Its diagonal consists of $n - \deg(d)$ entries equal to 1 (for the unchanged lower-degree basis elements) and $\deg(d)$ entries equal to $\frac{1}{p}$ (for the new elements). The determinant of this transformation matrix is $p^{-\deg(d)}$, proving that the index $[A' : A]$ is exactly $p^{\deg(d)}$. □

Since computing gcd or dividing polynomials in finite fields is very cheap, Dedekind's criterion is preferred when the order is monogenic. When the order isn't, computing round 2 on bad primes requires a lot of linear algebra.

Remark 7.21. A natural way to remember the Dedekind criterion is the following. Keeping the notation of Theorem 7.20:

- Consider only the $\bar{g}_i$ with $e_i \geq 2$. We say each such $\bar{g}_i$ passes the criterion if $\bar{g}_i \nmid \bar{k}$.
- If every such g_i passes, then A is p-maximal.
- Otherwise, let d be the product of the $\bar{g}_i$ which fail and let U be a monic lift of $\bar{f}/d$. Then
$$A' = (I : I) = A + A\frac{U(\alpha)}{p}$$
is a bigger order with index $[A' : A]$ equal to $p^{\deg d}$.

Algorithm 7.22. To compute with Dedekind's criterion, it is not necessary to factor f into irreducibles. Instead we can use squarefree factorization and the Euclidean algorithm over $\mathbb{F}_p$.

(a) Compute the squarefree part $\bar{g}$ of $\bar{f}$, meaning the product of its distinct monic irreducible factors. One can compute this recursively: for a monic polynomial F with $F' \neq 0$, set $D = \gcd(F, F')$ and use
$$\operatorname{rad}(F) = \operatorname{lcm}(F/D, \operatorname{rad}(D))$$
If $F' = 0$, write $F = V^p$ and use $\operatorname{rad}(F) = \operatorname{rad}(V)$, with $\operatorname{rad}(1) = 1$. Then choose monic lifts g, h of $\bar{g}$ and $\bar{h} = \bar{f}/\bar{g}$, and set
$$k = \frac{f - gh}{p}$$

(b) Compute $c = \gcd(\bar{g}, \bar{h})$. If $c = 1$, then A is p-maximal.

(c) Otherwise compute $d = \gcd(c, \bar{k})$. If $d = 1$, then A is p-maximal.

(d) If $d \neq 1$, let U be a monic lift of $\bar{f}/d$. Then
$$A' = A\left[\frac{U(\alpha)}{p}\right]$$
has index $[A' : A] = p^{\deg d}$ and basis
$$1, \alpha, \ldots, \alpha^{\deg U - 1}, \frac{U(\alpha)}{p}, \frac{\alpha U(\alpha)}{p}, \ldots, \frac{\alpha^{\deg d - 1} U(\alpha)}{p}$$

Algorithm 7.23. In practice one usually proceeds as follows.

(a) Start with a convenient monogenic order $A = \mathbb{Z}[\alpha]$. The power basis is the canonical basis we do our math in (so we finish with basis elements expressed as polynomials in α).

(b) Compute $\operatorname{disc}(f) = \operatorname{disc}(A/\mathbb{Z})$ using Algorithm 7.10. By Corollary 7.13, only the primes p with $p^2 \mid \operatorname{disc}(f)$ can be bad.

(c) For each such p, use Dedekind's criterion (Algorithm 7.22). If it fails, let A'_p be the new overorder.

(d) Send:

$$A \leftarrow A'_{p_1} + \cdots + A'_{p_r}$$

which is computed via HNF by adjoining all the $\mathbb{Z}$-basis elements we got in Algorithm 7.22. The new discriminant is the old divided by the product of the squares of the indices we got.

(e) For every prime for which Dedekind's criterion failed, and whose square divides the new discriminant, perform the round 2 algorithm (Algorithm 7.19) to expand A.

The round 2 is quite expensive, which is why this algorithm is almost designed to avoid it for as long as possible.

Problem 7.24. Let

$$K = \mathbb{Q}(\sqrt{5}), \qquad A = \mathbb{Z}[\sqrt{5}], \qquad f(x) = x^2 - 5$$

and perform this algorithm to obtain the ring of integers.

The algorithm is extreme overkill in the above case, however it is the first nontrivial quadratic example. Now that we have a general method, I'll list out some shortcuts that are often useful.

7.4. Computing Prime Behaviour

There are two complementary tools for computing how primes behave. When the condition in Eq. (1) holds, Dedekind–Kummer reads the primes directly from the factorization of the defining polynomial modulo the base prime. For $\mathcal{O}_K = \mathbb{Z}$, it suffices that $p \nmid [\mathcal{O}_L : \mathbb{Z}[\alpha]]$, which can be checked using Theorem 7.20.

At the remaining primes we can work over the completion, using Theorem 6.29. The Newton polygons developed in Section 6.2.2 separate roots with different valuations, and Hensel lifts coprime factors of rescaled or translated polynomials. A slope denominator gives a divisor of the local ramification index by Corollary 6.14; an Eisenstein factor determines a totally ramified extension by Theorem 6.22. For instance, Example 6.13 gives local factors of degrees 1 and 2, with the quadratic factor totally ramified. When several irreducible factors share a slope, the polygon alone does not determine their number or residue degrees, so further factorization is still needed.

7.5. Computing the Class Group

7.6. Pell Convergents

Now we will learn how to compute the units of quadratic fields. At a conceptual level, computing $\mathcal{O}_K^\times$ means finding a $\mathbb{Z}$-basis of the lattice

$$\iota(\mathcal{O}_K^\times) \subseteq H$$

In general this is difficult, but in the low-rank cases one can often do it by hand. The main tricks are:

- convert "being a unit" into a norm equation;
- in rank 1, a unit with the least positive logarithmic coordinate is fundamental;
- in real quadratic fields, continued fractions solve the relevant norm equation.

The first point is completely general by Theorem 3.7. So the first thing to do in practice is to write down the norm explicitly.

Lemma 7.25 (Rank-1 trick)**.** *Suppose that* $\operatorname{rk} \mathcal{O}_K^\times = 1$. *Let* $\lambda : \mathcal{O}_K^\times \to \mathbb{R}$ *be a homomorphism obtained by composing* ι *with a coordinate projection, and assume that* λ *is not identically zero on* $\mathcal{O}_K^\times/\mu_K$. *If* $\epsilon \in \mathcal{O}_K^\times$ *satisfies* $\lambda(\epsilon) > 0$ *and is minimal among all units with positive* λ*-value, then* ϵ *is a fundamental unit.*

Proof. By Theorem 5.3,

$$\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}$$

Since λ is nonzero on the free part, $\lambda(\mathcal{O}_K^\times)$ is a nonzero discrete subgroup of $\mathbb{R}$, hence of the form $\delta\mathbb{Z}$ for some $\delta > 0$. The least positive value in $\lambda(\mathcal{O}_K^\times)$ is therefore δ, and any unit attaining it generates the free part of $\mathcal{O}_K^\times$. □

This already settles the easiest case.

Proposition 7.26 (Imaginary quadratic fields)**.** *Let* $K = \mathbb{Q}(\sqrt{d})$ *with* $d < 0$ *squarefree. Then* $\mathcal{O}_K^\times = \mu_K$ *is finite. More precisely:*

$$\mathcal{O}_K^\times = \begin{cases} \{\pm 1, \pm i\} & K = \mathbb{Q}(i) \\ \{\pm 1, \pm\omega, \pm\omega^2\} & K = \mathbb{Q}(\sqrt{-3}) \\ \{\pm 1\} & \text{otherwise} \end{cases}$$

where $\omega = \frac{-1+\sqrt{-3}}{2}$.

Proof. By Theorem 5.3, imaginary quadratic fields have unit rank

$$r_1 + r_2 - 1 = 0 + 1 - 1 = 0$$

so every unit is a root of unity.

If $\zeta \in K$ has order m, then $\mathbb{Q}(\zeta) \subseteq K$, hence

$$\varphi(m) = [\mathbb{Q}(\zeta) : \mathbb{Q}] \leq [K : \mathbb{Q}] = 2$$

Thus $m \in \{1, 2, 3, 4, 6\}$. The cases $m = 1, 2$ give only ± 1. If $m = 4$ then $\mathbb{Q}(\zeta) = \mathbb{Q}(i)$. If $m = 3$ or 6 then $\mathbb{Q}(\zeta) = \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$. This proves the classification. □

So imaginary quadratic fields are done. The first interesting case is the real quadratic one.

Proposition 7.27. *Let $K = \mathbb{Q}(\sqrt{d})$ with $d > 1$ squarefree.*

(a) If $d \equiv 2, 3 \pmod 4$, then $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$, and for

$$u = x + y\sqrt{d} \in \mathcal{O}_K$$

one has

$$u \in \mathcal{O}_K^\times \qquad \iff \qquad x^2 - dy^2 = \pm 1$$

(b) If $d \equiv 1 \pmod 4$, then $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$, and every element of $\mathcal{O}_K$ is of the form

$$u = \frac{m + n\sqrt{d}}{2} \qquad (m \equiv n \pmod 2)$$

For such a u one has

$$u \in \mathcal{O}_K^\times \qquad \iff \qquad m^2 - dn^2 = \pm 4$$

Proof.

(a) The conjugate of $u = x + y\sqrt{d}$ is $\bar{u} = x - y\sqrt{d}$, so

$$\mathrm{N}_{K/\mathbb{Q}}(u) = u\bar{u} = x^2 - dy^2$$

Now apply Theorem 3.7.

(b) The norm is

$$\mathrm{N}_{K/\mathbb{Q}}(u) = \frac{m + n\sqrt{d}}{2} \cdot \frac{m - n\sqrt{d}}{2} = \frac{m^2 - dn^2}{4}$$

Again Theorem 3.7 gives the result.

□

Lemma 7.28. *Let K be a real quadratic field, and let $u > 1$ be a unit in the ring of integers $\mathcal{O}_K$. Then $u \geq \phi$, where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.*

Proof. Note u is a root of a monic quadratic polynomial with integer coefficients:

$$x^2 - Tx + N = 0$$

where $T = \mathrm{Tr}_{K/\mathbb{Q}}(u) = u + \bar{u}$ and $N = u\bar{u} = \pm 1$. Since $u > 1$ and $N = \pm 1$, its conjugate $\bar{u} = \frac{N}{u}$ strictly satisfies $-1 < \bar{u} < 1$. Now

$$T = u + \bar{u} > 1 - 1 = 0$$

but since T must be a strictly positive integer, we know $T \geq 1$.

Using the quadratic formula, and knowing $u > 1$, we take the positive root:

$$u = \frac{T + \sqrt{T^2 - 4N}}{2}$$

To find the absolute minimum value for u, we evaluate the two possible cases for the norm N:

Case 1, $N = 1$**:** The equation becomes $u = \frac{T+\sqrt{T^2-4}}{2}$. For u to be real and strictly greater than 1, we must have $T^2 - 4 > 0$. The smallest integer trace satisfying this condition is $T = 3$. This yields:

$$u = \frac{3 + \sqrt{3^2 - 4}}{2} = \frac{3 + \sqrt{5}}{2}$$

Because $\frac{3+\sqrt{5}}{2} > \phi$, this case does not produce the minimum. Comparing both cases, the absolute smallest possible unit $u > 1$ across all real quadratic fields is exactly ϕ.

Case 2, $N = -1$**:** The equation becomes $u = \frac{T+\sqrt{T^2+4}}{2}$. Since the term inside the square root is always positive, we minimize u by choosing the absolutely smallest allowable integer trace, $T = 1$. This yields:

$$u = \frac{1 + \sqrt{1^2 + 4}}{2} = \frac{1 + \sqrt{5}}{2} = \phi$$

□

Corollary 7.29. *In a real quadratic field, the solution with smallest positive first coordinate, and then smallest positive second coordinate, of the relevant norm equation in Proposition 7.27 gives a fundamental unit.*

Proof. Suppose that u is a fundamental unit. Since

$$N(u) = u \cdot \bar{u} = \pm 1 \implies |u| \cdot |\bar{u}| = 1$$

we may assume that $u > 1$ after replacing u by its negative or inverse as needed. Moreover its conjugate is $\bar{u} = \pm u^{-1}$. Because $u > 1$, it strictly follows that $\bar{u} \in (-1, 1)$.

By Dirichlet's Unit Theorem, any unit $v > 1$ in this field is a positive power of the fundamental unit, meaning $v = u^n$ for some integer $n \geq 1$. Write $u^n = (x_n + y_n\sqrt{d})/2$, so x_n and y_n are integers (both even when $d \equiv 2, 3 \pmod 4$). Then:

$$x_n = u^n + \bar{u}^n \quad \text{and} \quad y_n\sqrt{d} = u^n - \bar{u}^n$$

Since $u > 1$, its powers u^n are strictly greater than 1. Likewise, since $\bar{u} \in (-1, 1)$, its powers $\bar{u}^n \in (-1, 1)$. This guarantees that for all $n \geq 1$, both $x_n > 0$ and $y_n > 0$.

Substituting $\bar{u}^n = (\pm 1)^n u^{-n}$, we can express the first coefficient x_n explicitly as a function of the exponent n:

$$x_n = u^n + (\pm 1)^n u^{-n} \qquad y_n\sqrt{d} = u^n - (\pm 1)^n u^{-n}$$

Therefore

$$x_{n+1} - x_n = \begin{cases} (u-1)(u^n - u^{-n-1}) & \text{if } \bar{u} = u^{-1} \\ u^n(u-1) - (-1)^n u^{-n-1}(u+1) & \text{if } \bar{u} = -u^{-1} \end{cases}$$

$$\sqrt{d}(y_{n+1} - y_n) = \begin{cases} (u-1)(u^n + u^{-n-1}) & \text{if } \bar{u} = u^{-1} \\ u^n(u-1) + (-1)^n u^{-n-1}(u+1) & \text{if } \bar{u} = -u^{-1} \end{cases}$$

It is clear that in the first case x_n and y_n are both monotonically increasing since $u > 1 > u^{-1}$. In the other case we want

$$u^{2n+1}(u-1) > (u+1)$$

in order for x_n and y_n to monotonically increase for all n. Since the LHS increases with n, it suffices to check $n = 1$:

$$u^3(u-1) > (u+1)$$

We use Lemma 7.28 (u is at least the golden ratio). In the case that the equality is achieved (which can only happen when $K = \mathbb{Q}[\sqrt{5}]$) we can only show that x_n is strictly increasing and y_n is non-decreasing. Otherwise, our inequality is attained and both x_n and y_n must strictly increase. □

This provides a general method for real quadratic fields: test positive x and y as solutions to the norm equation until a solution is found. Once a solution is found, we need only check positive pairs with smaller first coordinate, or the same first coordinate and smaller second coordinate to guarantee we have found the fundamental unit, which is a finite and manageable task.

Remark 7.30 (A practical trick in any rank-1 field)**.** The real quadratic case is the cleanest, but Lemma 7.25 is more general. If K has unit rank 1 (for instance a complex cubic field), then after writing down an integral basis of $\mathcal{O}_K$ one can simply test small algebraic integers α by the norm criterion

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha) = \pm 1$$

Once one finds a non-torsion unit, the smallest one under any nonzero logarithmic coordinate is automatically fundamental. This is often the fastest by-hand computation in simple examples. For real quadratic fields there is a faster algorithm by computing *Pell-convergents*. I will define the algorithm and state why it works, but I will not prove a lot of things.

Definition 7.31 (Continued Fractions)**.** We define recursively

$$[a_0] = a_0 \qquad [a_0; a_1, a_2, \dots, a_n] = a_0 + \frac{1}{[a_1; a_2, \dots, a_n]}$$

which gives a good notation to express a continued fraction.

Definition 7.32 (Pell convergents)**.** Write

$$\sqrt{d} = [a_0; a_1, a_2, \dots], \qquad \frac{p_n}{q_n} = [a_0; a_1, \dots, a_n]$$

The fractions p_n/q_n are the *Pell convergents* of $\sqrt{d}$.

Theorem 7.33. *For $n \geq 1$, the Pell convergents are best approximations in the following sense*

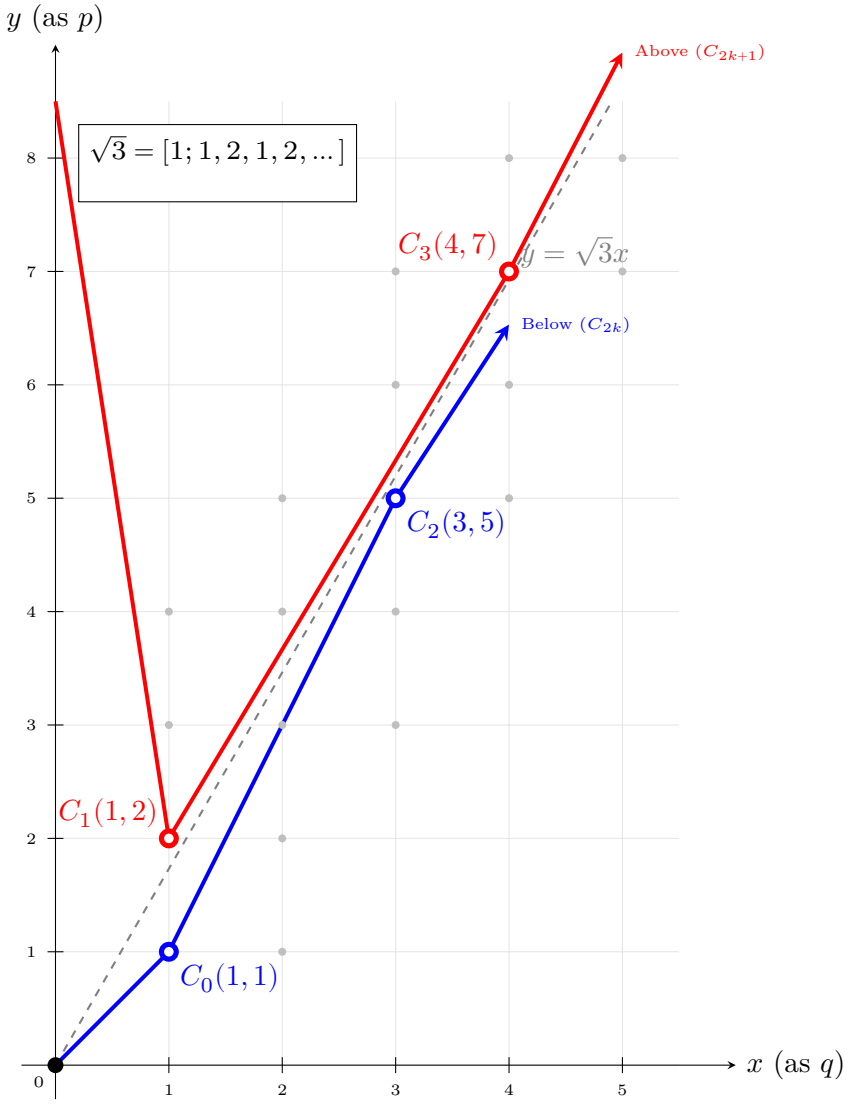
$$|q_n\sqrt{d} - p_n| < |b\sqrt{d} - a| \qquad \text{for } 0 < b \leq q_n \text{ and } a/b \neq p_n/q_n$$

Conversely, every reduced fraction which strictly minimizes this error among fractions with no larger positive denominator is a convergent. We will also use Legendre's criterion: if a/b is reduced, $b > 0$, and

$$\left|\sqrt{d} - \frac{a}{b}\right| < \frac{1}{2b^2}$$

then a/b is a convergent.

Geometrically, one plots the lattice points (q, p) and the line $y = \sqrt{d}\,x$. So in the Pell equation $p^2 - dq^2 = \pm 1$, the ratio p/q should be thought of as the lattice points trying to hug the irrational slope $\sqrt{d}$ as tightly as possible.



Proposition 7.34 (Why Pell solutions are convergents)**.** *Assume $d \equiv 2, 3 \pmod 4$. If $p, q > 0$ satisfy*

$$p^2 - dq^2 = \pm 1$$

then p/q is a Pell convergent of $\sqrt{d}$.

Proof. The equation implies $\gcd(p, q) = 1$. Moreover

$$\left|\sqrt{d} - \frac{p}{q}\right| = \frac{|dq^2 - p^2|}{q^2(\sqrt{d} + p/q)} = \frac{1}{q^2(\sqrt{d} + p/q)} < \frac{1}{2q^2}$$

where $\sqrt{d} + p/q > 2$ since $d \geq 2$ and $p^2 = dq^2 \pm 1$. Now apply Legendre's criterion. □

This is why continued fractions solve the unit problem in the real quadratic case: once the norm equation has been reduced to Pell, one no longer searches blindly through all lattice points. One walks through the Pell convergents and checks whether the norm equation is satisfied. Since the Pell-convergents increase in size exponentially, we check far fewer points and this is a significant speedup.

Remark 7.35. To compute the Pell convergents for $\sqrt{d}$, we first determine the continued fraction expansion $\sqrt{d} = [a_0; \overline{a_1, a_2, \ldots, a_k}]$.

The coefficients a_n are generated using the recursive auxiliary sequences P_n and Q_n:

$$\begin{aligned} P_0 &= 0 \quad Q_0 = 1 \quad a_0 = \lfloor \sqrt{d} \rfloor \\ P_n &= a_{n-1} Q_{n-1} - P_{n-1} \\ Q_n &= \frac{d - P_n^2}{Q_{n-1}} \\ a_n &= \left\lfloor \frac{a_0 + P_n}{Q_n} \right\rfloor \end{aligned}$$

The numerators p_n and denominators q_n of the convergents are defined recursively by:

$$\begin{aligned} p_{-2} &= 0 \quad p_{-1} = 1 \quad p_n = a_n p_{n-1} + p_{n-2} \\ q_{-2} &= 1 \quad q_{-1} = 0 \quad q_n = a_n q_{n-1} + q_{n-2} \end{aligned}$$

Equivalently, this recurrence can be expressed and computed efficiently via matrix multiplication. The convergents are obtained by the product of 2×2 matrices corresponding to the continued fraction coefficients:

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix}$$

By repeatedly computing the matrix product, the first column of the resulting matrix directly yields the numerator and denominator of the convergent $\frac{p_n}{q_n}$.

Remark 7.36. For $d \equiv 1 \pmod 4$, the equations $x^2 - dy^2 = \pm 1$ compute the units of $A = \mathbb{Z}[\sqrt{d}]$. Let $\eta > 1$ be the fundamental unit of A obtained from the Pell convergents. Since $A = \mathbb{Z} + 2\mathcal{O}_K$, a unit of $\mathcal{O}_K$ lies in A exactly when its reduction modulo $2\mathcal{O}_K$ is 1. If $d \equiv 1 \pmod 8$, then $\mathcal{O}_K/2\mathcal{O}_K \cong \mathbb{F}_2 \times \mathbb{F}_2$ has trivial unit group, so η is already fundamental in $\mathcal{O}_K$. If $d \equiv 5 \pmod 8$, then $\mathcal{O}_K/2\mathcal{O}_K \cong \mathbb{F}_4$ has unit group of order 3, so $[\mathcal{O}_K^\times : A^\times]$ is 1 or 3. Thus the fundamental unit of $\mathcal{O}_K$ is the positive real cube root of η if that root lies in $\mathcal{O}_K$, and is η otherwise. For example, when $d = 5$ the first Pell unit is $2 + \sqrt{5} = ((1 + \sqrt{5})/2)^3$.

8. Cyclotomic Fields

Cyclotomic fields are extremely useful, in this section we will understand them.

8.1. The Ring of Integers

If I asked you to guess what the ring of integers of $\mathbb{Q}[\zeta_m]$ is you would probably say $\mathbb{Z}[\zeta_m]$. This is completely correct, but proving it requires some work. Nevertheless, using the tools of Section 7.3 it is quite straightforward. We will first work with $m = p^n$, where p is any prime and $n \geq 1$.

Proposition 8.1. *Let ζ_{p^n} be a primitive p^n-th root of unity. Its minimal polynomial is*

$$\Phi_{p^n}(x) = \frac{x^{p^n} - 1}{x^{p^{n-1}} - 1} = \sum_{j=0}^{p-1} x^{jp^{n-1}} = \Phi_p(x^{p^{n-1}})$$

In particular

$$\deg \Phi_{p^n} = \varphi(p^n) = p^{n-1}(p-1)$$

Moreover $\Phi_{p^n}(x+1)$ is Eisenstein at p.

Proof. The quotient is a polynomial whose roots are exactly the p^n-th roots of unity which are not p^{n-1}-th roots, hence exactly the primitive p^n-th roots. Modulo p we have

$$x^{p^n} - 1 = (x-1)^{p^n} \qquad x^{p^{n-1}} - 1 = (x-1)^{p^{n-1}}$$

Cancelling in $\mathbb{F}_p[x]$ gives

$$\Phi_{p^n}(x) \equiv (x-1)^{p^{n-1}(p-1)} \pmod{p}$$

Thus $\Phi_{p^n}(x+1)$ is monic and every other coefficient is divisible by p. Its constant term is $\Phi_{p^n}(1) = p$, so it is Eisenstein. It follows that Φ_{p^n} is irreducible over $\mathbb{Q}$, proving the assertion about the minimal polynomial. □

Lemma 8.2. *Let $d = \varphi(p^n) = p^{n-1}(p-1)$. The discriminant of Φ_{p^n} is*

$$\operatorname{disc}(\Phi_{p^n}) = (-1)^{d(d-1)/2} p^{p^{n-1}(n(p-1)-1)}$$

Proof. The case $p^n = 2$ is immediate from $\Phi_2 = x + 1$, whose discriminant is 1. Otherwise let $K = \mathbb{Q}(\zeta)$ with $\zeta = \zeta_{p^n}$, so d is even. Differentiate the identity in Proposition 8.1 and evaluate at ζ to obtain

$$(\zeta^{p^{n-1}} - 1)\Phi'_{p^n}(\zeta) = p^n \zeta^{-1}$$

The element $\zeta^{p^{n-1}}$ is a primitive p-th root of unity, and

$$[K : \mathbb{Q}(\zeta_p)] = p^{n-1} \qquad \mathrm{N}_{K/\mathbb{Q}}(\zeta_p - 1) = p^{p^{n-1}}$$

For odd p, the last equality follows from $\Phi_p(1) = p$ and the even degree $p-1$. For $p = 2$ and $n \geq 2$, it follows from $\zeta_2 - 1 = -2$ and the even degree 2^{n-1}. Also $\mathrm{N}_{K/\mathbb{Q}}(\zeta) = 1$, since its conjugates occur in complex conjugate pairs. Taking norms therefore gives

$$\mathrm{N}_{K/\mathbb{Q}}(\Phi'_{p^n}(\zeta)) = p^{nd - p^{n-1}} = p^{p^{n-1}(n(p-1)-1)}$$

Now apply Theorems 2.26 and 2.29. □

Lemma 8.3. *Let $K = \mathbb{Q}(\alpha)$, where α is a root of a monic polynomial $f \in \mathbb{Z}[x]$ which is Eisenstein at a prime p. Then p does not divide the index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$.*

Proof. Write

$$f(x) = x^d + ph(x) \qquad h \in \mathbb{Z}[x] \qquad h(0) \not\equiv 0 \pmod{p}$$

Modulo p, the only irreducible factor of f is x. Since x does not divide $\bar{h}$, Dedekind's criterion gives the result. □

Theorem 8.4. *For every prime p and integer $n \geq 1$, let $K = \mathbb{Q}(\zeta_{p^n})$ and $d = p^{n-1}(p-1)$. Then*

$$\mathcal{O}_K = \mathbb{Z}[\zeta_{p^n}] \qquad d_K = (-1)^{d(d-1)/2} p^{p^{n-1}(n(p-1)-1)}$$

An integral basis is

$$1, \zeta_{p^n}, \dots, \zeta_{p^n}^{d-1}$$

Proof. Set $\lambda = \zeta_{p^n} - 1$. Then $\mathbb{Z}[\lambda] = \mathbb{Z}[\zeta_{p^n}]$, and the minimal polynomial of λ is $\Phi_{p^n}(x+1)$. This is Eisenstein at p by Proposition 8.1, so Lemma 8.3 gives

$$p \nmid [\mathcal{O}_K : \mathbb{Z}[\zeta_{p^n}]]$$

On the other hand, Lemma 8.2 and Proposition 7.11 show that only p can divide this index. Therefore the index is 1, and the polynomial discriminant is the field discriminant. □

It is useful to write out the cases separately.

Corollary 8.5.

(a) If p is odd and $n = 1$, then

$$\Phi_p(x) = 1 + x + \cdots + x^{p-1} \qquad \mathcal{O}_{\mathbb{Q}(\zeta_p)} = \mathbb{Z}[\zeta_p] \qquad d_{\mathbb{Q}(\zeta_p)} = (-1)^{(p-1)/2} p^{p-2}$$

(b) If p is odd and $n \geq 1$, then

$$\mathcal{O}_{\mathbb{Q}(\zeta_{p^n})} = \mathbb{Z}[\zeta_{p^n}] \qquad d_{\mathbb{Q}(\zeta_{p^n})} = (-1)^{p^{n-1}(p-1)/2} p^{p^{n-1}(n(p-1)-1)}$$

(c) If $p = 2$ and $n = 1$, then $\zeta_2 = -1$, so $K = \mathbb{Q}$, $\mathcal{O}_K = \mathbb{Z}$, and $d_K = 1$.

(d) If $p = 2$ and $n \geq 2$, then

$$\Phi_{2^n}(x) = x^{2^{n-1}} + 1 \qquad \mathcal{O}_{\mathbb{Q}(\zeta_{2^n})} = \mathbb{Z}[\zeta_{2^n}] \qquad d_{\mathbb{Q}(\zeta_{2^n})} = (-1)^{2^{n-2}} 2^{(n-1)2^{n-1}}$$

Thus $d_{\mathbb{Q}(i)} = -4$, and the discriminant is positive for $n \geq 3$.

Theorem 8.6. *For any $m \geq 1$, the ring of integers of $K = \mathbb{Q}(\zeta_m)$ is*

$$\mathcal{O}_K = \mathbb{Z}[\zeta_m]$$

Moreover $[K : \mathbb{Q}] = \varphi(m)$ and every prime dividing d_K divides m.

Proof. We proceed by induction on the number of distinct prime factors of m. The case $m = 1$ is immediate, and prime powers are covered by Theorem 8.4. Write $m = ap^n$ with $\gcd(a, p) = 1$, and let $E = \mathbb{Q}(\zeta_a)$ and $F = \mathbb{Q}(\zeta_{p^n})$. By induction their discriminants are coprime, so Corollary 3.27 and Theorem 3.29 give

$$[EF : \mathbb{Q}] = \varphi(a)\varphi(p^n) = \varphi(m) \qquad \mathcal{O}_{EF} = \mathbb{Z}[\zeta_a, \zeta_{p^n}]$$

Choose $u, v \in \mathbb{Z}$ with $up^n + va = 1$ and use compatible roots of unity. Then

$$\zeta_m = \zeta_a^u \zeta_{p^n}^v \qquad \zeta_a = \zeta_m^{p^n} \qquad \zeta_{p^n} = \zeta_m^a$$

Thus $EF = \mathbb{Q}(\zeta_m)$ and $\mathcal{O}_{EF} = \mathbb{Z}[\zeta_m]$. Finally Theorem 3.28 shows that the primes dividing its discriminant divide m, completing the induction. □

Corollary 8.7. *For $m \geq 3$, the discriminant is*

$$d_{\mathbb{Q}(\zeta_m)} = (-1)^{\varphi(m)/2} \frac{m^{\varphi(m)}}{\prod_{p \mid m} p^{\varphi(m)/(p-1)}}$$

Proof. Apply Theorem 3.28 to the prime-power factors of m and use Theorem 8.4. If p^n is the exact power of p dividing m, its exponent in the absolute discriminant is

$$\frac{\varphi(m)}{\varphi(p^n)} p^{n-1}(n(p-1) - 1) = n\varphi(m) - \frac{\varphi(m)}{p-1}$$

All embeddings are non-real when $m \geq 3$, so the sign is $(-1)^{\varphi(m)/2}$ by the embedding-matrix calculation in Proposition 4.3. □

8.2. Prime Behaviour in Prime-Power Fields

Now that we know the ring of integers, Dedekind-Kummer applies at every prime.

Theorem 8.8. *Let $K = \mathbb{Q}(\zeta_{p^n})$, $d = \varphi(p^n)$, and $\mathfrak{p} = (1 - \zeta_{p^n})$. Then*

$$p\mathcal{O}_K = \mathfrak{p}^d \qquad \mathcal{O}_K/\mathfrak{p} \cong \mathbb{F}_p$$

Thus $e = d$, $f = g = 1$, and $1 - \zeta_{p^n}$ is a uniformizer at the unique prime above p. No other rational prime ramifies. The different is

$$\mathcal{D}_{K/\mathbb{Q}} = \mathfrak{p}^{p^{n-1}(n(p-1)-1)}$$

Proof. By Proposition 8.1 and Theorem 8.4,

$$\mathcal{O}_K/(1 - \zeta_{p^n}) \cong \mathbb{Z}[x]/(\Phi_{p^n}, x - 1) \cong \mathbb{Z}/p\mathbb{Z}$$

so $\mathfrak{p}$ is prime of norm p. For every a prime to p, with $1 \leq a < p^n$, the quotient

$$\frac{1 - \zeta_{p^n}^a}{1 - \zeta_{p^n}} = 1 + \zeta_{p^n} + \cdots + \zeta_{p^n}^{a-1}$$

is an algebraic integer of norm 1, since numerator and denominator both have norm p. It is therefore a unit by Theorem 3.7. Evaluating the cyclotomic polynomial at 1 gives

$$p = \prod_{\substack{1 \le a < p^n \\ p \nmid a}} (1 - \zeta_{p^n}^a)$$

and taking ideals gives $p\mathcal{O}_K = \mathfrak{p}^d$. The discriminant formula shows that every other prime is unramified. The different is supported only at $\mathfrak{p}$, and its norm is the discriminant ideal. Since $\mathrm{N}(\mathfrak{p}) = p$, its exponent is exactly the exponent of p in Theorem 8.4. □

For $p^n = 2$ the degree is 1, so these formulas give $2\mathbb{Z} = (2)$ and different (1). For every other prime power, p is ramified with ramification index $\varphi(p^n)$.

Theorem 8.9. *Let $\ell \neq p$ be a rational prime and put*

$$f = \mathrm{ord}_{p^n}(\ell) = \min\{r \ge 1 : \ell^r \equiv 1 \pmod{p^n}\}$$

Then ℓ is unramified in $K = \mathbb{Q}(\zeta_{p^n})$ and

$$\ell\mathcal{O}_K = \mathfrak{q}_1 \cdots \mathfrak{q}_g \qquad g = \frac{\varphi(p^n)}{f} \qquad \mathrm{N}(\mathfrak{q}_i) = \ell^f$$

In particular, ℓ splits totally if and only if $\ell \equiv 1 \pmod{p^n}$, and it is inert if and only if $\mathrm{ord}_{p^n}(\ell) = \varphi(p^n)$.

Proof. Since $\ell \nmid p^n$, the polynomial $x^{p^n} - 1$ is squarefree modulo ℓ. The roots of $\bar{\Phi}_{p^n}$ therefore have exact order p^n. The irreducible factors over $\mathbb{F}_\ell$ correspond to orbits of the Frobenius map $x \mapsto x^\ell$ on these roots. Each orbit has length f, because for a primitive root $\bar{\zeta}$,

$$\bar{\zeta}^{\ell^r} = \bar{\zeta} \qquad \Longleftrightarrow \qquad \ell^r \equiv 1 \pmod{p^n}$$

Thus there are $\varphi(p^n)/f$ distinct factors, all of degree f. Now apply Dedekind-Kummer. □

Remark 8.10. The Galois group is

$$\mathrm{Gal}(\mathbb{Q}(\zeta_{p^n})/\mathbb{Q}) \cong (\mathbb{Z}/p^n\mathbb{Z})^\times \qquad a \longmapsto (\zeta_{p^n} \longmapsto \zeta_{p^n}^a)$$

The possible groups are

$$(\mathbb{Z}/p^n\mathbb{Z})^\times \cong \begin{cases} \mathbb{Z}/(p^{n-1}(p-1))\mathbb{Z} & p \text{ odd} \\ \{1\} & p = 2,\ n = 1 \\ \mathbb{Z}/2\mathbb{Z} & p = 2,\ n = 2 \\ \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{n-2}\mathbb{Z} & p = 2,\ n \ge 3 \end{cases}$$

For odd p this is the primitive root theorem. For $p = 2$ and $n \ge 3$, the factors are generated by -1 and 5 respectively. Thus no odd prime is inert in $\mathbb{Q}(\zeta_{2^n})$ when $n \ge 3$, since no element has order 2^{n-1}.

Example 8.11.

- In $\mathbb{Q}(\zeta_9)$ we have degree 6, discriminant -3^9, and $(3) = (1-\zeta_9)^6$ as ideals. The prime 2 is inert since its order modulo 9 is 6. The prime 19 splits totally since $19 \equiv 1 \pmod 9$.
- In $\mathbb{Q}(\zeta_8)$ we have degree 4, discriminant 2^8, and $(2) = (1-\zeta_8)^4$ as ideals. Odd primes congruent to 1 modulo 8 split totally. Each other odd prime splits into two primes of inertia degree 2.

9. The Galois Case

9.1. Decomposition and Inertia Groups

Let R be a Dedekind domain with fraction field K, let L/K be a finite *Galois* extension, and let S be the integral closure of R in L. Assume that the residue fields of R are perfect, as they are for rings of integers in number fields, and write

$$G := \mathrm{Gal}(L/K) \qquad n = [L:K]$$

For an intermediate field E, write $\mathcal{O}_E$ for the integral closure of R in E. Let $\mathfrak{p} \subset R$ be a nonzero prime and choose a prime $\mathfrak{q} \subset S$ with $\mathfrak{q} \cap R = \mathfrak{p}$. Write $\kappa(\mathfrak{p}) := R/\mathfrak{p}$ and $\kappa(\mathfrak{q}) := S/\mathfrak{q}$ for the residue fields.

Lemma 9.1. *G acts transitively on the primes above $\mathfrak{p}$.*

Proof. Suppose FTSOC that there are two distinct G orbits of primes above $\mathfrak{p}$. Call one of them $\{\mathfrak{q}_1, \dots, \mathfrak{q}_n\}$ and let $\{\mathfrak{r}_i\}$ be the primes above $\mathfrak{p}$ not in this orbit. Using CRT we may find $x \in S$ with:

$$x \equiv 1 \pmod{\mathfrak{q}_i} \qquad x \equiv 0 \pmod{\mathfrak{r}_i}$$

Note that $y = \prod_{\sigma \in G} \sigma(x)$ also has these residues and is an element of R. Therefore

$$y \in (R \cap \mathfrak{r}_1) = \mathfrak{p}$$

which is a contradiction with the fact that $y \notin \mathfrak{q}_1$, since it is an ideal that contains $\mathfrak{p}$. □

Corollary 9.2. *Every prime above $\mathfrak{p}$ has the same ramification index and inertia degree. Therefore Theorem 3.4 simplifies to*

$$n = efg$$

where g is the number of primes above $\mathfrak{p}$.

Definition 9.3 (Decomposition group)**.**

- The *decomposition group* of $\mathfrak{q}$ (over $\mathfrak{p}$) is the stabilizer

$$D_{\mathfrak{q}} := \{\sigma \in G : \sigma(\mathfrak{q}) = \mathfrak{q}\} \subseteq G$$

- Every $\sigma \in D_{\mathfrak{q}}$ induces a $\kappa(\mathfrak{p})$-automorphism of $\kappa(\mathfrak{q})$, hence a homomorphism:

$$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \longrightarrow \mathrm{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

  The *inertia group* is the kernel:

$$I_{\mathfrak{q}} := \ker(\rho_{\mathfrak{q}}) = \{\sigma \in D_{\mathfrak{q}} : \sigma(x) \equiv x \pmod{\mathfrak{q}} \text{ for all } x \in S\}$$

Proposition 9.4. *The reduction map $\rho_{\mathfrak{q}}$ fits into a short exact sequence*

$$1 \longrightarrow I_{\mathfrak{q}} \longrightarrow D_{\mathfrak{q}} \xrightarrow{\rho_{\mathfrak{q}}} \mathrm{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p})) \longrightarrow 1$$

Moreover $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is Galois, and $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ is Galois with group $\mathrm{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$.

Proof. Firstly, $I_{\mathfrak{q}}$ is normal in $D_{\mathfrak{q}}$ because it is the kernel of a homomorphism, so $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ is Galois. The main task is to show that $\rho_{\mathfrak{q}}$ is surjective. Because $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is finite separable we may choose a nonzero primitive element $\bar{\alpha}$ with $\kappa(\mathfrak{q}) = \kappa(\mathfrak{p})[\bar{\alpha}]$. By Chinese remainder theorem we may find an element $\alpha \in S$ with

$$\alpha \equiv \bar{\alpha} \pmod{\mathfrak{q}} \qquad \alpha \equiv 0 \pmod{\mathfrak{r}_i} \tag{1}$$

where $\mathfrak{r}_i$ are all the other primes lying over $\mathfrak{p}$. Let

$$f(x) = \prod_{\sigma \in G} (x - \sigma(\alpha)) \qquad \bar{f}(x) = \prod_{\sigma \in G} (x - \overline{\sigma(\alpha)})$$

be the characteristic polynomial of α and its reduction in the residue extension respectively. Since f has coefficients in R, $\bar{f}$ has coefficients in $R/\mathfrak{p}$. The minimal polynomial of $\bar{\alpha}$ divides $\bar{f}$, which splits in $\kappa(\mathfrak{q})$. Thus the residue extension is normal as well as separable. Therefore any $\tau \in \mathrm{Aut}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$ sends $\bar{\alpha}$ to another root of $\bar{f}(x)$ and:

$$\tau(\bar{\alpha}) = \overline{\sigma(\alpha)} \qquad \text{for some } \sigma \in G$$

We claim that $\rho_{\mathfrak{q}}(\sigma) = \tau$.

- Suppose for the sake of contradiction that $\sigma \notin D_{\mathfrak{q}}$. Then by Eq. (1) $\sigma(\alpha) \equiv 0 \pmod{\mathfrak{q}}$. Therefore $\overline{\sigma(\alpha)}$ is 0 in the residue extension, a contradiction since τ can't send a nonzero element to 0.
- Since α is primitive, $\tau(\bar{\alpha}) = \overline{\sigma(\alpha)}$ forces $\rho_{\mathfrak{q}}(\sigma) = \tau$.

□

Remark 9.5 (The efg-formula realized by subextensions)**.**

(a) $|D_{\mathfrak{q}}| = ef$, $|I_{\mathfrak{q}}| = e$, and $|D_{\mathfrak{q}}/I_{\mathfrak{q}}| = f$.

(b) Considering fixed fields

$$K \subseteq L^{D_{\mathfrak{q}}} \subseteq L^{I_{\mathfrak{q}}} \subseteq L$$

and the degrees in this tower recover the splitting invariants:

$$[L^{D_{\mathfrak{q}}} : K] = g \qquad [L^{I_{\mathfrak{q}}} : L^{D_{\mathfrak{q}}}] = f \qquad [L : L^{I_{\mathfrak{q}}}] = e$$

Proof.

(a) By orbit stabilizer theorem $|D_{\mathfrak{q}}| = |G|/g$. Then plugging in $|G| = efg$ gives $|D_{\mathfrak{q}}| = ef$ as desired. We must have:

$$ef = |D_{\mathfrak{q}}| = |I_{\mathfrak{q}}||D_{\mathfrak{q}}/I_{\mathfrak{q}}|$$

Note by Proposition 9.4 that

$$|D_{\mathfrak{q}}/I_{\mathfrak{q}}| = [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})] = f$$

which forces $|I_{\mathfrak{q}}| = e$.

(b) This is now a direct translation via the Galois correspondence:

$$[L^{D_{\mathfrak{q}}} : K] = [G : D_{\mathfrak{q}}] = g \qquad [L^{I_{\mathfrak{q}}} : L^{D_{\mathfrak{q}}}] = [D_{\mathfrak{q}} : I_{\mathfrak{q}}] = f \qquad [L : L^{I_{\mathfrak{q}}}] = |I_{\mathfrak{q}}| = e$$

□

Theorem 9.6 (How to read the tower)**.** *Define:*

$$\mathfrak{q}_I = \mathcal{O}_{L^{I_{\mathfrak{q}}}} \cap \mathfrak{q} \qquad \mathfrak{q}_D = \mathcal{O}_{L^{D_{\mathfrak{q}}}} \cap \mathfrak{q}$$

Then we can draw the diagram

Fields	***Primes***	***Galois Groups***
L	$\mathfrak{q}$	$\{1\}$
$\Big\vert e$	$\Big\vert e(\mathfrak{q}\vert\mathfrak{q}_I)=e$	$\Big\vert e$
$L^{I_{\mathfrak{q}}}$	$\mathfrak{q}_I$	$I_{\mathfrak{q}}$
$\Big\vert f$	$\Big\vert f(\mathfrak{q}_I\vert\mathfrak{q}_D)=f$	$\Big\vert f$
$L^{D_{\mathfrak{q}}}$	$\mathfrak{q}_D$	$D_{\mathfrak{q}}$
$\Big\vert g$	$\Big\vert$	$\Big\vert g$
K	$\mathfrak{p}$	$G = Gal(L/K)$

where the inertia and ramification degrees are assumed 1 *unless indicated.*

Proof. We will make some counting arguments with Theorem 3.4.

- Consider the extension $L/L^{D_{\mathfrak{q}}}$. Since the Galois group of this extension fixes $\mathfrak{q}$ there are no other primes above $\mathfrak{q}_D$ by Lemma 9.1. Therefore by Corollary 9.2 $\mathfrak{q}_D$ does not split in this extension. Now since $L/L^{D_{\mathfrak{q}}}$ has degree ef we must have $f(\mathfrak{q}|\mathfrak{q}_D) = f$ and $e(\mathfrak{q}|\mathfrak{q}_D) = e$.

- Consider the extension $L/L^{I_\mathfrak{q}}$. By definition the Galois group acts trivially on the residue field extension. On the other hand the Galois group surjects onto the residue field automorphism group by the argument in Proposition 9.4. Therefore this extension has residue degree 1, since the residue extension is Galois with trivial automorphism group. It has a unique prime above $\mathfrak{q}_I$ and degree e, so $e(\mathfrak{q}|\mathfrak{q}_I) = e$.
- By the previous results $L^{I_\mathfrak{q}}/L^{D_\mathfrak{q}}$ must account for the inertia. Moreover by counting degrees $f(\mathfrak{q}_I|\mathfrak{q}_D) = f$.

□

Corollary 9.7. *In the case that both $D_\mathfrak{q}$ and $I_\mathfrak{q}$ are normal in G (for instance when G is abelian)*

- *$\mathfrak{p}$ is totally split in $L^{D_\mathfrak{q}}/K$;*
- *every prime over $\mathfrak{p}$ in $L^{D_\mathfrak{q}}$ is inert in $L^{I_\mathfrak{q}}/L^{D_\mathfrak{q}}$;*
- *every prime over $\mathfrak{p}$ in $L^{I_\mathfrak{q}}$ is totally ramified in $L/L^{I_\mathfrak{q}}$.*

Theorem 9.8. *Suppose that $K \subset E \subset L$ is an intermediate extension with $\mathrm{Gal}(L/E) := H \subset G$. Then*

$$I_\mathfrak{q}(L/E) = I_\mathfrak{q}(L/K) \cap H \qquad D_\mathfrak{q}(L/E) = D_\mathfrak{q}(L/K) \cap H$$

If E/K is also Galois, set $\mathfrak{q}_E = \mathfrak{q} \cap \mathcal{O}_E$, where $\mathcal{O}_E$ is the integral closure of R in E. Then there is the following commutative diagram of exact sequences:

$$\begin{array}{ccccccccc}
 & & 1 & & 1 & & 1 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & I_\mathfrak{q}(L/E) & \longrightarrow & I_\mathfrak{q}(L/K) & \longrightarrow & I_{\mathfrak{q}_E}(E/K) & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & D_\mathfrak{q}(L/E) & \longrightarrow & D_\mathfrak{q}(L/K) & \longrightarrow & D_{\mathfrak{q}_E}(E/K) & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \overline{G}_\mathfrak{q}(L/E) & \longrightarrow & \overline{G}_\mathfrak{q}(L/K) & \longrightarrow & \overline{G}_{\mathfrak{q}_E}(E/K) & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 1 & & 1 & & 1 & &
\end{array}$$

where we have used $\overline{G}$ to denote the automorphisms of the respective residue field.

Proof. The decomposition group consists of the automorphisms stabilizing $\mathfrak{q}$, and the inertia group consists of those acting trivially on its residue field. Restricting to H therefore intersects both groups with H.

Now assume E/K is Galois. The columns of the diagram are exact by Proposition 9.4. Given $\sigma \in D_{\mathfrak{q}_E}(E/K)$, lift it to $\tilde{\sigma} \in G$. The primes $\tilde{\sigma}(\mathfrak{q})$ and $\mathfrak{q}$ lie over the same prime of

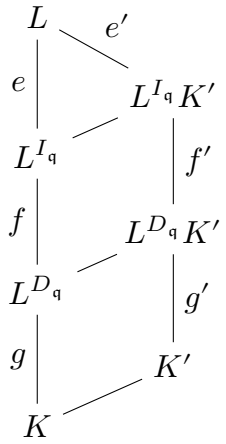
E. By transitivity there is $h \in H$ carrying $\tilde{\sigma}(\mathfrak{q})$ to $\mathfrak{q}$, so $h\tilde{\sigma} \in D_{\mathfrak{q}}(L/K)$ lifts σ. This proves surjectivity in the decomposition row.

Restriction carries $I_{\mathfrak{q}}(L/K)$ into $I_{\mathfrak{q}_E}(E/K)$, with kernel $I_{\mathfrak{q}}(L/E)$. Its image has order $e(L/K)/e(L/E) = e(E/K)$ at the chosen primes, by multiplicativity of ramification indices. This is the full order of $I_{\mathfrak{q}_E}(E/K)$, so the inertia row is exact. The residue-field row is the usual restriction sequence for the corresponding tower of Galois residue extensions. □

Remark 9.9. The most natural way to remember this is that:

- $L^{D_{\mathfrak{q}}}$ is the largest subextension of L/K such that $e(\mathfrak{q} \cap L^{D_{\mathfrak{q}}} | \mathfrak{p}) = f(\mathfrak{q} \cap L^{D_{\mathfrak{q}}} | \mathfrak{p}) = 1$.
- $L^{I_{\mathfrak{q}}}$ is the largest subextension of L/K such that $e(\mathfrak{q} \cap L^{I_{\mathfrak{q}}} | \mathfrak{p}) = 1$.

Proof. Suppose that $K \subset K' \subset L$ is an intermediate extension and consider L/K'. Theorem 9.8 tells us that the inertial and decomposition groups are intersections with $\mathrm{Gal}(L/K')$. Since group intersection corresponds to field multiplication in the Galois correspondence we obtain the diagram



Since ramification indices and residue degrees multiply along the chosen chain of primes:

- The chosen prime in K' having $e = f = 1$ over $\mathfrak{p}$ is exactly the condition $e' = e$ and $f' = f$. This occurs when $L^{D_{\mathfrak{q}}} K' = L^{D_{\mathfrak{q}}}$ or equivalently $K' \subseteq L^{D_{\mathfrak{q}}}$.
- The chosen prime in K' being unramified over $\mathfrak{p}$ is exactly the condition $e' = e$. This occurs when $L^{I_{\mathfrak{q}}} K' = L^{I_{\mathfrak{q}}}$ or equivalently $K' \subseteq L^{I_{\mathfrak{q}}}$.

□

While this stuff is extremely beautiful, it is also very useful even when things aren't Galois. For instance consider how this understanding enables the following argument.

Theorem 9.10. *Let E_1 and E_2 be finite separable extensions of K. Then:*

- *If $\mathfrak{p}$ is unramified in E_1/K and E_2/K it is unramified in the compositum E_1E_2/K.*
- *If $\mathfrak{p}$ is totally split in E_1/K and E_2/K it is totally split in the compositum E_1E_2/K.*

Proof. Let L be the normal closure of E_1E_2 so that L/K is Galois. By Remark 9.9:

- If $\mathfrak{p}$ is unramified in both extensions then $E_1, E_2 \subseteq L^{I_\mathfrak{q}}$ for every prime $\mathfrak{q}$ above $\mathfrak{p}$ in L. Therefore $E_1E_2 \subseteq L^{I_\mathfrak{q}}$ exactly meaning $\mathfrak{p}$ is unramified in the compositum.
- If $\mathfrak{p}$ is totally split in both extensions then $E_1, E_2 \subseteq L^{D_\mathfrak{q}}$ for every prime $\mathfrak{q}$ above $\mathfrak{p}$ in L. Therefore $E_1E_2 \subseteq L^{D_\mathfrak{q}}$ exactly meaning $\mathfrak{p}$ is totally split in the compositum.

□

Corollary 9.11. *A prime $\mathfrak{p}$ is ramified (resp not totally split) in an extension E/K if and only if it is ramified (resp not totally split) in the Galois closure of E.*

Proof. The Galois closure of a field is the compositum of its conjugate fields. Now apply Theorem 9.10. □

Remark 9.12. Suppose that $K = \mathbb{Q}$. Then the residue field $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is of the form $\mathbb{F}_{p^n}/\mathbb{F}_p$ for some prime p. Now

$$D_\mathfrak{q}/I_\mathfrak{q} \cong \mathrm{Gal}(L^{I_\mathfrak{q}}/L^{D_\mathfrak{q}}) \cong \mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$$

is a cyclic group of order n generated by the *Frobenius element* $x \mapsto x^p$ on the right hand side. In particular when $\mathfrak{p}$ is unramified then $I_\mathfrak{q} = 1$ and we may find a unique element $\sigma \in D_\mathfrak{q}$ with

$$\sigma(\alpha) \equiv \alpha^p \pmod{\mathfrak{q}} \qquad \text{for every } \alpha \in S$$

The Frobenius element is extremely useful, and will make many appearances when studying more number theory.

Problem 9.13. Suppose that $K \subseteq E \subseteq L$ is an intermediate field with $\mathrm{Gal}(L/E) = H$. Then there is a bijective correspondence:

$$\text{primes of } E \text{ above } \mathfrak{p} \longleftrightarrow \text{double cosets } H\sigma D_\mathfrak{q}$$

The Frobenius element is defined only with respect to a single prime over $\mathfrak{p}$. However the next theorem claims it somehow "knows" something about the other primes over $\mathfrak{p}$. This is very non-intuitive but interesting.

Theorem 9.14. *Suppose that $\mathbb{Q} \subseteq E$ and let L be the normal closure of E. The Frobenius element has a natural action on the embeddings of E into L. If p is unramified in E then the cycle type of the Frobenius element is exactly the inertial degrees of the primes above p in E.*

Proof. Let $H = \mathrm{Gal}(L/E)$ and choose a prime $\mathfrak{q}$ of L above p. By Corollary 9.11, p is also unramified in L, so $I_\mathfrak{q} = 1$ and the Frobenius element generates $D_\mathfrak{q}$. The embeddings of E into L are the cosets σH. Their Frobenius orbits are the double cosets $D_\mathfrak{q}\sigma H$, which correspond under inversion to $H\sigma^{-1}D_\mathfrak{q}$. By Problem 9.13, this orbit corresponds to the prime $\mathfrak{q}_E = \sigma^{-1}(\mathfrak{q}) \cap \mathcal{O}_E$. Put $\mathfrak{q}' = \sigma^{-1}(\mathfrak{q})$, so $D_{\mathfrak{q}'} = \sigma^{-1}D_\mathfrak{q}\sigma$. The orbit has size

$$[D_\mathfrak{q} : D_\mathfrak{q} \cap \sigma H \sigma^{-1}] = \frac{|D_{\mathfrak{q}'}|}{|D_{\mathfrak{q}'} \cap H|} = \frac{f(\mathfrak{q}'|p)}{f(\mathfrak{q}'|\mathfrak{q}_E)} = f(\mathfrak{q}_E|p)$$

where the middle equality uses the absence of ramification, and the last is the tower law for residue degrees. □

As an extra treat, here is a really nice proof of quadratic reciprocity using this theory.

Theorem 9.15 (Reciprocity)**.** *Let p and q be distinct odd primes. Then:*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$$

Proof. Let $K = \mathbb{Q}[\zeta_p]$ where ζ_p is a primitive pth root of unity. Since

$$\mathrm{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times \cong \mathbb{Z}/(p-1)\mathbb{Z}$$

it contains a unique subgroup of order $(p-1)/2$ containing the elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ that are squares. Therefore K contains a unique quadratic extension F. On the other hand, only p is ramified in $K/\mathbb{Q}$, so F must have this same property. Now Problem 3.6 allows us to deduce F:

- If $p \equiv 1 \pmod 4$ then p is the only prime ramifying in $\mathbb{Q}[\sqrt{p}]$ and this is the only quadratic field for which this is true.
- If $p \equiv 3 \pmod 4$ then p is the only prime ramifying in $\mathbb{Q}[\sqrt{-p}]$ and this is the only quadratic field for which this is true.

Therefore $F = \mathbb{Q}[\sqrt{d}]$ where $d = (-1)^{(p-1)/2} \cdot p$. Since the Galois group of F has order 2 we identify it with $\{+1, -1\}$. Choose a prime $\mathfrak{q}$ of K above q, and let $\tau \in \mathrm{Gal}(K/\mathbb{Q})$ be its Frobenius element. Then

$$\tau(\alpha) \equiv \alpha^q \pmod{\mathfrak{q}} \qquad \tau(\zeta_p) = \zeta_p^q$$

The restriction $\tau|_F$ is the identity exactly when q is a square in $(\mathbb{Z}/p\mathbb{Z})^\times$, so

$$\tau|_F = \left(\frac{q}{p}\right)$$

On the other hand, this restriction is the Frobenius of F at q. It generates the decomposition group, which is nontrivial exactly when q is inert in F. By Problem 3.6, this gives

$$\tau|_F = \left(\frac{d}{q}\right)$$

Now by equating

$$\left(\frac{q}{p}\right) = \left(\frac{-1}{q}\right)^{(p-1)/2} \cdot \left(\frac{p}{q}\right) = (-1)^{\frac{(p-1)(q-1)}{4}} \cdot \left(\frac{p}{q}\right)$$

where we have used that -1 is a square iff $q \equiv 1 \pmod 4$. □

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